

Abstract

The next generation of gravitational-wave (GW) observatories, Laser Interferometer Space Antenna (LISA) and the Einstein Telescope (ET), will jointly probe compact objects with masses ranging from a few to $\sim 10^7 M_\odot$ over the frequency interval extending from 10^{-4} to 10^4 Hz. This unprecedented observational window makes it essential to identify promising GW sources, determine which of them may emit a detectable signal and characterize its physical properties. Among the most compelling candidates are intermediate-mass black hole (IMBH) binaries, whose inspiral GW emission would offer unique insights into this still poorly constrained population of massive black holes (MBHs).

This thesis investigates the dynamical evolution of IMBH binaries embedded in stellar backgrounds, with particular emphasis on the coalescence timescale and the time evolution of the binary separation and eccentricity. To this end, both the Quinlan–Sesana hybrid model and Ketju direct N-body simulations are employed. While the former approach provides an efficient framework for reconstructing the evolution of BH binaries once the stellar background properties and the binary initial conditions are specified, the Ketju module, implemented into GADGET-4, modeling SMBHs and stars as point particles during close encounters through an algorithmically regularized integrator, accurately captures the orbital decay driven by stellar scattering with unprecedented spatial resolution. The analysis explores different stellar backgrounds, including nucleated dwarf galaxies and observationally motivated high-redshift stellar clumps and galaxy models with enhanced stellar densities resulting from cosmic size evolution. The obtained results confirm that IMBH binaries embedded in nucleated dwarf galaxies can merge within a few Myr. More remarkably, they show that even non-nucleated high-redshift dwarf galaxies and stellar clumps may drive binary coalescence within a Hubble time, identifying these systems as promising sources of detectable gravitational waves. Finally, the reconstructed binary evolution is finally exploited to compute the associated GW signals, providing observational predictions for the LISA and Einstein Telescope era.

Contents

Introduction	3
CHAPTER. 1 Physical Mechanisms of Binary Hardening	7
1.1 Galaxy Merging and Violent Relaxation	8
1.2 Two-Body Relaxation and Dynamical Friction	12
1.3 Black Hole Binary–Star Three-Body Scattering	16
1.4 Stellar Scattering and Loss Cone Problem	19
1.5 Gravitational Radiation from Black Hole Binaries	23
1.5.1 Linearized Gravitational Field Equation	23
1.5.2 Generation and Propagation of Gravitational Waves	24
1.5.3 Gravitational Waves from Black Hole Binaries and Back-Reaction	27
CHAPTER. 2 A Hybrid Approach to IMBH Binary Evolution	31
2.1 Quinlan-Sesana Hybrid Model	32
2.2 Stellar Cusps IMBH Binary Evolution	37
2.3 NGC 205 IMBH Binary evolution	43
2.4 Cosmic Size Evolution Down to NGC 205	48
CHAPTER. 3 IMBH Binary Evolution with KETJU N-Body Simulations	57
3.1 Dynamical Equilibrium of Stellar Systems	58
3.1.1 Equilibrium Theory of Stellar Systems	58
3.1.2 Generation of N-Body ICs with OpOpGadget	61
3.1.3 Dynamical Evolution with KETJU	66
3.2 Analysis of the Generated Initial Conditions	69
3.2.1 NCG 205 N-body Initial Conditions	69
3.2.2 NCG 5206 N-body Initial Conditions	70
3.3 Ketju Numerical Simulations	70

INTRODUCTION

Massive black holes play a fundamental role in the formation and evolution of galaxies. Observations have established that they are ubiquitously found at the centers of nearby galaxies (Kormendy and Richstone 1995), while tight empirical correlations between their masses and the properties of the host galactic bulges suggest a close co-evolution of black holes and galaxies (Magorrian et al. 1998). Moreover, MBHs are believed to be common in galaxy centres at all cosmic epochs; a piece of evidence for their existence, from low to high redshift, comes from the transient activity of galactic nuclei (AGN) over the hosts lifetime (see, e.g., Jahnke (2025)).

Black holes (BHs) span a wide range of masses ($\sim 10 - 10^9 M_\odot$) and various formation and accretion models have been proposed based on their mass. At the two extremes of the black-hole mass spectrum lie stellar-mass black holes and supermassive black holes (SMBHs). Stellar evolution theory predicts that stellar-mass BHs may be the remnants of massive stars with either Zero Age Main Sequence mass $M_{ZAMS} \lesssim 140 M_\odot$ or $M_{ZAMS} \gtrsim 260 M_\odot$; otherwise, they undergo fully disruption and end their life as Pair Instability Super Novae, leaving no remnant behind (Farmer et al. 2019). In recent years, the properties of this population have been significantly constrained through the detection of gravitational waves from merging compact binaries (Abbott et al. 2016), as well as by advances in microlensing techniques capable of identifying isolated BH candidates (Bennett et al. 2002). On the opposite side ($\sim 10^6 - 10^9 M_\odot$), instead, Pulsar timing arrays has already probed the upper mass range of SMBHs population (Taylor 2021), whose most accepted formation model interpret them as the final stage of a secular accretion onto initially lighter black hole seeds (see, e.g., Peterson (1997)).

While substantial progress has been made in detecting and characterizing both stellar-mass BHs and SMBHs, the observational confirmation and detailed investigation of intermediate-mass black holes (IMBHs), lying in the mass range $\sim 10^2 - 10^5 M_\odot$, remain significantly more challenging. Although an increasing number of candidates have been identified in the local Universe, particularly in dwarf galaxies, through accretion signatures (see, e.g., Reines et al. (2013)), their sphere of influence—conventionally defined as the sphere enclosing a stellar mass equal to twice that of the central BH (Sesana and Khan 2015), within which the BH gravitational potential dominates—is typically too small for reliable stellar or gas dynamical measurements, especially at higher redshift. Nevertheless, the discovery of high-redshift quasars (see, e.g., Bañados et al. (2018)) points to the existence of BH seeds with masses comparable to those of IMBHs already in the early Universe, strengthening the idea that even the nuclei of less massive galaxies harbor MBHs. Therefore, it is common practice to infer the central BH mass by extrapolating the empirical correlation between BH mass and the stellar mass of the host galaxy’s bulge, even for quiescent galactic nuclei (see, e.g., Schutte et al. (2019); Maiolino et al. (2024)).

Figure 1 illustrates three distinct theoretical pathways for IMBH formation, highlighting the number of BHs expected to form through each channel, their corresponding mass ranges, and the redshift range over which each formation mechanism is expected to be most efficient. Each formation channel proceeds as follows (Greene et al. 2020):

Pop III. Sufficiently massive primordial stars, commonly referred to as Population III (Pop III) stars, may possess enough gravitational binding energy to withstand the energy released during the photodisintegration-induced collapse of their cores, ultimately leading to the formation of a BH remnant (Fryer et al. 2001).

Direct collapse. At high redshift, metal-free gas clouds may undergo nearly monolithic gravitational collapse without significant fragmentation, in contrast to what is typically observed in the present-day Universe, owing to the inefficient cooling of neutral hydrogen, and lead to the formation of IMBHs (Lodato and Natarajan 2006).

Gravitational runaway. Runaway stellar collisions in high-density clusters may trigger a sequence of stellar mergers, progressively producing more massive stars that can undergo further collisions. The final product of this repeated mergers process may eventually collapse into an IMBH (see, e.g., Rodriguez et al. (2019); Devecchi and Volonteri (2009)).

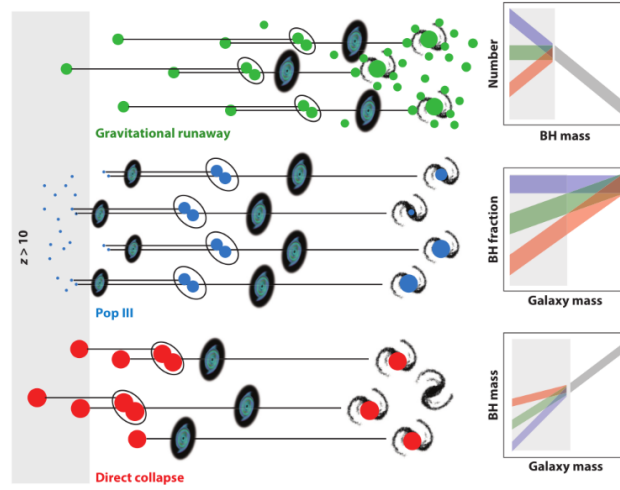


Figure 1: Schematic representation of different BH seeding channels, from their formation and subsequent growth through mergers and accretion to the present day: Pop III stars (blue), direct collapse (red), and gravitational runaway (green). The corresponding present-day BH mass functions, occupation fractions, and BH–galaxy scaling relations are shown on the right. Figure taken from Greene et al. (2020).

Within the framework of the hierarchical growth scenario of galaxies (White and Rees 1978), the formation of IMBH binaries is expected to occur naturally as a consequence of mergers between their host stellar systems (Maiolino et al. 2024). These systems are among the most powerful sources of gravitational waves (GWs) in the Universe (Maggiore 2007), making them prime targets for current and future GW observatories. In particular, the upcoming Laser Interferometer Space Antenna (LISA), a joint ESA/NASA space mission, will probe the low-frequency GW regime, from $\sim 10^{-4}$ to $\sim 10^{-1}$ Hz (Amaro-Seoane, Audley, et al. 2017; Amaro-Seoane, Andrews, et al. 2023), providing access to compact-object binaries with masses of $\sim 10^4 - 10^7, M_\odot$ and enabling the detection of their mergers out to redshift $z \sim 20$ (Colpi et al. 2024). At higher frequencies, the ground-based Einstein Telescope (ET) will complement LISA by extending the accessible frequency and mass ranges towards lower-mass systems, being expected to detect IMBH binaries with masses of $\sim 10^2 - 10^3, M_\odot$ out to redshifts of a few (Abac et al. 2026). Therefore, current and future GW observatories may provide a unique opportunity to unveil the population of IMBHs through their gravitational-wave emission, which may be detectable even when their electromagnetic signatures are too faint to be observed.

In light of this perspective, this thesis investigates the time evolution of the main orbital parameters of IMBH binaries, with particular focus on the semimajor axis a and eccentricity e , and examines how BH masses, stellar background properties, and the binary initial condition influence their dynamical evolution and coalescence timescale through a series of numerical experiments. To this end, the evolution of IMBH binaries is studied using two complementary approaches: the numerical integration of the differential equations for the binary separation and eccentricity provided by the hybrid model developed by Quinlan (1996) and extended by Sesana (2010), and high-resolution, gas-free N-body simulations of mergers between BHs and their stellar hosts performed with the hybrid tree-regularized code KETJU (Mannerkoski et al. 2023).

The suite of numerical experiments presented in Khan and Holley-Bockelmann (2021) and Khan, Javed, et al. (2024) on both nucleated and non-nucleated dwarf galaxies have highlighted the crucial role of the stellar background density, particularly in the vicinity of the BHB sphere of influence, in driving fast and efficient mergers. In particular, high-density stellar environments ensure an adequate supply of low-angular-momentum stars, thereby sustaining efficient hardening via stellar slingshot (see, e.g., Mikkola and Valtonen (1992)), while in low-density environments, the insufficient replenishment of such orbits may lead to binary stalling, giving rise to the so-called "last-parsec problem" (Milosavljević 2003). The evidence that different stellar environments produce peculiar pathways to coalescence highlights the need to investigate the dynamical evolution of IMBH binaries across a representative range of realistic stellar hosts. The first class considered here comprises dwarf galaxies, whose most commonly adopted definition relies on an upper limit of $M_V > -18$ in absolute (V)-band magnitude Grebel et al. (2003). The observation of these systems even at high redshifts (see, e.g., Nguyen et al. (2018); (Kikuchihara et al. 2020)) makes them plausible stellar hosts at the time of IMBH pairing, whose GW emission may be probed by present-day observatories. Although N-body simulations have shown that IMBH binaries embedded in non-nucleated dwarf galaxies have coalescence times exceeding the Hubble time, and despite the fact that most dwarf galaxies harbor a nuclear star cluster (see, e.g., Neumayer et al. (2020)), we artificially evolve non-nucleated dwarf galaxies back to redshift z according to the cosmic size evolution law derived by Ono et al. (2023). Together with the observed increase in the BH-to-bulge mass ratio towards higher redshifts as shown in Figure 2, this effect is expected to partially compensate for the absence of a nuclear star cluster, potentially allowing the binary to coalesce within the present-day Universe.

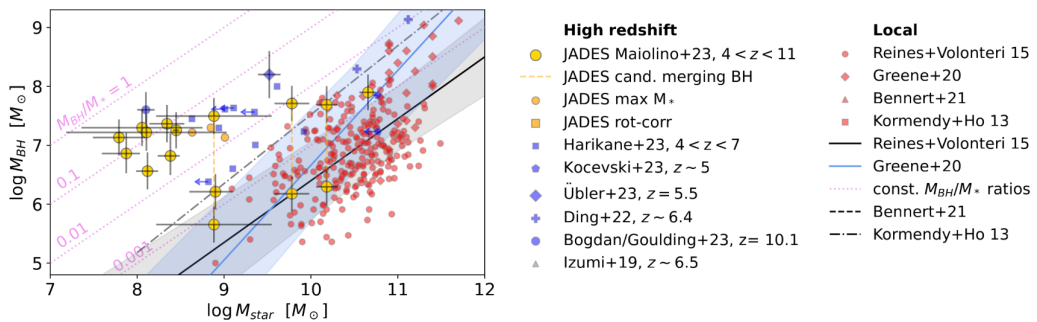


Figure 2: Black hole mass as a function of host galaxy stellar mass. Golden circles denote the JADES results, blue symbols high- z JWST measurements, grey triangles high- z QSOs, and red symbols local galaxies. Dashed orange lines connect candidate merging BHs. Figure taken from Maiolino et al. (2024).

In recent years, several claims have been made regarding the possible presence of IMBHs in Galactic and extragalactic globular clusters (GCs). Although these detections are not always conclusive, dynamical measurements and signatures of accretion in the X-ray band have

provided indications of central IMBHs in a number of old stellar clusters (e.g., Gebhardt et al. (2005); H. Baumgardt (2017)). At the same time, observations of galaxies at very high redshift are revealing stellar systems with properties that may provide insights into the possible progenitors of present-day GCs. An example of this is constituted by the "Misty Moons" (Nakane et al. 2026) and the "Bullet arc" (Bradač et al. 2025), two strongly lensed galaxies at a photometric redshift of $z_{\text{phot}} \sim 11 - 12$ and spectroscopic redshift $z_{\text{spec}} \sim 11$ respectively, which are resolved into several massive stellar clumps. This suggests that star formation in the early universe may have been clump-dominated, with such stellar clumps potentially representing the progenitors of the more compact and less massive GCs observed in the local Universe, before undergoing substantial mass loss due to tidal interactions with their host galaxies. Figure 3 shows the size–mass relation of local star clusters, star-forming clumps, and dwarf galaxies, providing a direct comparison between the properties of local GCs and high- z stellar clumps.

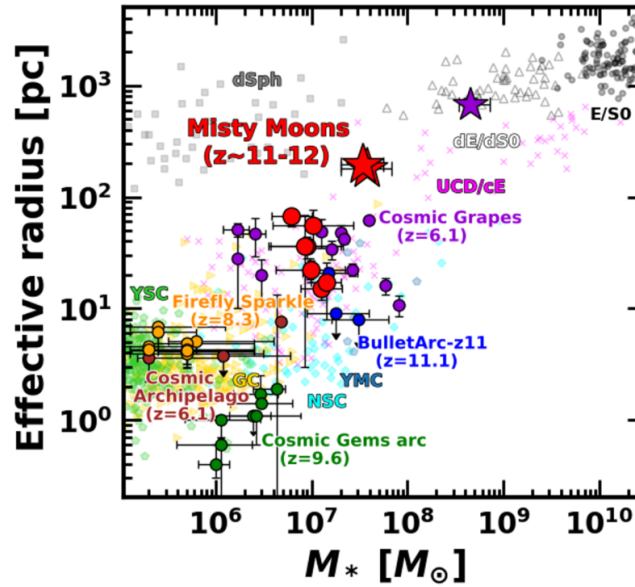


Figure 3: Size–mass relation. Stars denote the entire systems, while circles denote individual stellar clumps. Other symbols denote local Globular Clusters, young massive star clusters, and Nuclear Star Clusters. Figure taken from Nakane et al. (2026).

Mindful of these considerations, we also include stellar-clump-like backgrounds in our suite of numerical experiments, adopting IMBH masses consistent with the empirical scaling relation shown in Figure 2.

The structure of this thesis is outlined as follows.

In chapter 1, we provide a brief overview of the main physical mechanisms promoting the shrinking of BH binaries, including violent relaxation, dynamical friction, stellar scattering by low-angular-momentum stars, and GW emission.

In chapter 2, it will be introduced the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010). We then present the evolutionary tracks obtained for the adopted sample of host galaxies and explore the relevant parameter space, highlighting how different environmental properties and initial conditions affect the binary evolution.

In chapter 3 are reported the main features of the codes used to generate equilibrium N-body initial conditions and carry out the numerical simulations. The chapter then presents the results obtained by combining the KETJU evolution, which follows the BH pair from early merger to the stellar-scattering-dominated regime, with the hybrid model used to track its subsequent evolution towards final coalescence.

Finally, the main results of this work are summarized, and possible future improvements are reported in the concluding chapter.

PHYSICAL MECHANISMS OF BINARY HARDENING

In the "bottom-up" scenario, in which low-mass structures form first and progressively more massive systems assemble through hierarchical clustering and mergers, massive black hole binaries are expected to form as a consequence of mergers involving stellar systems or galaxies embedding such compact objects.

This chapter reviews the main physical mechanisms driving the hardening of MBH binaries, from the initial merger of their hosts to the final coalescence. We first discuss violent relaxation, which efficiently converts the orbital energy of the merging systems into disordered kinetic energy of the newly formed system. Then, dynamical friction is considered. It acts on massive components such as MBHs and drags them toward the centre of the merger remnant, eventually leading to the formation of a gravitationally bound binary. The subsequent evolution is governed by three-body interactions with low-angular momentum stars: we discuss the gravitational slingshot mechanism, through which stellar ejections extract energy and angular momentum from the binary and promote its hardening, as well as the role of loss-cone refilling. Finally, we derive the key equations describing the binary orbital evolution once gravitational-wave emission becomes the dominant mechanism governing the orbital decay.

1.1 Galaxy Merging and Violent Relaxation

When two galaxies approach each other with sufficiently low relative velocity and small impact parameter, the encounter may result in a galaxy merger (see, e.g., A. Toomre and J. Toomre 1972; Cimatti et al. 2019). Galaxy mergers can be broadly classified according to the physical properties of the systems involved. Of particular relevance here are dissipationless, or dry, mergers, which involve gas-poor galaxies and are characterized by negligible radiative energy losses, so that the total energy of the system is approximately conserved (Khochfar and Silk 2009). Numerical N -body simulations have shown that collisions between galaxies of comparable mass, commonly referred to as "major mergers", generally result in a single, virialized remnant (see, e.g. Barnes and Hernquist 1992; Hilz et al. 2012). The formation of this remnant proceeds through a series of relaxation processes driven by the rapidly varying gravitational potential during the encounter, which redistributes the energy and angular momentum of the system and drives the initially out-of-equilibrium configuration toward a new equilibrium state (see, e.g. Lynden-Bell (1967); William C. Saslaw (1970)).

In the scientific literature, several approaches have been proposed to define and derive the timescale of violent relaxation. Here, we follow the approach of Lynden-Bell (1967) and define it as the characteristic oscillation period of the newly formed galaxy around the equilibrium configuration established by the virial condition

$$2K + W = 0, \quad (1.1)$$

holding in its barycentric inertial reference system S_{CM} , with K and W standing for the total kinetic and gravitational potential energy of the galaxy remnant respectively. We therefore first characterize the equilibrium state attained by the system following the relaxation process.

Let us consider two galaxies, each with mass M_i and total energy $E_i = K_i + W_i$. In an ideal dissipationless merger, the total mass and energy are conserved, so that the resulting remnant has mass $M = M_1 + M_2$ and energy $E = E_1 + E_2 + E_{orb}$, where E_{orb} accounts for the orbital energy of the pair, including both its ordered kinetic and mutual gravitational contributions. Following violent relaxation on a timescale T_r , the remnant settles into an equilibrium configuration that can be characterized by the virial velocity dispersion σ_v and gravitational radius r_g , defined by

$$\sigma_v^2 = \frac{2K}{M}, \quad r_g = -\frac{GM^2}{W}, \quad (1.2)$$

which provide proxies for the characteristic stellar velocity dispersion and size of the galaxy in equilibrium, thus representing key information on the final stage reached by the merging galaxies.

By combining the virial condition (1.1)—which applies both to the progenitor galaxies prior to the interaction and to the remnant after the violent relaxation timescale T_r —with energy conservation, one obtains

$$\sigma_v^2 = \frac{M_1\sigma_{v,1}^2 + M_2\sigma_{v,2}^2 - 2E_{orb}}{M_1 + M_2} \quad (1.3)$$

and

$$\frac{(M_1 + M_2)^2}{r_g} = \frac{M_1^2}{r_{g,1}} + \frac{M_2^2}{r_{g,2}} - \frac{2E_{orb}}{G}, \quad (1.4)$$

where $\sigma_{v,i}$ and $r_{g,i}$ denote the virial velocity dispersion and the gravitational radius, respectively, of the colliding systems (Cimatti et al. 2019). From Equations (1.3) and (1.4), it follows that parabolic encounters between identical progenitors, defined by $E_{orb} = 0$, which are the

configuration considered in this thesis to carry out N -body numerical experiments as shown in chapter 3, give rise to a galaxy remnant having

$$M = 2M_1 = 2M_2, \quad r_g = 2r_{g,1} = 2r_{g,2}, \quad \sigma_v = \sigma_{v,1} = \sigma_{v,2}. \quad (1.5)$$

Therefore, before the merger to occur, each progenitor galaxy, which is assumed to be made of N stars of masses m_i and velocities $\mathbf{v}_i$, is individually virialized and satisfies the virial equilibrium condition

$$2 \sum_{i=1}^N \frac{1}{2} m_i v_i^2 - \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N \frac{G m_i m_j}{|\mathbf{x}_i - \mathbf{x}_j|} = 2K + W = 0. \quad (1.6)$$

Nevertheless, this condition is temporarily violated once the collision begins, as the newly formed remnant is driven far from equilibrium by the coherent motion inherited from the relative orbital motion of the progenitors. Consequently, during violent relaxation, Equation (1.1) is no longer satisfied at every instant; rather, the system obeys the time-dependent virial theorem,

$$\frac{1}{2} \ddot{I} = 2K + W, \quad (1.7)$$

where I denotes the polar moment of inertia.

To quantitatively characterize the level of virialization of any system evolving towards equilibrium, it is useful to introduce the so-called virial ratio

$$Q \equiv \frac{2K}{|W|}. \quad (1.8)$$

Figure 1.1 shows the results of N -body simulations of dissipationless galaxy mergers (see Nipoti et al. 2007), focusing on the evolution of the virial ratio Q . The oscillations of Q have been reconstructed both within the framework of Modified Newtonian Dynamics (MOND) and under Newtonian gravity. In the former case, the standard Poisson equation is replaced by the non-relativistic field equation (Bekenstein and Milgrom 1984)

$$\nabla \cdot \left[\mu \left(\frac{\|\nabla\phi\|}{a_0} \right) \nabla\phi \right] = 4\pi G\rho, \quad (1.9)$$

where ϕ denotes the MOND gravitational potential, ρ is the mass density, $a_0 \simeq 1.2 \times 10^{-10} \text{ m/s}^2$ is the characteristic acceleration scale of MOND, and $\mu(y) = y/\sqrt{1+y^2}$ (Milgrom 1983).

Although the merging timescales are significantly longer in MOND than in Newtonian gravity with dark matter, in the Newtonian case the amplitude of the virial ratio oscillations rapidly decreases with time as Q approaches its equilibrium value, $Q = 1$, owing to the progressive conversion of ordered kinetic energy into random stellar motions. As illustrated by the green curve, the system approaches virial equilibrium through a series of progressively damped oscillations, with most of the virialization process occurring in the vicinity of the equilibrium value. We will take advantage of this observed behaviour for estimating the timescale T_r over which violent relaxation takes place.

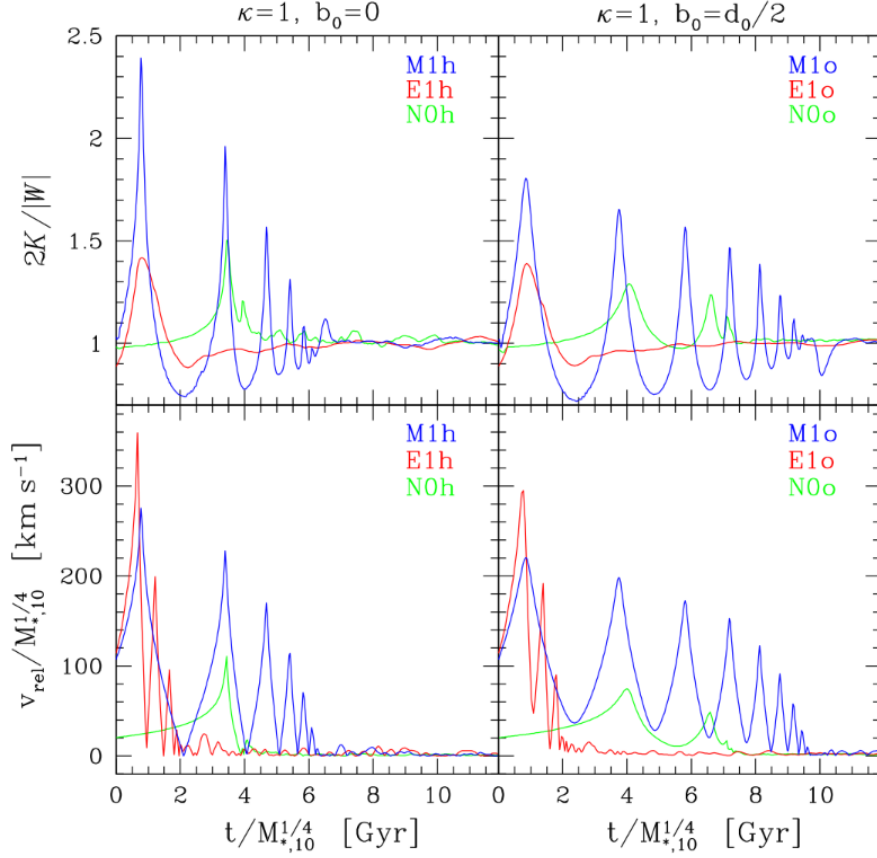


Figure 1.1: Time evolution of the virial ratio (top panels) and of the barycentric relative speed (bottom panels) for the two sets of galaxy merger simulations: head-on (left panels) and off-centre (right panels). Blue, red, and green curves refer to MOND, equivalent Newtonian, and purely baryonic Newtonian simulations, respectively. Figure taken from Nipoti et al. (2007).

To this end, it is useful to start from the physical mechanism underlying violent relaxation and consider the evolution of the energy ϵ_* of an individual star moving in a time-dependent gravitational potential $\psi(\mathbf{x}, t)$. We first show that the stellar energy can change only as a consequence of an explicit time dependence of the gravitational potential. Thus, if the potential is chosen to vanish at infinity, the energy ϵ_* of a star with mass m_* and velocity $\mathbf{v}$ is given by

$$\epsilon_* = \frac{1}{2}m_*v^2 + m_*\psi(\mathbf{x}, t). \quad (1.10)$$

Differentiating both sides of Equation (1.10) with respect to time yields

$$\frac{d\epsilon_*}{dt} = m_* \left[\mathbf{v} \cdot \dot{\mathbf{v}} + \nabla\psi \cdot \mathbf{v} + \frac{\partial\psi}{\partial t} \right]. \quad (1.11)$$

For a star moving under the gravitational potential ψ , its acceleration satisfies $\dot{\mathbf{v}} = -\nabla\psi$ and the previous equation reduces to

$$\frac{d\epsilon}{dt} = \frac{\partial\psi}{\partial t}, \quad (1.12)$$

where $\epsilon = \epsilon_*/m_*$ is the stellar energy per unit mass. Equation (1.12) shows that individual stars continuously gain and lose orbital energy as a consequence of the rapidly varying gravitational field. This naturally motivates defining the violent relaxation timescale T_r as the timescale over which the energy change of a typical star becomes comparable to its initial energy (Lynden-Bell 1967), i.e.

$$T_r = \left\langle \frac{(d\epsilon/dt)^2}{\epsilon^2} \right\rangle^{-1/2} = \left\langle \frac{(\partial\psi/\partial t)^2}{\epsilon^2} \right\rangle^{-1/2}. \quad (1.13)$$

As discussed above, during violent relaxation, the remnant progressively approaches virial equilibrium set by $Q = 1$, as indicated by the damped oscillations of the virial ratio around this value. However, in this regime $\dot{I}$ remains small but non-zero and a mean star experiences changes in both kinetic and gravitational energies in a coupled manner, approximately satisfying

$$\frac{1}{2}m_*v^2 \sim -\frac{1}{4}m_*\psi, \quad (1.14)$$

as dictated by Equation (1.6).

By substituting Equation (1.14) into Equation (1.10), we can express the specific stellar energy solely in terms of the gravitational potential and, mindful of Equation (1.13), obtain the following equivalent form for the violent relaxation timescale:

$$T_r \equiv \frac{3}{4} \left\langle \frac{\psi^2}{\dot{\psi}^2} \right\rangle^{1/2}. \quad (1.15)$$

Dimensional analysis suggests that the polar moment of inertia can be expressed in terms of the total mass M and the square of the instantaneous characteristic radius $R(t)$ as

$$I(t) = \lambda^2 M R^2(t), \quad (1.16)$$

where λ^2 is a dimensionless structural factor of order unity that encodes the instantaneous mass distribution and is determined by the geometrical configuration of the masses m_i .

Let us consider small radial perturbations of the system about its virial equilibrium configuration. In this regime, the instantaneous characteristic radius of the system can be expressed as a small deviation from its equilibrium value, which we identify with the gravitational radius. Therefore, to the first order,

$$R(t) \approx r_g + \delta R, \quad (1.17)$$

with $|\delta R| \ll r_g$ representing the small radial displacement from equilibrium. In this regime, linearizing Equation (1.1) with respect to the perturbation gives

$$2\lambda^2 r_g \delta \ddot{R} - \frac{GM}{r_g^2} \delta R, \quad (1.18)$$

which we recognize to be that of a simple harmonic oscillator, with angular frequency

$$\omega = \left(\frac{GM}{2\lambda^2 r_g^3} \right)^{1/2}. \quad (1.19)$$

It is worth noticing that, introducing the galaxy mean density $\bar{\rho} = 3M/(4\pi r_g^3)$, the latter quantity can be expressed as:

$$\omega = (2\pi G \bar{\rho})^{1/2}. \quad (1.20)$$

Since the gravitational potential scales as $\psi \sim 1/r$, its time variation is governed by the same characteristic timescale as the radial oscillations, i.e. $\sim \omega^{-1}$. Plugging this scaling into the definition of the violent relaxation timescale in Equation (1.15) returns

$$T_r \simeq \frac{3}{4\omega} = \frac{3}{\sqrt{32\pi G \bar{\rho}}}. \quad (1.21)$$

Thus, the violent relaxation timescale is directly related to the characteristic dynamical timescale of the newly formed system and is expected to be of the order of its free-fall timescale $t_{ff} \propto 1/\sqrt{G\bar{\rho}}$.

To obtain an order-of-magnitude estimate of this timescale for the merger of two identical

non-nucleated dwarf galaxies, we may consider the non-nucleated galaxy model built by Khan, Javed, et al. (2024) from a sample of non-nucleated dwarf galaxies of the Fornax cluster. For a galaxy of stellar mass $M_* = 10^8 \times M_\odot$, effective radius $R_e = 0.56 \text{ kpc}$ and Sersic index $n = 0.81$ (see Section 2.3), the violent relaxation timescale is expected to be of the order of $T_r = 21.2 \text{ Myr}$. This estimate has been obtained from Equation (1.21) by considering the average stellar density of the mentioned reference galaxy within its effective radius. This result demonstrates that this mechanism can redistribute the orbital energy of the progenitor galaxies on a much shorter timescale than that associated with two-body relaxation t_{2b} . The latter will be introduced in § 1.2 as a key process responsible for replenishing the so-called binary "loss cone".

1.2 Two-Body Relaxation and Dynamical Friction

Galaxies are multi-component systems in dynamical equilibrium, where different stellar populations, gaseous phases, dust, and dark matter coexist (Cimatti et al. 2019). These environments host an enormous number of stars, which makes it impossible to track the orbit of each element. Nevertheless, by abandoning the idea of characterizing the time evolution of each degree of freedom in the phase-space (PhS), a probabilistic description of the stellar population is still possible in terms of the one-particle distribution function (DF) $f(\mathbf{x}, \mathbf{v}, t)$. This can be interpreted as the probability density of finding a star at a given point $\mathbf{w} = (\mathbf{x}, \mathbf{v})$ in PhS at time t . It is conventionally normalized so that

$$\rho(\mathbf{x}, t) = \int_{\mathbb{R}^3} d^3\mathbf{v} f(\mathbf{x}, \mathbf{v}, t), \quad (1.22)$$

where $\rho(\mathbf{x}, t)$ is the stellar density providing a smooth approximation to the actual, discrete distribution of N stars.

In the collisionless approximation, DF is conserved along the orbits of the gravitational potential $\Phi(\mathbf{x}, t)$ associated with the stellar density $\rho(\mathbf{x}, t)$. In this regime, it thus obeys the so-called Collisionless Boltzmann Equation (CBE),

$$\frac{df}{dt} \equiv \frac{\partial f}{\partial t} + \frac{d\mathbf{x}}{dt} \cdot \frac{\partial f}{\partial \mathbf{x}} + \frac{d\mathbf{v}}{dt} \cdot \frac{\partial f}{\partial \mathbf{v}} = 0. \quad (1.23)$$

Equation (1.23) is expected to hold as long as the effects of repeated gravitational encounters between stars remain negligible over the age of the system. Although individual encounters are typically weak, their cumulative effect can progressively redistribute energy and angular momentum among stars, causing their trajectories to deviate from those drawn by the smooth potential Φ and DF to diffuse in PS (Binney and Tremaine 2011).

The characteristic timescale over which these cumulative encounters significantly alter stellar trajectories is the two-body relaxation time t_{2b} , which provides a natural criterion for assessing the validity of the collisionless approximation. Systems whose age is shorter than their two-body relaxation time, i.e. $t_{age} < t_{2b}$, are conventionally referred to as collisionless; otherwise, they will be referred to as collisional (Ciotti 2021).

The approach of Chandrasekhar (1943) for estimating t_{2b} relies on the impulsive approximation. It consists of two main assumptions:

1. Stellar encounters are treated as independent events, so that correlations between consecutive encounters and simultaneous multiple interactions are neglected.
2. The gravitational influence of the remaining $N-2$ stars is neglected during each encounter, reducing the problem to a sequence of isolated hyperbolic two-body interactions.

Let us work under the simplifying assumption of a spatially homogeneous distribution of field stars with isotropic velocity field. The DF $f(\mathbf{x}_f, \mathbf{v}_f)$ of such system can be expressed as the product of the position-independent number density n_f and probability density function $g(v_f)$ in the velocity space,

$$f(\mathbf{x}_f, \mathbf{v}_f) = n_f g(v_f). \quad (1.24)$$

Two-body encounters between a test particle of mass m_t and a field star of mass m_f induce small changes in an orbital property P of the test particle. Although each interaction produces only a small perturbation, the cumulative effect of many encounters causes P to diffuse. This process is quantified by the diffusion coefficient $D(P)$, which measures the rate at which P changes per unit time. Owing to the Impulsive Approximation, it can be written as

$$D(P) \equiv 2\pi n_f \int_{\mathbb{R}^3} d^3\mathbf{v}_f \|\mathbf{v}_t - \mathbf{v}_f\| g(v_f) \int_0^\infty db \langle P \rangle b, \quad (1.25)$$

where $\langle P \rangle$ denotes the angle-averaged value of the orbital property P , and b is the impact parameter of the hyperbolic encounter (Ciotti 2021). Since significant deflections of the original orbits are expected beyond t_{2b} , corresponding to the onset of the collisional regime, a useful indicator for determining t_{2b} may be the orbital property $P = \Delta\mathbf{v}_{t\perp}$, that is the component of the test-particle velocity change perpendicular to the initial relative velocity with respect to the field particle. However, under the assumption of isotropy, the first-order diffusion coefficients of the velocity components vanish (Binney and Tremaine 2011). Consequently, the leading non-vanishing contribution is represented by the second-order diffusion coefficient, hence, making the quantity $P = \|\Delta\mathbf{v}_{t\perp}\|^2$ the relevant orbital parameter for our analysis. By taking advantage of the small-angle scattering approximation (see, e.g., Chandrasekhar (1943)),

$$\|\Delta\mathbf{v}_{t\perp}\| \simeq \frac{2Gm_f}{vb}, \quad (1.26)$$

with v denoting the modulus of the relative velocity between m_t and m_f at infinity, and Equation (1.25) becomes

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \simeq 8\pi G^2 m_f^2 n_f \int_{\mathbb{R}^3} d^3\mathbf{v}_f \frac{g(v_f)}{v} \int_0^\infty \frac{db}{b}. \quad (1.27)$$

The integral in Equation (1.27) exhibits two logarithmic divergences. The former, known as ultraviolet (UV) divergence, arises in the limit $b \rightarrow 0$ and results from extending the small-angle scattering approximation to encounters with sufficiently small impact parameters, where it breaks down. The latter, the infrared (IR) divergence, arises in the limit $b \rightarrow \infty$ and is a consequence of modelling the stellar system as infinite and homogeneous. The UV divergence is avoided by replacing the small-angle approximated solution in Equation (1.26) with the exact two-body scattering solution; mathematically, this operation is equivalent to impose a lower cutoff to the impact parameter equals $b_{\pi/2}(v)$, corresponding to the impact parameter associated with a deflection angle of $\pi/2$. The IR divergence cannot be eliminated and thus we have to set a fiducial upper cutoff $b_{\max}$.

The resulting diffusion coefficient can then be written as

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \sim 8\pi G^2 n_f m_f^2 \int_{\mathbb{R}^3} \frac{g(v_f) \ln \Lambda}{v} d^3\mathbf{v}_f, \quad (1.28)$$

where $\ln(\Lambda) = \ln(b_{\max}/b_{\pi/2})$, is the so-called Coulomb logarithm. Although it depends on test-field particle relative velocity through $b_{\pi/2}(v)$, corresponding to a deflection angle of $\pi/2$, by defining the velocity-weighted Coulomb logarithm $\ln(\bar{\Lambda})$ so that

$$\int_{\mathbb{R}^3} \frac{g(v_f) \ln \Lambda}{\|\mathbf{v}_t - \mathbf{v}_f\|} d^3\mathbf{v}_f \equiv \Psi(v_t) \ln \bar{\Lambda}, \quad (1.29)$$

with

$$\Psi(v_t) = \int_{\mathcal{R}^3} \frac{g(v_f)}{\|\mathbf{v}_t - \mathbf{v}_f\|} d^3\mathbf{v}_f, \quad (1.30)$$

being the Rosenbluth potential, Equation (1.28) turns into

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \sim 8\pi G^2 n_f m_f^2 \Psi(v_t) \ln \bar{\Lambda}. \quad (1.31)$$

One of the major simplifications introduced by the assumption of isotropy is that $\Psi(v_t)$ can be expressed as

$$\Psi(v_t) = \int_{\mathcal{R}^3} \frac{g(v_f)}{\|\mathbf{v}_t - \mathbf{v}_f\|} d^3\mathbf{v}_f \equiv \frac{\Xi(v_t)}{v_t} + 4\pi \int_{v_t}^{\infty} g(v_f) v_f dv_f, \quad (1.32)$$

where

$$\Xi(v_t) = 4\pi \int_0^{v_t} g(v_f) v_f^2 dv_f \quad (1.33)$$

is normalized to unity as a consequence of the normalization of velocity-space distribution function $g(v_f)$.

Finally, if one defines the two-body relaxation time t_{2b} as the ratio between the orbital parameter $P = \|\Delta\mathbf{v}_{t\perp}\|^2$ and the associated diffusion coefficient, it turns out that

$$t_{2b} \equiv \frac{v_t^2(-\infty)}{D(\Delta E_{t\perp})} \sim \frac{v_t^3}{8\pi G^2 n_f m_f^2 \ln \bar{\Lambda}}, \quad (1.34)$$

holding in the high velocity limit from Equation (1.32). Keeping in mind the estimate of T_r obtained in § 1.1 for a representative non-nucleated dwarf galaxy, we now compare it with the corresponding two-body relaxation time, t_{2b} , of the same system. An order-of-magnitude estimate can be obtained from

$$t_{2b} \approx t_{cross} \frac{N}{12 \log N}, \quad (1.35)$$

where N denotes the number of stars and $t_{cross} = 2\sigma_v/R_e$ (Ciotti 2021). Assuming equal-mass stars with $m_* = 1, M_\odot$, Equation (1.35) yields a t_{2b} of the order of tens of Gyr, thus significantly exceeding the corresponding value of T_r .

As discussed above, two-body encounters redistribute the test particle's kinetic energy, increasing its velocity component perpendicular to the initial direction of motion. In this sense, two-body interactions act as a heating mechanism in the transverse direction. Conversely, dynamical friction produces a systematic loss of kinetic energy along the direction of motion and behaving as cooling process (see, e.g., Chandrasekhar 1943; Binney and Tremaine 2011). This results in the sinking of massive objects, such as an intruding secondary BH, and N -body simulations by Baumgardt et al. (2006) have demonstrated that dynamical friction efficiently drives the secondary BH down to a_0 , defined as the radius enclosing a stellar mass equal to twice that of M_2 , that is

$$M_*(< a_0) = 2M_2. \quad (1.36)$$

In order to derive an estimate for the dynamical friction timescale t_{df} , we follow the same approach and notation adopted above and first calculate the diffusion coefficient associated with the orbital property $P = \Delta\mathbf{v}_t$ for a test particle of mass m_t moving through a stellar field composed of particles of mass m_f , i.e.

$$\frac{d\mathbf{v}_t}{dt} = D[\Delta\mathbf{v}_t] = -4\pi G^2 m_t m_f \ln \Lambda \int d^3\mathbf{v}_f f(\mathbf{v}_f) \frac{\mathbf{v}_t - \mathbf{v}_f}{|\mathbf{v}_t - \mathbf{v}_f|^3}. \quad (1.37)$$

Assuming an isotropic velocity field, Equation (1.37) simplifies to

$$\frac{d\mathbf{v}_t}{dt} = -16\pi^2 G^2 m_t m_f \ln \Lambda \left[\int_0^{v_t} dv_f v_f^2 f(v_f) \right] \frac{\mathbf{v}_t}{v_t^3}, \quad (1.38)$$

known as Chandrasekhar's dynamical friction formula (Chandrasekhar 1943). Equation (1.38) shows that the effect of dynamical friction is to systematically decrease the test particle's velocity through an acceleration directed antiparallel to its velocity vector.

By following an analogous derivation to that used to estimate the two-body relaxation timescale t_{2b} in Equation (1.34), the dynamical friction timescale t_{df} can be expressed as

$$t_{df} \equiv \frac{\|\mathbf{v}_t(-\infty)\|}{\|D(\Delta\mathbf{v}_t)\|} \sim \frac{v_t^3}{4\pi G^2 n_f m_f (m_f + m_t) \ln(\bar{\Lambda}) \Xi(v_t)}, \quad (1.39)$$

We conclude this Section by deriving, under physically motivated assumptions, an analytic expression for the dynamical friction timescale t_{df} of an intruding secondary BH of mass m_2 and velocity $\mathbf{v}_2$, which sinks towards the center of a stellar background hosting a primary BH of mass m_1 . Let us assume the stellar distribution to possess a Maxwellian velocity distribution and a power-law density profile

$$\rho(r) = \rho_0 \left(\frac{r}{r_0} \right)^{-\gamma}. \quad (1.40)$$

We begin by noting that for a Maxwellian velocity distribution, the integral over the stellar velocity distribution appearing in Equation (1.38) can be evaluated analytically, yielding

$$\frac{d\mathbf{v}_2}{dt} = -4\pi \ln \Lambda \left[\operatorname{erf}(X) - \frac{2X}{\sqrt{\pi}} e^{-X^2} \right] \frac{G^2 m_2 \rho(r)}{v_2^3} \mathbf{v}_2 \quad (1.41)$$

where

$$\operatorname{erf}(X) = \frac{2}{\sqrt{\pi}} \int_0^X dt e^{-t^2} \quad (1.42)$$

is the so-called error function. Here, $X \equiv v_c/(\sqrt{2}\sigma)$ is a dimensionless parameter that combines the circular velocity v_c of the test particle with the velocity dispersion σ of the stellar background.

For an intruder BH moving along quasi-circular orbits within the sphere of influence of the primary BH, the angular momentum and the dynamical friction force acting on it are given by

$$\tilde{L} = m_2 \sqrt{G m_1 r} \quad (1.43)$$

and

$$\tilde{F} = m_2 \frac{dv_2}{dt} = -C(\Lambda, X) \frac{G^2 m_2^2 \rho(r)}{v_c^2} \quad (1.44)$$

respectively. Here, $C(\Lambda, X)$ is the dimensionless factor implicitly defined through the Equations (1.41) and (1.44).

At this point, we can exploit the fact that, within the primary BH's sphere of influence, the circular velocity can be approximated as

$$v_c^2 = \frac{G m_1}{r}. \quad (1.45)$$

Under these assumptions, the second cardinal equation of dynamics takes the form

$$\frac{d\tilde{L}}{dt} = \tilde{F} r, \quad (1.46)$$

and leads to the differential equation for $r = r(t)$ upon substituting the expressions given in Equations (1.44), (1.43) and (1.45) (see, e.g., Baumgardt et al. 2006; Gualandris et al. 2022):

$$r^{-5/2} \frac{dr}{dt} = -2C(\Lambda, X(r)) \frac{\sqrt{G} m_t}{m_f^{3/2}} \rho(r). \quad (1.47)$$

Equation (1.47) admits an analytic solution for an isotropic power-law density profile. Indeed, in this case

$$X = \sqrt{\gamma - 1} \quad (1.48)$$

and is therefore independent of radius (Madigan et al. 2011). Integrating the differential equation subject to the initial condition $r(t = 0) = r_0 < r_{inf}$, we obtain

$$r(t) = \begin{cases} r_0 \left[1 - 2C(\Lambda, X) \frac{\sqrt{G}m_2}{m_1^{3/2}} \left(\gamma - \frac{3}{2} \right) \rho_0 r_0^{3/2} t \right]^{\frac{1}{\gamma-3/2}} & \text{if } 1 < \gamma \leq 3 \wedge \gamma \neq \frac{3}{2} \\ r_0 \exp \left[-2C(\Lambda, X) \frac{\sqrt{G}m_2}{m_1^{3/2}} \rho_0 r_0^{3/2} t \right] & \text{if } \gamma = \frac{3}{2} \end{cases} \quad (1.49)$$

Equation (1.49) applies to an isotropic, power-law stellar background, or more generally to a stellar distribution that approaches such a density profile within the sphere of influence of the central BH and provides an estimate of the duration of the final stage of dynamical friction, from r_{inf} down to a_0 . Indeed, setting the reference radius r_0 and density ρ_0 to their respective values at the influence radius, and defining t_{fric} such that $r(t_{fric}) = a_0$, we obtain

$$t_{fric} = \begin{cases} \frac{m_1^{3/2}}{2C(\Lambda, X) \sqrt{G} m_2 \rho_{inf} r_{inf}^\gamma} \frac{r_{inf}^{\gamma-3/2} - a_0^{\gamma-3/2}}{\gamma - \frac{3}{2}}, & \text{if } \gamma \neq \frac{3}{2}, \\ \frac{m_1^{3/2}}{2C(\Lambda, X) \sqrt{G} m_2 \rho_{inf} r_{inf}^{3/2}} \ln \left(\frac{r_{inf}}{a_0} \right), & \text{if } \gamma = \frac{3}{2}. \end{cases} \quad (1.50)$$

We now seek to determine both r_{inf} and a_0 for a power-law stellar distribution hosting a BH binary of total mass $M_{12} = m_1 + m_2$. In this case, the influence radius r_{inf} , defined as

$$M_*(< r_{inf}) = 2M_{12}, \quad (1.51)$$

and a_0 , defined by Equation (1.36), admit an analytical expression. They turn to be

$$r_{inf} = r_0 \left[\frac{(3 - \gamma)M_{12}}{2\pi\rho_0 r_0^3} \right]^{1/(3-\gamma)} \quad (1.52)$$

and

$$a_0 = r_{inf} \left(\frac{q}{1+q} \right)^{1/(3-\gamma)}, \quad (1.53)$$

respectively, with $q = m_2/m_1 < 1$ denoting the BH masses ratio. Equation (1.53) shows that, at fixed power-law slope γ , a_0 spans the range from 0 to $r_{inf} 2^{1/\gamma-3}$ as q varies from 0 to 1.

From a_0 onward, dynamical friction, which has driven the orbital decay and pairing of the BHs up to this point, becomes inefficient and the further hardening of the binary becomes primarily driven by individual encounters with low-angular-momentum stars that are unbound to the binary and can extract energy and angular momentum from it. We address this shrinking mechanism in § 1.3.

1.3 Black Hole Binary–Star Three-Body Scattering

As discussed in § 1.1 and § 1.2, during a dry, major merger, the remnant evolves towards equilibrium on the violent relaxation timescale T_r , while the MBHs initially located at the centers of the progenitor galaxies sink towards the center of the remnant through dynamical friction on a timescale t_{fric} . As their separation decreases, the MBHs eventually cross each other sphere of influence and form a gravitationally bound pair at a separation of order a_0 ,

defined as the binary separation at which the enclosed stellar mass equals the secondary BH mass M_2 (Baumgardt et al. 2006). At this stage, the binary separation is sufficiently small that distant stars mainly perturb the binary center of mass, with negligible effect on its orbital energy (Quinlan 1996). Further hardening is therefore driven predominantly by low angular momentum stars passing in the vicinity of the binary and extracting both energy and angular momentum from it. Each time such a star—belonging to the so-called binary "loss cone", as we will discuss in more detail in § 1.4—passes sufficiently close to the binary, a three-body encounter takes place. Under suitable conditions, this interaction produces hyper-velocity stars (see, e.g., Quinlan 1996; Sesana, Haardt, et al. 2006) through the so-called stellar slingshot mechanism (Mikkola and Valtonen 1992). In this section, we review the main physical aspects of three-body scattering and the stellar slingshot mechanism, which are key processes driving the hardening of MBHBs.

We begin by considering the so-called "two-body problem" and reviewing some of its key properties, which will be useful in understanding the physics of the stellar slingshot. Adopting the same notation as in the previous Section, let us consider a test particle and a field particle, with masses m_t and m_f , respectively, interacting solely through gravity. Introducing the total mass of the system, $M = m_t + m_f$, and the reduced mass, $\mu = m_t m_f / M$, the total energy in the mass centre frame S_{CM} is given by

$$E = \frac{1}{2}\mu\|\mathbf{v}\|^2 - \frac{Gm_f m_t}{r}, \quad (1.54)$$

where $r(t) \equiv \|\mathbf{x}_t - \mathbf{x}_f\|$ and $\mathbf{v}(t) \equiv \mathbf{v}_t - \mathbf{v}_f$ denote particles relative separation and velocity. For two unbound particles moving along hyperbolic orbits in S_{CM} , energy conservation, as expressed by Equation (1.54), evaluated in the asymptotic limit $r(-\infty) = r(+\infty) \rightarrow +\infty$, yields

$$\|\mathbf{v}(-\infty)\| = \|\mathbf{v}(+\infty)\|. \quad (1.55)$$

Although it is not conserved along the entire orbit, Equation (1.55) shows that the magnitude of the relative velocity—i.e., the velocity of the test particle as measured by an observer comoving with the field particle—is the same at the two asymptotic limits. Therefore, the encounter results only in a rotation of the velocity vector $\mathbf{v}$ and the corresponding velocity change $\Delta\mathbf{v} = \mathbf{v}(+\infty) - \mathbf{v}(-\infty)$ can be conveniently decomposed as the sum of the components perpendicular and parallel to the initial relative velocity $\mathbf{v}(-\infty)$ as follows:

$$\mathbf{v}(+\infty) = \mathbf{v}(-\infty) + \Delta\mathbf{v}_{\parallel} + \Delta\mathbf{v}_{\perp}. \quad (1.56)$$

Substituting the expression for $\mathbf{v}(+\infty)$ given in Equation (1.56) into Equation (1.55) leads to the identity

$$\|\Delta\mathbf{v}_{\parallel}\|^2 + \|\Delta\mathbf{v}_{\perp}\|^2 + \langle\Delta\mathbf{v}_{\parallel}, \mathbf{v}(-\infty)\rangle = 0. \quad (1.57)$$

A first inspection of Equation (1.57) reveals that a necessary condition for it to hold is that the scalar product be negative. This observation will be crucial for assessing whether a binary BH–star interaction results in binary hardening or softening (Ciotti 2021).

The results derived above impose a constraint on the relative velocity of the interacting particles in a specific inertial frame. However, the interaction can be described in any inertial frame S_0 . Indeed, while Equation (1.55) shows that both the total energy and the individual mechanical energies of the particles are conserved between the two asymptotic limits, the close encounter generally induces changes in the individual energies E_t and E_f in S_0 .

Between the two configurations at infinite relative separation, the test particle undergoes a net variation in mechanical energy,

$$\Delta E_t = \frac{1}{2}m_t[|\mathbf{v}_t(+\infty)|^2 - |\mathbf{v}_t(-\infty)|^2] = -\Delta E_f, \quad (1.58)$$

which is exactly compensated by the corresponding variation in the mechanical energy of the field particle, such that

$$\Delta E = \Delta E_t + \Delta E_f = 0. \quad (1.59)$$

By combining the velocity composition law

$$\mathbf{v}_t(t) = \mathbf{v}_{\text{CM}} + \frac{\mu}{m_t} \mathbf{v}(t) \quad (1.60)$$

with the conservation of the center-of-mass velocity vector $\mathbf{v}_{\text{CM}}$, which holds for isolated systems, Equation (1.58) can be written as (Ciotti 2021)

$$\Delta E_t = \mu \langle \Delta \mathbf{v}, \mathbf{v}_{\text{CM}} \rangle. \quad (1.61)$$

We are now in a position to address three-body scattering involving a gravitationally bound BH binary and an incoming star. The dynamics of this encounter can be understood by considering the limiting case of a strongly asymmetric binary in which $M_1 \gg M_2 \gg m_*$ so that both the star and the secondary BH, having masses m_* and M_2 , can travel along nearly Keplerian orbits around M_1 with slightly different radii and, consequently, possess slightly different Keplerian velocities. The secondary BH acts as the field particle, while the star plays the role of the test particle. The primary BH serves only to establish the relative velocity between the two bodies. As a result, the center of mass motion of this gravitationally unbound pair is essentially determined by the orbit of M_2 around M_1 . Therefore, the approximation

$$\mathbf{x}_{\text{CM}} \simeq \mathbf{x}_f, \quad \mathbf{v}_{\text{CM}} \simeq \mathbf{v}_f, \quad (1.62)$$

applies and the center of mass will move locally at constant velocity if the observation time is short compared to the orbital period, thus making negligible its acceleration.

Hereafter, by analogy with the previous notation, we identify $M_2 \equiv m_f$ and $m_* \equiv m_t$.

Binary hardening. When the star is outside the secondary BH orbit, the Keplerian velocity profile imposes $\|\mathbf{v}_f(-\infty)\| = \|\mathbf{v}_{\text{CM}}\| > \|\mathbf{v}_t(-\infty)\|$. Therefore, at point where m_f surpasses m_t and they are both located on the same radius, $\mathbf{v}(-\infty)$ is nearly antiparallel to $\mathbf{v}_{\text{CM}}$ and $\Delta \mathbf{v}_{\parallel}$ can be read as the projected vector along $\mathbf{v}_{\text{CM}}$ rather than $\mathbf{v}(-\infty)$. Taking advantage of Equation (1.57), it follows that

$$0 < \langle \Delta \mathbf{v}_{\parallel}, \mathbf{v}_{\text{CM}} \rangle = \langle \Delta \mathbf{v}, \mathbf{v}_{\text{CM}} \rangle = \Delta E_t / \mu. \quad (1.63)$$

Equation (1.63) shows that in the reference system centered on M_1 the star gains orbital energy and is scattered onto a wider orbit. On the other hand, the secondary BH experiences an equal and opposite change in orbital energy, leading to a reduction of the binary separation and, consequently, to binary hardening.

It is worth noticing that if $\Delta E_t > |E_t(-\infty)|$, which occurs at very high circular velocity of the binary BHs, stellar ejection occurs; in this regime “hypervelocity stars” (HVSs) travelling along nearly radial orbits are produced (see, e.g., Quinlan 1996; Sesana, Haardt, et al. 2006).

Binary softening. Contrary to the previous case, binary softening takes place when the star is inside the secondary BH orbit. Now $\mathbf{v}(-\infty)$ is nearly parallel to $\mathbf{v}_{\text{CM}}$ at the crossover point and by repeating the same arguments used above it follows

$$E_t / \mu = \langle \Delta \mathbf{v}, \mathbf{v}_{\text{CM}} \rangle = \langle \Delta \mathbf{v}_{\parallel}, \mathbf{v}_{\text{CM}} \rangle < 0. \quad (1.64)$$

On one hand the star loses energy to the binary, it is pushed inward and may undergo tidal disruption due to primary BH (Lightman and Shapiro 1977); on the other hand the binary gains energy and its semimajor axis increases, resulting in binary softening.

Although our discussion has so far focused on a specific three-body scattering configuration, the results can be readily generalized by observing that the distinction between binary hardening and softening is determined by the relative magnitude of the BH binary's orbital velocity and the velocity of the incoming loss-cone star.

The binary BH embedded in an isotropic stellar background with constant velocity dispersion σ is expected to undergo three-body scatterings with loss-cone stars that will approach the galactic centre where the binary lives with velocity of the order of σ . Because the specific binding energy of a Keplerian with reduced mass μ and semimajor axis a is

$$E_b = \frac{G\mu}{2a} \quad (1.65)$$

and the specific energy of the loss-cone star is of the order of σ^2 , the velocity criterion can be conveniently reformulated in terms of the specific energy. When $E_b \ll \sigma^2$, the binary is said to be soft, and repeated stellar encounters tend to further decrease its binding energy. On the other hand, when $E_b \gg \sigma^2$, the binary is hard, and stellar encounters, on average, increase its binding energy. This behavior is well summarized by Heggie's law (see, e.g., D. C. Heggie (1975); Douglas C. Heggie and Hut (1993).

Heggie's law. – *On average, hard binaries get harder and soft binaries get softer.*

By extrapolation of the two limiting cases, one can define the "hardening radius" a_h as the binary separation at which binary passes from the soft to the hard regime. Throughout this thesis we adopt the definition proposed by Quinlan (1996), unless otherwise specified:

$$a_h = \frac{GM_2}{4\sigma^2}. \quad (1.66)$$

At this stage, it should be clear that stellar scattering plays a fundamental role in driving the binary toward the regime in which GWs emission dominates its evolution. Nevertheless, one may wonder how a binary can evolve from the soft to the hard regime. The key point is that this transition is driven primarily by dynamical friction against the stellar background, rather than by individual stellar encounters, as discussed in § 1.2.

The discussion then turns to the loss-cone refilling problem, whose efficient replenishment is essential to ensure a sustained supply of low angular momentum stars capable of driving the continued hardening of the binary through stellar ejection.

1.4 Stellar Scattering and Loss Cone Problem

In § 1.3 we have been concerned in describing the physics of three-body scattering, pointing out its relevance to the orbital decay of the BH binary. In the following Section, instead, we will be explaining which stars can effectively exchange energy and angular momentum with the binary, the rate at which these close encounters take place and how all of this enters the self consistent model for MBHBs evolution developed by Quinlan (1996) and Sesana (2010) that will be presented in Chapter 2.

Let $a(t)$ be the binary semimajor axis. Numerical experiments demonstrated that only stars intersecting the binary orbit can efficiently extract energy and angular momentum from the binary when ejected at higher velocity (Mikkola and Valtonen 1992). This class of stars of the stellar background populates the so-called "binary loss cone" and comprises both low-energy stars and high-energy, low angular momentum stars. Under the assumption of Keplerian orbits, the former have small mean radius ($\langle r \rangle < a$) and specific energy $E_* < -GM/a$, while latter spend most of their orbital period $P(E_*)$ far from the binary but a close encounter can occur at the pericenter r_P whenever $r_P < a$.

The size of the binary loss cone, J_{lc} , in specific angular momentum space, hereafter J-space, is a function of E_* and can be determined from the maximum and minimum angular momentum values allowed at a given specific energy. At the configuration of closest approach, the star radial velocity vanishes and the specific energy in terms of r_P is

$$E_* = \frac{J^2}{2r_P^2} - \frac{GM}{r_P}. \quad (1.67)$$

From Equation (1.67) it follows that there exists a value of J below which the star enters the binary loss cone. Hereafter, we refer to it as $J_{lc}(E_*)$ and is given by

$$J_{lc}(E_*) \equiv \left[2 \left(E_* + \frac{GM}{a} \right) \right]^{1/2} a, \quad (1.68)$$

which can be approximated to $\sim \sqrt{2GMa}$ as the vast majority of stars orbit far from the binary and have a specific angular energy $|E_*| \ll GM/a$.

Besides, stars of fixed specific energy E_* , other than a minimum of the specific angular momentum $J_{min} = J_{lc}(E_*)$, possess a maximum set by the chain of inequalities

$$E_* \geq \frac{J^2}{2r^2} - \frac{GM}{r} \equiv V_{eff}(r) \geq V_{eff,min} = -\frac{G^2 M^2}{2J^2}, \quad (1.69)$$

holding because of the non-negative behavior of the radial kinetic term. This constrain poses an upper limit to J , that is

$$J_{max}(E_*) = \frac{GM}{\sqrt{2|E_*|}}, \quad (1.70)$$

corresponding to the specific angular momentum of a circular orbit of specific energy E_* (Capozziello and De Laurentis 2008).

Once the binary has depleted, on a dynamical timescale $P(E_*)$, the population of stars that are bound to it and intersect its orbit—most of which, hence, reside within the binar sphere of influence—further hardening through stellar slingshot interactions depends on the rate at which the binary loss cone is replenished by stellar diffusion in both energy and angular momentum space, allowing new stars to enter the loss cone (Lightman and Shapiro 1977). This diffusion arises from stellar encounters that have been proven to be efficient in exchanging angular momentum and results in a random walk (RW) in J-space (see, e.g., Milosavljević 2003; Merritt 2013). Determining the rate at which this process occurs is in order to understand how it enters the self-consistent hybrid model devoted by Quinlan (1996) and Sesana (2010), discussed in chapter 2.

To this end, let us define j_2 the rms angular momentum transferred in one orbital period and $J_{max}(E_*)$ the same quantity associated with the timescale t_{2b} . Only stars with $J_{lc} \leq J \leq J_{lc} + j_2$ can be scattered into the loss cone over one orbital period and the kind of loss cone replenishing mechanism, along with the associated timescale, are set by the dimensionless quantity

$$q(E_*) \equiv \frac{j_2^2}{J_{min}^2}. \quad (1.71)$$

Taking Figure 1.2 as a reference, $q(E_*)$ enables us to distinguish between two extreme scenarios:

Empty Loss Cone. This condition is represented by picture on the left in Figure 1.2(b).

When $q(E_*) \ll 1$ the RW step size is very small compared to the loss cone size, making the process ascribable to a diffusion problem with timescale t_{2b} . Therefore, stars are removed from the loss cone faster than the rate at which it is fed, hence, remaining nearly empty.

Full Loss Cone. This condition is represented by the picture on the right in Figure 1.2(b). The loss cone is smaller than the RW step size and even high J stars can be scattered into the loss cone by many, weak stellar encounters taking place during one orbital period. Thus, the rate at which loss cone stars are expelled is lower than the rate of refilling, causing the loss cone to remain almost full.

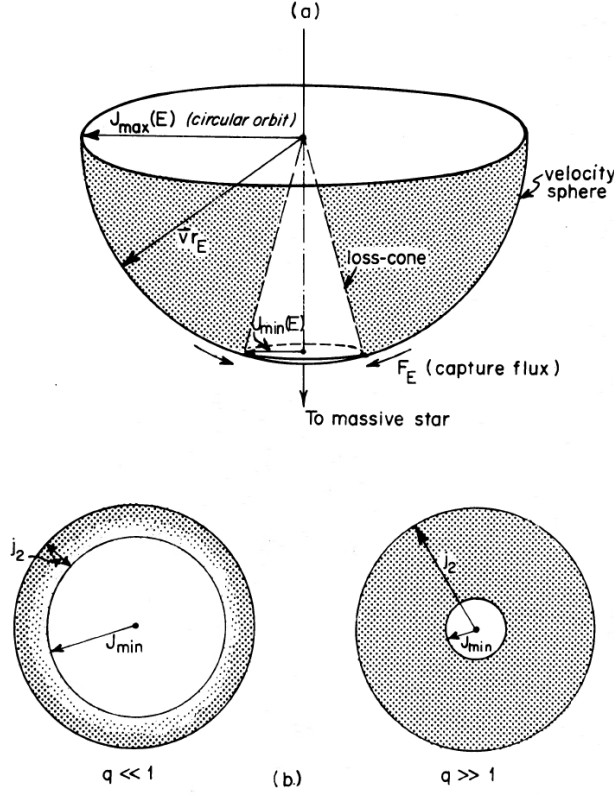


Figure 1.2: Panel (a): Velocity distribution of stars with fixed energy $E_* \equiv E$ and angular momentum J . The latter varies accordingly to Equation (1.69). The vertical axis points towards the compact object of mass M .

Panel (b): The velocity sphere viewed from below. On the left the empty loss cone scenario. On the right the full loss cone regime. Figure taken from Lightman and Shapiro (1977).

By employing the same dimensional analysis carried out by Lightman and Shapiro (1977), we expect that the stellar capture rate per unit energy $\Gamma_E(E_*)$ could be written as

$$\Gamma_E(E_*) \equiv \frac{d\Gamma}{dE_*} \approx \epsilon |E_*|^{-1} F_{max}(E_*) \quad (1.72)$$

where $F_{max}(E_*) \sim N(E_*)/P(E_*)$ denotes the maximum rate of loss cone replenishing from stars with specific energy E_* and ϵ is a correction factor helping us to account for the fraction of stars within the loss cone.

Owing to the 2-D nature of the loss-cone problem in the J -space, ϵ is expected to be an area correction factor, thus scaling as

$$\epsilon \sim \frac{J_{lc}^2}{(J_{lc} + j_2)^2}. \quad (1.73)$$

Therefore, in the full loss cone regime, where $J_{lc} \ll j_2 \approx J_{max}$, one easily finds that $\sim J_{lc}^2/J_{max}^2 \sim a/r$ and, by substitution into Equation (1.72),

$$\Gamma_E(E_*) \approx |E_*|^{-1} \frac{a}{r} \frac{N(E_*)}{P(E_*)}. \quad (1.74)$$

Equation (1.74) allows us to finally discuss how loss-cone refilling is accounted for in the Quinlan–Sesana hybrid model.

In his scattering experiments, Quinlan (1996) quantified the binary dimensionless hardening rate H and eccentricity growth rate K through the flux of stars

$$\Gamma_Q = \left(\frac{\rho}{m_*} \right) \Sigma v \quad (1.75)$$

across the binary effective cross section Σ , where ρ is the stellar mass density, m_* is the stellar mass, and v is the stellar velocity. Let b denote the impact parameter of a star initially unbound to the binary of mass M and undergoing a three-body interaction with it. The corresponding interaction cross section can then be defined as

$$\Sigma \equiv \pi b_{\max}^2, \quad (1.76)$$

where $b_{\max}$ is the maximum impact parameter for which stellar capture occurs. If we assume this to take place whenever $r_p < a$, mindful of the gravitational focusing

$$b^2 = \frac{2GM r_P}{v^2} \quad (1.77)$$

relating impact parameter and pericenter distance, then we can pose $b_{\max} = b(a)$ and express Γ_Q in Equation (1.75) as

$$\Gamma_Q = 2\pi \left(\frac{M}{m_*} \right) \frac{G\rho a}{v}, \quad (1.78)$$

with ρ standing for the mass density of the homogeneous stellar background.

On the other hand, Sesana (2010) estimated the rate Γ in Equation (1.72) under the assumption of a full loss cone, adopting an isothermal stellar sphere of total mass M_* , influence radius $r_{\inf}$ and constant stellar velocity dispersion σ .

If all stars have the same mass m_* , the total number of stars inside r is simply proportional to the stellar mass enclosed within this sphere, that is

$$N(< r) = \frac{M}{m_*} \frac{r}{r_{\inf}}, \quad (1.79)$$

while, under the assumption of Keplerian orbits bound to the binary, the energy of each star depends on the semimajor axis only through the relation

$$E_*(r) = -\frac{GMm_*}{2r}. \quad (1.80)$$

In this regime, by exploiting the relation $dE = |E| d \ln r$, we can rewrite Equation (1.74) as (see, e.g., Perets, Hopman, et al. (2007); Perets and Alexander (2008))

$$\frac{d\Gamma}{d \log r} = \frac{a}{r} \frac{N(< r)}{P(r)}. \quad (1.81)$$

Finally, approximating the orbital period as $P(r) \sim r/\sigma$ and substituting the estimates for $N(< r)$ and $P(r)$ derived above into Equation (1.81), integration yields

$$\Gamma = 2\pi \frac{M}{m_*} \frac{G\rho_{\inf} a}{\sigma}. \quad (1.82)$$

A comparison between Equations (1.75) and (1.82) shows that, by associating the velocity v of an unbound star at infinity with the velocity dispersion σ of the isothermal sphere, the prescriptions adopted for H and K effectively describe a system in which the loss cone is full at $r_{\inf}$. However, by introducing the dimensionless factor LCN , it is possible to describe, within the hybrid model formulation, different regimes of the loss-cone simply by replacing $\rho_{\inf}$ with $LCN \cdot \rho_{\inf}$ as it will be shown in chapter 3.

1.5 Gravitational Radiation from Black Hole Binaries

This Section reviews the physics of gravitational-wave emission from a binary system within the framework of linearized general relativity, following the treatment presented by Maggiore (2007), and examines their role as the final mechanism driving a binary black hole (BH) system towards progressively smaller separations and, ultimately, coalescence.

The Section is organized into three Subsections. The first focuses on the linearized Einstein equations in the harmonic gauge. The second discusses the generation of GWs in the quadrupole approximation. Finally, the third derives the expected GW signal from a binary system and examines the associated radiation-reaction effects on its orbital evolution.

1.5.1 Linearized Gravitational Field Equation

The Einstein equations are the fundamental field equations of general relativity (GR). They are given by

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}, \quad (1.83)$$

where $R_{\mu\nu}$, $g_{\mu\nu}$, and $T_{\mu\nu}$ denote the components of the Ricci, metric, and energy-momentum tensors, respectively, while R is the Ricci scalar, obtained by contracting the Ricci tensor.

Equations (1.83) constitute ten coupled, non-linear, second-order partial differential equations for the metric tensor. Nevertheless, owing to the four Bianchi identities, they are not sufficient to uniquely set the ten independent components of $g_{\mu\nu}$, hence, resulting in six independent equations only. These four degrees of freedom are intimately connected to gauge symmetries, reflecting the arbitrariness of the four coordinates x^μ used to represent and solve the tensor Equations (1.83).

To examine how gravitational radiation arises in linearized GR within the quadrupole approximation, we first consider the weak-field limit, in which the metric tensor deviates only slightly from the Minkowski metric $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$, which describes the flat spacetime. We then derive the corresponding form of the Einstein equations in this limit. Therefore we assume

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \quad (1.84)$$

with $|h_{\mu\nu}| \ll 1$ being a symmetric tensor representing a small perturbation of the flat spacetime. Linearizing Equations (1.83) requires performing this operation on the various terms appearing on the left-hand side. This side encodes the geometry and curvature of spacetime and contains all terms that are functions of the metric and its derivatives. Introducing the affine connections

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}) \quad (1.85)$$

and the Riemann tensor

$$R_{\lambda\mu}{}^\rho{}_\nu = \partial_\lambda \Gamma_{\mu\nu}^\rho - \partial_\mu \Gamma_{\lambda\nu}^\rho + \Gamma_{\lambda\sigma}^\rho \Gamma_{\mu\nu}^\sigma - \Gamma_{\mu\sigma}^\rho \Gamma_{\lambda\nu}^\sigma, \quad (1.86)$$

they can be expressed indeed as

$$R_{\mu\nu} = R_{\sigma\mu}{}^\sigma{}_\nu \quad (1.87)$$

and

$$R = g^{\alpha\beta}R_{\alpha\beta} = R^\sigma{}_\sigma. \quad (1.88)$$

Bearing in mind that, owing to Equation (1.84), the metric derivatives reduce to derivatives of the perturbation $h_{\mu\nu}$ and that at linear order the inverse metric tensor can be approximated as

$$g^{\mu\nu} \approx \eta^{\mu\nu} - h^{\mu\nu}, \quad (1.89)$$

it straightforward to prove that, to the first order in the metric perturbation, these terms reduce to

$$R_{\mu\nu} = -\frac{1}{2}(\square h_{\mu\nu} + \partial_\mu \partial_\nu h - \partial_\lambda \partial_\mu h_\nu^\lambda - \partial_\nu \partial_\lambda h_\mu^\lambda) \quad (1.90)$$

and

$$R = \partial_\mu \partial_\nu h^{\mu\nu} - \square h, \quad (1.91)$$

respectively, with h denoting the trace of $h_{\mu\nu}$.

The linearized Einstein equations are greatly simplified if they are expressed in terms of the so-called "trace reverse" of $h_{\mu\nu}$, defined as

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h. \quad (1.92)$$

In this way, writing the original perturbation $h_{\mu\nu}$ as a function of the associated trace reverse $\bar{h}_{\mu\nu}$ and plugging the terms of Equations (1.90) and (1.91) into Equation (1.83), the field equations, finally, take the form

$$\square \bar{h}_{\mu\nu} + \eta_{\mu\nu} \partial_\alpha \partial_\beta \bar{h}^{\alpha\beta} - \partial_\mu \partial_\lambda \bar{h}_\nu^\lambda - \partial_\nu \partial_\lambda \bar{h}_\mu^\lambda = -\frac{16\pi G}{c^4} T_{\mu\nu}. \quad (1.93)$$

Equations (1.93) are already linearized in the metric perturbation, however they can be further simplified if we exploit the gauge freedom holding for the Einstein equations. This reflects the possibility of an infinitesimal coordinate transformation of the kind

$$x^\mu \rightarrow x'^\mu = x^\mu + \xi^\mu(x), \quad (1.94)$$

with $\xi^\mu(x)$ four arbitrary functions of the spacetime.

As tensors obey to precise law of transformations under change of coordinates, the coordinates transformation in Equation (1.94) induces a variation of the trace reverse $\bar{h}_{\mu\nu}$ components, specifically:

$$\bar{h}'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \bar{h}_{\alpha\beta} \approx \bar{h}_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu + \eta_{\mu\nu} \partial_\sigma \xi^\sigma. \quad (1.95)$$

A direct inspection of Equations (1.93) and (1.95) shows that, among the possible transformations, choosing the four functions ξ^μ such that they satisfy the so-called De Donder gauge condition, i.e. $\square \xi^\mu = \partial_\sigma \bar{h}^{\sigma\mu}$, leads to $\partial_\sigma \bar{h}'^{\sigma\mu} = 0$. Dropping the prime thereafter, Equation (1.93) then simplifies to

$$\begin{cases} \square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu} \\ \partial_\mu \bar{h}^{\mu\nu} = 0. \end{cases} \quad (1.96)$$

We are now ready to address the problem of determining the general solution of the linearized Einstein's equations coupled with the De Donder gauge condition.

1.5.2 Generation and Propagation of Gravitational Waves

To study the formation of GWs we initially focus on the solution of the problem defined in Equation (1.96). Let us analyze first how GWs propagate in vacuo.

Outside the GWs source, $T_{\mu\nu}$ vanishes and, owing to the linearity of the d'Alembert operator, the solution can be expressed as superposition of plane waves without loss of generality, that is

$$\bar{h}_{\mu\nu}(x) = \int d^3k (A_{\mu\nu}(\mathbf{k}) e^{ik_\rho x^\rho} + c.c.), \quad (1.97)$$

with "c.c" being a short for complex conjugate of the first term.

It is worth noting that not all plane waves satisfy the linearized field equation in vacuo along with the gauge condition. However, linearity enables us to study the properties of a single plane wave separately; consequently, inserting the Ansatz $\bar{h}_{\mu\nu} \sim A_{\mu\nu} e^{ik_\rho x^\rho}$ into Equations (1.96) yields the following two constraints:

1. $(k_\rho k^\rho = 0)$: the four-vector $k^\mu = (\omega/c, \mathbf{k})$ is a light-like vector (GWs travel at light speed c in vacuo).
2. $(k_\mu A^{\mu\nu} = 0)$: it implies $A^{0\nu} = A^{3\nu}$ if we deal with GWs propagating along the x^3 direction, hence $k^\mu = (k, 0, 0, k)$.

At this stage, the amplitude tensor $A^{\mu\nu}$ possesses six independent components and can be written as

$$A^{\mu\nu} = \begin{pmatrix} A^{00} & A^{10} & A^{20} & A^{00} \\ A^{10} & A^{11} & A^{12} & A^{10} \\ A^{20} & A^{12} & A^{22} & A^{20} \\ A^{00} & A^{10} & A^{20} & A^{00} \end{pmatrix}. \quad (1.98)$$

Out of the six polarizations shown in Equation (1.98), four are unphysical because they vanish when the residual gauge freedom associated with four arbitrary harmonic functions ξ^μ is exploited. Indeed, one is free to adopt functions proportional to the plane wave solutions of the vacuo problem, e.g. $\xi^\mu = \epsilon^\mu e^{ik_\rho x^\rho}$, preserving the validity of the De Donder gauge, and to set their amplitudes ϵ^μ so that

$$A_{TT}^{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_x & 0 \\ 0 & h_x & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (1.99)$$

The coordinate system where $A^{\mu\nu}$ is in the form of Equation (1.99), with two independent polarizations only, and defined by

$$\partial_\mu h^{\mu\nu} = 0, \quad h^{0\mu} = 0, \quad h^i_i = 0, \quad (1.100)$$

is referred to as transverse, traceless, or "TT", gauge¹.

We now focus on the solutions of Equations (1.96) in presence of the source term $T_{\mu\nu} \neq 0$. It can be read as the source term of the gravitational field equations and turns out to be

$$T^{\mu\nu} = (\rho + p/c^2)u^\mu u^\nu + pg^{\mu\nu} \quad (1.101)$$

for a perfect fluid of density ρ , pressure p and four-velocity v^μ .

Solutions for linear partial differential equations can be obtained by means of the Green's function method. By definition, the Green's function $G(x - x')$ is the solution of the associated equation in case of a point-like source, i.e.

$$\square_x G(x - x') = \delta^{(4)}(x - x'), \quad (1.102)$$

with $\delta^{(4)}(x - x')$ being the four-dimensional Dirac function.

It is relevant to note that, apart from the B.C. of the problem, $G(x - x')$ depends only on the linear operator. Therefore, from the radiation problem in electromagnetism, we already know that the proper solution is the retarded Green's function

$$G(x - x') = -\frac{1}{4\pi|\mathbf{x} - \mathbf{x}'|} \delta^{(4)}(ct_{ret} - ct'), \quad (1.103)$$

¹In the TT gauge, according to the definition provided, $\bar{h}_{ij}^{TT} = h_{ij}^{TT}$.

where

$$t_{ret} = t - \frac{|\mathbf{x} - \mathbf{x}'|}{c} \quad (1.104)$$

is the time in past at which the radiation was emitted at $\mathbf{x}'$ for it to be observed at $\mathbf{x}$ at time t . Once the Green's function is found, the general solution will be the convolution between it and source term:

$$\bar{h}_{\mu\nu}(t, \mathbf{x}) = \frac{4G}{c^4} \int d^3x' \frac{T_{\mu\nu}(t_{ret}, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}. \quad (1.105)$$

One of the greatest simplifications to this integral consists of the so-called "Compact-source approximation"; it applies outside the source whenever its size is negligible with respect to the source-detector separation. In this regime, we can retain the leading term of the multipole expansion only, thus

$$h_{ij}^{TT}(t, \mathbf{x}) = \frac{4G}{c^4 r} \int d^3x' T_{ij}(t_{ret}, \mathbf{x}'). \quad (1.106)$$

Furthermore, far from the source, the conservation of the Energy-Momentum tensor, namely $\nabla_\mu T^{\mu\nu} \approx \partial_\mu T^{\mu\nu} = 0$, yields

$$h_{ij}^{TT}(t, \mathbf{x}) = \frac{2G}{r c^4} \ddot{Q}_{ij}^{TT}(t - r/c) \quad (1.107)$$

at lowest order in the low-velocity expansion.

The term Q_{ij}^{TT} introduced in Equation (1.107)² is the representation of the quadrupole moment in the class of coordinate systems satisfying the TT gauge conditions. Defining the quadrupole moment of $T^{00}/c^2 = \rho$ as

$$M^{ij} \equiv \frac{1}{c^2} \int d^3x T^{00}(t, \mathbf{x}) x^i x^j, \quad (1.108)$$

it is thus possible to write

$$Q^{ij} \equiv M^{ij} - \frac{1}{3} \delta^{ij} M_{kk} = \int d^3x \rho(t, \mathbf{x}) \left(x^i x^j - \frac{1}{3} r^2 \delta^{ij} \right). \quad (1.109)$$

Equation (1.107) shows that, unlike electromagnetic radiation, at lowest order the gravitational radiation arises from changes in the quadrupole moment. We conclude by reporting other important observables such as the energy and angular momentum carried by GWs, needed to estimate the back-reaction and related to the time derivatives of quadrupole moment tensor defined in Equation (1.109).

The conservation of energy and angular momentum requires that the rates at which GWs carry these quantities away from the source match the corresponding rates of orbital energy and angular momentum loss. The former are measured at a large distance r from the source and at time t , while the latter must be evaluated at the appropriate retarded time. By neglecting the contributions from possible spins of the two black holes, it can be proved that the energy and angular momentum transported by GWs per unit time are

$$P_{quad} = \frac{dE_{gw}}{dt} = \frac{G}{5c^5} \langle \ddot{Q}_{ij} \ddot{Q}_{ij} \rangle = -\frac{dE_{source}}{dt}, \quad (1.110)$$

or

$$P_{quad} = \frac{G}{5c^5} \left\langle \ddot{M}_{ij} \ddot{M}_{ij} - \frac{1}{3} (\ddot{M}_{kk})^2 \right\rangle \quad (1.111)$$

in terms of M_{ij} , and

$$\frac{dL_{gw}^i}{dt} = \frac{2G}{5c^5} \epsilon^{ijk} \langle \ddot{Q}_{jl} \ddot{Q}_{kl} \rangle = -\frac{dL_{source}^i}{dt}. \quad (1.112)$$

respectively. Equations (1.110) and (1.112) will be of fundamental importance for estimating the back-reaction induced by GWs emission on the black hole binary.

²The components with at least one time index are excluded from our analysis because they are constant and therefore do not contribute to GWs formation.

1.5.3 Gravitational Waves from Black Hole Binaries and Back-Reaction

We now focus on the gravitational radiation emitted by two masses m_1 and m_2 moving along elliptic Keplerian orbit. Let $m = m_1 + m_2$ and $\mu = m_1 m_2 / m$ be the total mass and the reduced mass, respectively.

Whether the orbit is bound or unbound, the motion of the two bodies is confined to a plane by the conservation of the angular momentum $\mathbf{L}$. It is therefore convenient to choose this plane as the x - y plane, perpendicular to $\mathbf{L}$, and introduce polar coordinates (r, ψ) centered on the center of mass. In this reference frame, denoted by S_{CM} ,

$$L_z = \mu r^2 \dot{\psi} \quad (1.113)$$

and, owing to the K oenig theorem, the total binary energy

$$E = \frac{1}{2} \mu \dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{Gm\mu}{r} \quad (1.114)$$

exhibits only the term of kinetic energy corresponding to the relative motion of the system, as in S_{CM} holds $K_{CM} = 0$ by definition.

The integration of Equation (1.114) requires the introduction of the auxiliary variable $w(\psi) = 1/r(\psi)$ (Capozziello and De Laurentis 2008). Plugging it into the energy conservation identity, Equation (1.114) becomes

$$w'^2 + w^2 - \frac{2Gm_1 m_2 \mu}{L^2} w = \frac{2\mu E}{L^2} \quad (1.115)$$

where $w' \equiv dw/d\psi$.

Differentiating both sides of the last identity with respect to ψ leads to a new differential equation, which is

$$w' \left(w'' + w - \frac{Gm_1 m_2 \mu}{L^2} \right) = 0. \quad (1.116)$$

Apart from the solution $w = \text{const}$, corresponding to a circular binary, the problem is formally identical to that of a harmonic oscillator. Therefore, upon solving for $r(\psi)$, it is found

$$r(\psi) = \frac{R}{1 + e \cos \psi}, \quad (1.117)$$

with $R = L^2/(Gm\mu^2)$ being the semi-latus rectum.³

Being interested in the GW emission process by a BH binary, which is by definition a gravitationally bound system, from this point onward it will be assumed $E < 0$. In this energy range, the trajectory is an ellipse semiaxes given by

$$a = \frac{R}{1 - e^2} \quad b = \frac{R}{\sqrt{1 - e^2}}, \quad (1.118)$$

with

$$e^2 = 1 - \frac{R}{a} = 1 + \frac{2EL^2}{G^2 m^2 \mu^3}. \quad (1.119)$$

denoting the eccentricity $0 \leq e < 1$ of a binary.

Inserting Equation (1.119) into the first identity of Equation (1.118) yields the semi-major axis a in terms of E , i.e.

$$a = -\frac{Gm\mu}{2E}, \quad (1.120)$$

³The polar axis is aligned with the major axis $2a$, so that R corresponds to the distance between the origin and the point of the conic at $\psi = \pi/2$.

implying that fixing the energy means setting the orbit size.

The functions $r(t)$ and $\psi(t)$ can be obtained by direct integration of Equations (1.113) and (1.114); in the parametric form they can be written as

$$r(t) = a[1 - e \cos u(t)] \quad \psi(t) = \frac{\cos u(t) - e}{1 - e \cos u(t)}, \quad (1.121)$$

involving the so-called "eccentric anomaly"

$$u(t) - e \sin u(t) = \left(\frac{Gm}{a^3} \right)^{1/2} t \equiv \omega_0 t. \quad (1.122)$$

Once the motion of the Keplerian binary has been characterised, we have all the necessary ingredients to compute the rates at which energy and angular momentum are carried away by GWs. These quantities are essential for determining the back-reaction exerted on the binary system.

In the reference system S_{CM} , the mass density distribution $\rho(\mathbf{x}, t)$ of two point-like masses is

$$\rho(\mathbf{x}, t) = \mu \delta(x - r \cos \psi) \delta(y - r \sin \psi) \delta(z). \quad (1.123)$$

Therefore, according to the prescription provided by Equation (1.108), it is immediate to obtain

$$M_{ij} = \mu r^2(t) \begin{bmatrix} \cos^2(\psi) & \sin(\psi) \cos(\psi) \\ \sin(\psi) \cos(\psi) & \sin^2(\psi) \end{bmatrix}. \quad (1.124)$$

The time derivatives of M_{ij} , required to compute the energy and angular momentum transfer rates via GW emission, are more easily computed when r is expressed as a function of ψ . This is accomplished by replacing this variable with the general solution of the two-body problem provided by Equation (1.117), thus finding

$$\ddot{M}_{11} = \beta(1 + e \cos \psi)^2 [2 \sin 2\psi + 3e \sin \psi \cos^2 \psi], \quad (1.125)$$

$$\ddot{M}_{22} = \beta(1 + e \cos \psi)^2 [-2 \sin 2\psi - e \sin \psi (1 + 3 \cos^2 \psi)], \quad (1.126)$$

and

$$\ddot{M}_{12} = \beta(1 + e \cos \psi)^2 [-2 \cos 2\psi + e \cos \psi (1 - 3 \cos^2 \psi)], \quad (1.127)$$

where the factor β is defined by

$$\beta^2 \equiv \frac{4G^3 \mu^2 m^3}{a^5 (1 - e^2)^5}. \quad (1.128)$$

Plugging Equations (1.125)-(1.126)-(1.127) into Equation (1.110) yields

$$P(\psi) = \frac{8G^4}{15c^5} \frac{\mu^2 m^3}{a^5 (1 - e^2)^5} (1 + e \cos(\psi))^4 [12(1 + e \cos(\psi))^2 + e^2 \sin^2(\psi)]. \quad (1.129)$$

However, we recall that this instantaneous quantity is not constant over one orbital period for elliptic Keplerian orbits as in general M_{ij} will suffer from different rate of variation from pericenter to apocenter; therefore, taking the temporal average over one orbital period, $T \equiv 2\pi/\omega_0$ returns

$$P \equiv \frac{1}{T} \int_0^T dt P(\psi) = \frac{\omega_0}{2\pi} \int_0^{2\pi} d\psi \frac{P(\psi)}{\dot{\psi}} = \frac{32G^4 \mu^2 m^3}{5c^5 a^5} F(e), \quad (1.130)$$

with

$$F(e) = \frac{1}{(1 - e^2)^{7/2}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \quad (1.131)$$

being a function strongly depending on the eccentricity, which reduces to unity in case of circular binary.

Throughout the inspiral phase, the binary radiates not only energy but angular momentum as well, carried away by GWs, and undergoes secular changes in both semimajor axis and ellipticity until coalescence eventually occurs. Therefore, another key ingredient in evaluating the back-reaction on the binary evolution is the determination of the angular momentum loss rate. It was quantified in Equation (1.112), which now becomes

$$\frac{dL_{source}^i}{dt} = -\frac{2G}{5c^5} \epsilon^{ikl} \langle \ddot{M}_{ka} \ddot{M}_{la} \rangle \quad (1.132)$$

as the terms of the form $\epsilon^{ijk} \delta_{jl} \ddot{Q}_{kl}$ vanish, since they correspond to the contraction of an antisymmetric tensor with a symmetric one.

Taking into account that, in S_{CM} , all components of kind M_{i3} are identically zero and that $L = L_z = L_{source}^3$, one can rewrite Equation (1.132) as

$$\frac{dL}{dt} = \frac{4G}{5c^5} \langle \ddot{M}_{12} (\ddot{M}_{11} - \ddot{M}_{22}) \rangle. \quad (1.133)$$

The last identity provides dL/dt in terms of the second and third derivatives of the quadrupole moment tensor M_{ij} . Although the second derivative $\ddot{M}_{12}$ has not been computed explicitly yet, it follows straightforwardly from the same procedure used to derive $\ddot{M}_{ij}$ that

$$\ddot{M}_{12} = \frac{G\mu m}{a(1-e^2)} \sin \psi \left[-4(1 + e \cos \psi)^2 \cos \psi + 2e (3 \cos^2 \psi - 1 + 2e \cos^3 \psi) \right]. \quad (1.134)$$

Yet, upon substituting the expressions given in Equations (1.125), (1.126), and (1.134), respectively, we obtain the angular momentum loss, corresponding to the negative of the angular momentum carried away by gravitational radiation in the quadrupole approximation. Taking the temporal average over one orbital period T and expressing it as an integral over the angular position ψ then yields

$$\frac{dL}{dt} = -\frac{32}{5} \frac{G^{7/2} \mu^2 m^{5/2}}{c^5 a^{7/2}} \frac{1}{(1-e^2)^2} \left(1 + \frac{7}{8} e^2 \right) \quad (1.135)$$

Finally, the secular evolution of the binary orbit can be obtained by relating the losses of orbital energy and angular momentum to the corresponding changes in the semi-major axis and eccentricity. In particular, applying the corresponding conservation laws to the combined system of the BH binary and the emitted GWs, and differentiating Equations (1.119) and (1.120) with respect to time, we get

$$\frac{da}{dt} = \frac{Gm\mu}{2E^2(a)} (-P) \quad (1.136)$$

and

$$2e \frac{de}{dt} = \frac{2}{G^2 m^2 \mu^3} \left[-PL^2(a, e) + 2E(a)L(a, e) \frac{dL}{dt} \right]. \quad (1.137)$$

Combining the resulting expressions and eliminating the intermediate quantities, Peters and Mathews (1963) derived

$$\frac{da}{dt} = -\frac{64}{5} \frac{G^3 \mu m^2}{c^5 a^3} \frac{1}{(1-e^2)^{7/2}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \quad (1.138)$$

along with

$$\frac{de}{dt} = -\frac{304}{15} \frac{G^3 \mu m^2}{c^5 a^4} \frac{e}{(1-e^2)^{5/2}} \left(1 + \frac{121}{304} e^2 \right). \quad (1.139)$$

Equations (1.138) and (1.139) show that GW emission becomes increasingly effective as the binary separation decreases, as reflected by the steep a^{-3} and a^{-4} dependences of the semi-major axis and eccentricity evolution, respectively. Therefore, at sufficiently small separations, GW emission is expected to become the dominant mechanism governing the binary evolution, superseding the effects of stellar scattering discussed in § 1.3.

These terms were incorporated into the self-consistent hybrid model developed independently by Quinlan (1996) and Sesana (2010), which combines stellar dynamical interactions and GW emission to describe the evolution of massive black hole binaries across the different stages of their inspiral. A detailed discussion of this model, together with the characteristic transition lengthscales that ideally separate the regimes in which the different evolutionary mechanisms dominate, is presented in Chapter 2.

2

A HYBRID APPROACH TO IMBH BINARY EVOLUTION

In light of the forthcoming GW observatories, LISA and ET, investigating the properties and evolution of GW-loud sources is of particular importance. Among these, IMBH binaries are especially compelling, and studying their dynamical evolution, together with their associated coalescence times, is essential for assessing their prospects as GW sources.

This chapter aims to review the physical framework underlying the Quinlan-Sesana hybrid model for the dynamical evolution of black hole binaries and to present the main results of the numerical experiments performed using this approach. The chapter begins with a detailed description of the hybrid model, focusing on its main parameters and on the numerical integration of the differential equations governing the evolution of the binary semimajor axis and eccentricity. The model is then applied to physically motivated stellar backgrounds. The second section is therefore devoted to numerical experiments in stellar-cusp environments, while the third section presents the main results obtained by applying the model to nucleated dwarf galaxies. Finally, the chapter concludes with a study of IMBH binaries evolving within either present-day or artificially evolved high- z non-nucleated dwarf galaxies, according to the observed cosmic size evolution of galaxies size.

2.1 Quinlan-Sesana Hybrid Model

As already discussed in Chapter 1, the coalescence time of a BH binary driven solely by GW emission depends strongly on several parameters, including the initial binary separation a_0 and eccentricity e_0 . However, the GW-driven regime is reached only after a sequence of earlier hardening mechanisms, primarily dynamical friction and three-body stellar scattering, which dominate the early stages of the binary evolution. It is therefore essential to model the dynamical evolution of the binary by consistently accounting for the different hardening processes from its formation to final coalescence in order to obtain a reliable description of the binary evolutionary track, i.e. $a(t)$ and $e(t)$, and its coalescence time.

Although direct N -body simulations provide a self-consistent approach to carrying out this analysis, they remain computationally too demanding to efficiently explore the broad parameter space governing binary evolution. This parameter space, for our purposes, includes the BH masses, the binary separation and eccentricity at pairing, and the properties of the stellar background. In this context, the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010) provides an efficient alternative, overcoming the computational limitations and long runtimes associated with direct N -body experiments while retaining the key physical processes driving the binary evolution.

Let us begin by introducing the Quinlan formalism and the main model parameters were determined through restricted three-body scattering experiment, eventually leading to the coupled differential equation for $(a(t), e(t))$. Adopting the original notation, we denote by m_1 and $m_2 < m_1$ the primary and secondary BH masses, respectively, by $M_{12} = m_1 + m_2$ the total binary mass, by $q = m_2/m_1$ the BHs mass ratio, and by $m_* \ll m_2$ the mass of an individual star, assuming all stars have the same mass. In his work, Quinlan (1996) studied the dynamical evolution of a BH binary embedded in a fixed, homogeneous background of unbound stars with density ρ and a Maxwellian velocity distribution characterised by a constant velocity dispersion σ . The description relies on three dimensionless quantities, H , K , and J , which denote the dimensionless binary hardening rate, eccentricity growth rate, and stellar mass ejection rate, respectively. They are defined as follows:

$$H = \frac{\sigma}{G\rho} \frac{d}{dt} \left(\frac{1}{a} \right), \quad (2.1)$$

$$K = \frac{de}{d \ln(1/a)} \quad (2.2)$$

and

$$J = \frac{1}{M_{12}} \frac{dM_{ej}}{d \ln(1/a)}. \quad (2.3)$$

Let us focus on Equations (2.1) and (2.2). Both dimensionless rates and the derivatives of the binary parameters account for the various, independent mechanisms involved during its evolution, thus

$$\begin{cases} \frac{da}{dt} = \sum_i \left. \frac{da}{dt} \right|_i = -\frac{a^2 G \rho}{\sigma} \sum_i H_i \\ \frac{de}{dt} = \sum_i \left. \frac{de}{dt} \right|_i = -\frac{a G \rho}{\sigma} \sum_i H_i K_i \end{cases} \quad (2.4)$$

and

$$\begin{cases} H_i = \frac{\sigma}{a^2 G \rho} \frac{da}{dt} \Big|_i \\ K_i = a \frac{de}{dt} \Big|_i \left(\sum_j \frac{da}{dt} \Big|_j \right)^{-1} \end{cases} \quad (2.5)$$

where the index i identifies a precise dynamical process (Sesana 2010).

As in Quinlan (1996) and Sesana, Haardt, et al. (2006), in our numerical experiments, we account for the binary evolution due to close encounters with unbound, loss-cone stars and by GW emission, as described by the equations derived by Peters and Mathews (1963). Therefore, we assume

$$\begin{cases} \frac{da}{dt} = \frac{da}{dt} \Big|_u + \frac{da}{dt} \Big|_{GW} = -\frac{a^2 G \rho}{\sigma} (H_u + H_{GW}) \\ \frac{de}{dt} = \frac{de}{dt} \Big|_u + \frac{de}{dt} \Big|_{GW} = -\frac{a G \rho}{\sigma} (H_u K_u + H_{GW} K_{GW}) \end{cases} \quad (2.6)$$

The latter, derived in § 1.5, is reported here for completeness and to ensure consistency with the notation adopted in this chapter:

$$\frac{da}{dt} \Big|_{GW} = -\frac{64}{5} \frac{G^3 \mu m^2}{c^5 a^3} \frac{1}{(1-e^2)^{7/2}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right), \quad (2.7)$$

$$\frac{de}{dt} \Big|_{GW} = -\frac{304}{15} \frac{G^3 \mu m^2}{c^5 a^4} \frac{e}{(1-e^2)^{5/2}} \left(1 + \frac{121}{304} e^2 \right). \quad (2.8)$$

The former, instead, was independently derived by Quinlan (1996) and Sesana, Haardt, et al. (2006) using two different approaches, as shown below. Indeed, with the aim of deriving the dimensionless rates, both works employ suites of three-body scattering experiments designed to characterize the response of a BH binary to repeated close encounters with stars over its lifetime. However, while Quinlan (1996) fixes the incoming stellar velocity v and varies the binary separation a , Sesana, Haardt, et al. (2006) adopt the opposite approach, varying v at fixed a . It is worth of noting that the two approaches are physically equivalent, but lead to different parametrizations of the functions H_u and K_u . When numerically integrating Equations (2.6), the parametrization provided by Sesana, Haardt, et al. (2006) will be adopted, as it directly involves the quantities being integrated.

Let us now briefly review how H_u and K_u were originally estimated. The underlying idea is to infer the variations in the binary orbital energy and angular momentum from the corresponding changes experienced by the scattered star, which are equal in magnitude and opposite in sign as a consequence of energy and angular momentum conservation. The first step is to define the dimensionless energy change

$$C \equiv \frac{a \Delta E_*}{G \mu}, \quad (2.9)$$

introduced by Hills (1983), and the dimensionless impact parameter

$$x \equiv \frac{b}{b_0} \quad (2.10)$$

with $b_0^2 = 2GM_{12}a/v^2$.

Denoting by $\langle C \rangle$ the average over all angular variables, the “single-velocity” hardening rate $H_1(v)$, derived from restricted three-body scattering experiments in which all stars have the same velocity at infinity, is given by

$$H_1(v) = \frac{1}{\pi b_0^2} \int db, (2\pi b) 4\pi \langle C \rangle = 8\pi \int dx, \langle C \rangle x \equiv 8\pi I_x(C). \quad (2.11)$$

where the operator $I_x(C)$ is implicitly defined by Equation (2.11). Finally, the hardening rate $H_u(\sigma)$ is obtained by averaging the single-velocity hardening rate $H_1(v)$ over the Maxwellian distribution of stellar velocities:

$$H_u(\sigma) = \int_0^\infty dv 4\pi v^2 H_1(v) \frac{\sigma}{v} f(v, \sigma) \quad (2.12)$$

where

$$f(v, \sigma) = \frac{1}{(2\pi\sigma^2)^{3/2}} e^{-\frac{v^2}{2\sigma^2}} \quad (2.13)$$

is the Maxwellian velocity distribution associated with the constant velocity dispersion σ .

Similarly, one finds K_u from the “single-velocity” dimensionless eccentricity growth rate K_1 , which has the same meaning of H_1 . It can be derived starting from the expression for the binary eccentricity,

$$e^2 = 1 + \frac{2EL^2}{\mu(GM_{12})^2}, \quad (2.14)$$

which relates the eccentricity to the binary orbital energy E and angular momentum L . Differentiating Equation (2.14) and expressing the resulting variations in terms of finite differences yields

$$\frac{\Delta e}{\Delta \ln(1/a)} = \frac{1 - e^2}{2e} \left[-2 \left(\frac{E}{\Delta E} \right) \left(\frac{\Delta L_z}{L_z} \right) - 1 \right] = \frac{1 - e^2}{2e} (B - C), \quad (2.15)$$

where

$$B \equiv -\frac{M_{12}}{m_*} \frac{\Delta L_z}{L_z}. \quad (2.16)$$

is the dimensionless angular momentum change. Finally, by integrating over all possible impact parameters and angular variables and normalizing the result to B , the single-velocity eccentricity growth rate K_1 turns out to be

$$K_1 = \frac{\int dx, xC(x) \left\langle \frac{\Delta e}{\Delta \ln(1/a)} \right\rangle}{\int dx, xC(x)} = \frac{1 - e^2}{2e} \left[\frac{I_x(B - C)}{I_x(C)} \right]. \quad (2.17)$$

The best-fitting functions found by Sesana, Haardt, et al. (2006) from numerical experiments to H_u and K_u are

$$H(a) = A_H \left(1 + \frac{a}{a_{0,H}} \right)^{\gamma_H} \quad (2.18)$$

and

$$K(a, e) = A_K \left(1 + \frac{a}{a_{0,K}} \right)^{\gamma_K} + B_K, \quad (2.19)$$

with best-fitting parameters listed in Tables 2.1 and 2.2.

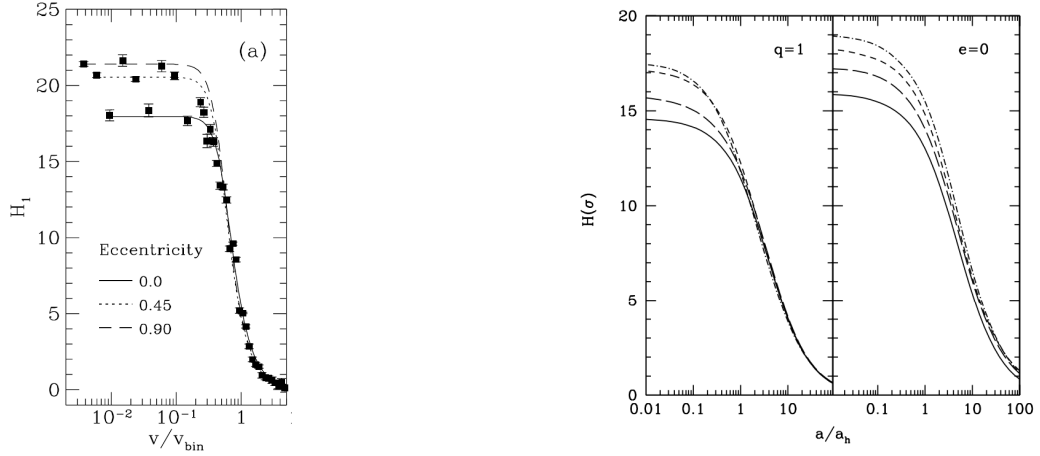
q	A_H	$a_{0,H}$	γ_H
1	14.55	3.48	-0.95
1/3	15.82	4.18	-0.90
1/9	17.17	3.59	-0.79
1/27	18.15	3.32	-0.77
1/81	18.81	3.87	-0.82
1/243	19.16	4.16	-0.86

Table 2.1: Best-fit parameters for the dimensionless stellar hardening function H_u (see Equation (2.18)) as a function of the BH mass ratio q . The parameter $a_{0,H}$ is given in units of the hardening radius. Values from Sesana, Haardt, et al. (2006).

e	A_K	$a_{0,K}$	γ_K	B_K
$q = 1$				
0.15	0.037	0.339	-3.335	-0.012
0.3	0.075	0.151	-1.548	-0.008
0.45	0.105	0.088	-0.893	-0.005
0.6	0.121	0.090	-0.895	-0.008
0.75	0.134	0.064	-0.544	-0.006
0.9	0.082	0.085	-0.663	-0.004
$q = 1/3$				
0.15	0.082	0.042	-0.168	-0.048
0.3	0.095	0.213	-1.152	-0.012
0.45	0.129	0.137	-0.655	-0.006
0.6	0.166	0.081	-0.546	-0.006
0.75	0.159	0.079	-0.497	-0.010
0.9	0.095	0.122	-0.716	-0.008
$q = 1/9$				
0.15	0.051	0.385	-0.891	-0.011
0.3	0.111	0.307	-1.107	-0.007
0.45	0.172	0.526	-1.174	-0.016
0.6	0.181	0.251	-1.169	-0.007
0.75	0.179	0.195	-0.846	-0.004
0.9	0.117	0.400	-1.170	-0.001
$q = 1/27$				
0.15	0.064	0.284	-1.206	0.021
0.3	0.143	1.033	-1.537	-0.021
0.45	0.212	0.722	-1.257	-0.022
0.6	0.216	0.430	-1.163	-0.014
0.75	0.173	0.771	-1.934	-0.014
0.9	0.129	0.329	-1.125	-0.020

Table 2.2: Best-fit parameters for the dimensionless eccentricity growth rate function K_u (see Equation (2.19)) as a function of the BH mass ratio q and binary eccentricity e . The parameter $a_{0,K}$ is given in units of the hardening radius. Values from Sesana, Haardt, et al. (2006).

The absence of a column dedicated to the eccentricity sampling in Table 2.2 reflects one of the main findings of the scattering experiments by Quinlan (1996) and Sesana, Haardt, et al. (2006): the dimensionless hardening rate H_u is almost independent of the BH binary eccentricity. This result, shown in Figure 2.1, is in excellent agreement with the main statement of Heggie’s law, discussed in § 1.3.



(a) "Single-velocity" binary hardening rate H_1 (see Equation 2.11) as a function of the initial stellar velocity V , normalized to the binary associated circular velocity $V_{bin} = \sqrt{GM_{12}/a}$, for three different values of the BH binary eccentricities, namely $e = (0, 0.45, 0.90)$, at fixed BH binary masses ratio $q = 1$. Picture from Quinlan (1996).

(b) Binary hardening rate H_U as a function of binary separation a , normalized to its hardening radius a_h for different eccentricities e (left) and mass ratios q (right). In both panels, solid, long-dashed, short-dashed, and dot-dashed lines correspond to $e = (0, 0.3, 0.6, 0.9)$ and $q = (1/3, 1/9, 1/27, 1/81)$, respectively. Picture from Sesana, Haardt, et al. (2006).

Figure 2.1: Comparison of the dimensionless stellar hardening rates obtained by Quinlan (1996) and Sesana, Haardt, et al. (2006) from independent three-body scattering experiments.

In contrast, the dimensionless eccentricity growth rate K_u depends on both the binary eccentricity e and the separation a . Following the procedure illustrated by Sesana (2010), at a given binary separation, $K_u(a, e)$ is obtained from Equation (2.19) by linearly interpolating in the eccentricity domain the best-fitting parameters reported in Table 2.2. The interpolation is supplemented by the boundary conditions $K(a, e = 0) = 0$, as theoretically predicted in the test-particle limit for circular orbits, since circular binaries remain circular, and $K(a, e = 1) = 0$. Thus far, this Section has focused on introducing the main functions and equations that characterize the self-consistent hybrid model for the dynamical evolution of BH binaries in a fixed stellar environment. We are now in a position to illustrate how the model can be applied to an arbitrary stellar background and how the corresponding evolution of the binary separation and eccentricity, $a(t)$ and $e(t)$, is obtained numerically.

As reported by Quinlan (1996) and Sesana, Haardt, et al. (2006), the three-body scattering experiments underlying the hybrid model consider stars that are initially unbound to the binary and whose velocities follow a Maxwellian distribution. Since these calculations are performed in a specific stellar background, a prescription is required to extend their results to the diverse stellar distributions hosting MBHs. A useful calibration is provided by Sesana and Khan (2015), who showed that the hardening rate measured from direct N -body simulations of a MBHB embedded in a spherically symmetric and isotropic stellar system, characterized by a mass density $\rho(r)$ and velocity dispersion $\sigma(r)$, agrees to within approximately 30% with that predicted by the hybrid model for a homogeneous stellar background characterized by a constant density ρ_{inf} and velocity dispersion σ_{inf} , denoting the stellar mass density and velocity dispersion values at the binary influence radius respectively. Indeed, once the binary has ejected most of the stars initially bound within its radius of influence, the loss cone is predominantly replenished by stars coming from just outside r_{inf} , thus having locally a nearly homogeneous spatial distribution with $\rho = \rho_{inf}$ and velocity dispersion $\sigma = \sigma_{inf}$.

In summary, once the BH masses and the properties of the stellar background are specified, the dynamical evolution of the binary is governed by the following system of coupled differential

equations:

$$\begin{cases} \frac{da}{dt} = -\frac{a^2 G \rho_{inf}}{\sigma_{inf}} H_u(a) - \frac{64 G^3 \mu M_{12}^2}{5 c^5 a^3} \frac{1}{(1-e^2)^{7/2}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4\right) \\ \frac{de}{dt} = -\frac{a G \rho_{inf}}{\sigma_{inf}} H_u(a) K_u(a, e) - \frac{304 G^3 \mu M_{12}^2}{15 c^5 a^4} \frac{e}{(1-e^2)^{5/2}} \left(1 + \frac{121}{304} e^2\right) \end{cases} \quad (2.20)$$

which can be solved numerically for a given set of initial conditions (a_0, e_0) .

The system of Equations (2.20) constitutes a set of first-order autonomous ordinary differential equations (ODEs), which can be written in the general form

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}), \quad (2.21)$$

where $\mathbf{u} = (a, e)$ is the vector of dynamical variables.

The numerical integration of this system with the adopted fourth-order Runge–Kutta (RK4) method requires the time step to satisfy an appropriate stability condition. The stability analysis to the first order of RK4 applied to a system of N ODEs in the N -dimensional vector $\mathbf{u}$ yields the following criterion (Butcher 2016):

$$\Delta t \leq \min \left\{ \frac{2.79}{|\lambda_i|} \right\} \quad \forall i = 1, N \quad (2.22)$$

where λ_i denotes the i -th eigenvalue of the Jacobian matrix $\mathbf{J} = \partial \mathbf{f} / \partial \mathbf{u}$.

For the system considered here, the Jacobian elements can be derived directly from Equations (2.20) as

$$J_{11} = \left. \frac{da}{dt} \right|_u \left(\frac{2}{a} + \frac{\gamma}{a + a_0 a_h} \right) - \left. \frac{da}{dt} \right|_{gw} \left(\frac{3}{a} \right), \quad (2.23)$$

$$J_{12} = \left. \frac{da}{dt} \right|_{GW} \left(\frac{7e}{1-e^2} + \frac{73e/12 + 37e^3/24}{1 + 73e^2/24 + 37e^4/96} \right), \quad (2.24)$$

$$J_{21} = \left. \frac{de}{dt} \right|_u \left(\frac{1}{a} + \frac{\gamma_H}{a + a_{0,H} a_h} + \frac{\gamma_K}{a + a_{0,K} a_h} \right) - \left. \frac{de}{dt} \right|_{GW} \left(\frac{4}{a} \right), \quad (2.25)$$

$$J_{22} = \left. \frac{de}{dt} \right|_u \left(\frac{\partial \ln K}{\partial e} \right) + \left. \frac{de}{dt} \right|_{GW} \left(\frac{5e}{1-e^2} + \frac{1 + 363e^2/304}{e + 121e^3/304} \right). \quad (2.26)$$

These expressions allow the two eigenvalues, λ_1 and λ_2 , of the Jacobian matrix to be evaluated at each point along the binary evolutionary track and the adaptive time-step to be calculated as

$$\Delta t \leq 2.79 \cdot \min \left\{ \frac{1}{|\lambda_1|}; \frac{1}{|\lambda_2|} \right\}. \quad (2.27)$$

In the following sections, § 2.2-2.3, will be presented the results of the numerical experiments performed with the hybrid model described above, considering different IMBH binaries masses, initial conditions and host stellar systems.

2.2 Stellar Cusps IMBH Binary Evolution

Observational evidences and direct N -body simulations suggest that galaxies hosting dynamically evolved BH binaries may undergo stellar scouring during binary hardening, significantly altering the original stellar distribution and leading to the formation of a shallower inner cusp (see, e.g., Milosavljević and Merritt (2001); Merritt and Milosavljević (2005)). As the binary

orbit shrinks within the galactic core, it transfers energy to the surrounding stars, progressively depleting the central stellar distribution and producing a stellar mass deficit in the innermost region of the host galaxy.

In light of these results, the first stellar background considered here for studying the dynamical evolution of an IMBH binary, from pairing to coalescence, is thus a power-law density profile within the binary influence radius, matched to a singular isothermal sphere (SIS) outside it (Sesana 2010). The stellar density $\rho(r)$ of this spherically symmetric distribution is given by

$$\begin{cases} \rho(r) = \rho_{inf} \left(\frac{r}{r_{inf}} \right)^{-\gamma} & \text{if } r < r_{inf} \\ \rho(r) = \frac{\sigma^2}{2\pi G r^2} & \text{if } r > r_{inf} \end{cases} \quad (2.28)$$

where γ is the exponent of the inner power-law stellar cusp and σ denotes the position-independent velocity dispersion of the SIS ($\gamma = 2$) profile.

Adopting the notation introduced in the previous Section, we assume that σ is uniquely determined by the binary mass M_{12} through the $M - \sigma$ relation

$$M_6 = 0.84\sigma_{70}^4. \quad (2.29)$$

provided by Tremaine et al. (2002), with σ_{70} indicating the stellar velocity dispersion in units of 70 km/s and M_6 the binary mass in units of $10^6 M_\odot$.

Imposing the continuity of the stellar profile in Equation (2.28) at the binary influence radius r_{inf} returns the following analytic expressions for r_{inf} and ρ_{inf} :

$$r_{inf} = (3 - \gamma) \frac{GM_{12}}{\sigma^2}, \quad (2.30)$$

$$\rho_{inf} = \frac{\sigma^2}{2\pi G r_{inf}^2}. \quad (2.31)$$

As already mentioned in § 1.2, dynamical friction exerted by the stellar background on an intruding secondary BH efficiently drags it towards the galactic center, where it eventually forms a gravitationally bound binary with the primary BH, with semimajor axis a_0 . This separation is defined as the radius enclosing a stellar mass equal to the secondary BH mass m_2 and, for the adopted stellar cusp profile, can be expressed as

$$a_0 = (3 - \gamma) \frac{GM_{12}}{\sigma^2} \left(\frac{q}{1 + q} \right)^{1/(3-\gamma)} = r_{inf} \left(\frac{q}{1 + q} \right)^{1/(3-\gamma)}, \quad (2.32)$$

In addition to a_0 , it is useful to define a set of characteristic distances associated with different binary separations, which allows us to distinguish the various stages of binary evolution and identify the dominant physical mechanism governing each stage. Starting the binary evolution from a_0 , the binary is initially too “soft”, in the sense of Heggie’s law, for scattering with unbound stars to significantly contribute to its hardening. Therefore, once dynamical friction has become inefficient at a_0 , the subsequent binary evolution is driven by interactions with bound stars orbiting within the binary sphere of influence (Sesana, Haardt, et al. 2008). As the binary hardens, these stars are progressively ejected from the central region, carrying away the orbital energy extracted from the binary and causing its separation to decrease towards the hardening radius

$$a_h = \frac{1}{4(3 - \gamma)} \left(\frac{q}{1 + q} \right)^{(2-\gamma)/(3-\gamma)}, \quad (2.33)$$

below which the stellar-scattering term included in Equations (2.20) becomes dominant (Sesana 2010).

Finally, provided that a sufficient supply of loss-cone stars is available, the binary eventually enters the GW-dominated regime, in which GW emission drives the orbital decay towards final coalescence. For a stellar cusp like stellar background, the transition occurs at the binary separation a_{GW} defined by the condition $H_u = H_{GW}$ (Sesana 2010), which is

$$a_{GW} = 4 \left[\frac{128\pi(3-\gamma)F(e)}{5H_u} \right]^{1/5} \frac{\sigma}{c} q^{-4/5} (1+q)^{3/5} a_h \quad (2.34)$$

and $F(e)$ is a function of the eccentricity e introduced in Equation (1.131).

In conclusion, for the equal-mass BH binaries considered in this work, the initial binary separation a_0 , given by Equation (2.32), and the hardening radius a_h , defined in Equation (2.33), are comparable. In particular, their separation in logarithmic space is small compared with the overall range of binary separations spanned during the subsequent evolution. Consequently, the binary is already close to the hard-binary regime at its birth, and the additional hardening associated with the erosion of the bound stellar cusp can safely be omitted from our analysis (Sesana and Khan 2015).

To investigate the role played by the central stellar density in the overall evolution of the binary, while keeping all other relevant conditions fixed, we consider three numerical experiments, one for each value of the stellar cusp slope $\gamma = (1, 1.5, 2)$. Using the hybrid model described above, we follow the evolutionary tracks of the same equal-mass IMBH binary, with $m_1 = m_2 = 10^5 M_\odot$, initially placed at the same initial conditions (a_0, e_0) in all cases, but embedded in stellar backgrounds characterized by different cusp slopes. Although the value of a_0 depends on the adopted cusp slope γ , as determined from Equation (2.32), we adopt a common initial configuration for all binaries, $(\bar{a}_0, 0.5)$, where $\bar{a}_0$ is the average of the a_0 values obtained for the different stellar cusps. All stellar cusps are instead characterized by the same velocity dispersion, $\sigma = 48.9 \text{ km/s}$, owing to Equation (2.29). Combined with the different cusp slopes, this common value of σ yields the corresponding stellar densities at the influence radius, $\rho_{inf} = (1.71, 3.04, 6.83) \times 10^5 M_\odot/\text{pc}^3$ for $\gamma = (1, 1.5, 2)$, respectively.

Following the prescriptions given by Equations (2.30), (2.32), (2.33), and (2.34), the characteristic transition lengthscales are computed for each stellar background, or equivalently for each value of γ , and reported in Table 2.3.

γ	$r_{inf} [\text{pc}]$	$a_0 [\text{pc}]$	$a_h [\text{pc}]$	$a_{GW} [10^{-4} \text{pc}]$
1	0.72	0.51	0.016	0.97
1.5	0.54	0.34	0.034	0.91
2	0.36	0.18	0.045	0.84

Table 2.3: Hybrid model parameters for three different stellar cusp slopes γ . Column 1: Stellar cusp slope. Column 2: Influence radius of the stellar cusp. Column 3: Initial binary separation. Column 4: Binary hardening radius. Column 5: Stellar scattering-GW hardening transition lengthscale.

A first inspection of Table 2.3 and Figure 2.2 shows that, for fixed BH masses, increasing γ leads to an increasingly concentrated stellar distribution and, consequently, to a higher stellar density within the binary sphere of influence. As a result, the characteristic lengthscales r_{inf} , a_0 , and a_{GW} progressively decrease with increasing γ . This reflects the fact that, in a steeper cusp, the stellar mass reaches twice the binary mass and twice the secondary BH mass at smaller radii, while the transition to the GW-dominated regime occurs at a smaller binary separation.

Figure 2.2 shows the mass density profile of each stellar cusp profile considered here, along with the corresponding influence radius.

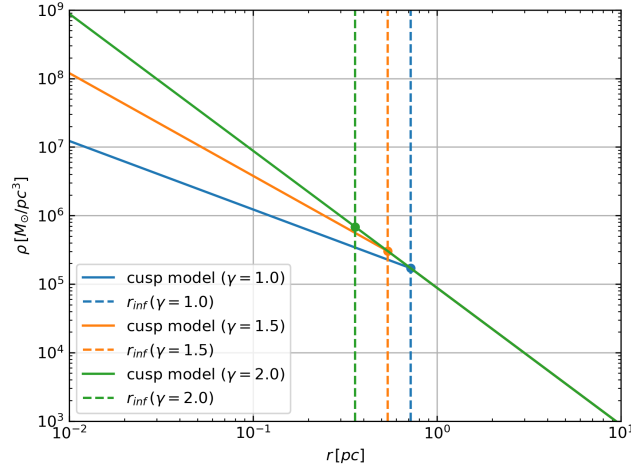


Figure 2.2: Stellar density profiles for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively), with the corresponding influence radii r_{inf} marked by vertical dashed lines of the same colour.

As all three numerical experiments involve equal-mass binaries, the mass ratio is fixed to $q = 1$ for all values of γ . The dimensionless hardening rate H_u is therefore determined directly from Equation (2.18), using the hardening radii a_h listed in Table 2.3 for each cusp slope and the fitting parameters reported in the first row of Table 2.1. The resulting hardening rates for the three stellar cusps are shown in Figure 2.3.

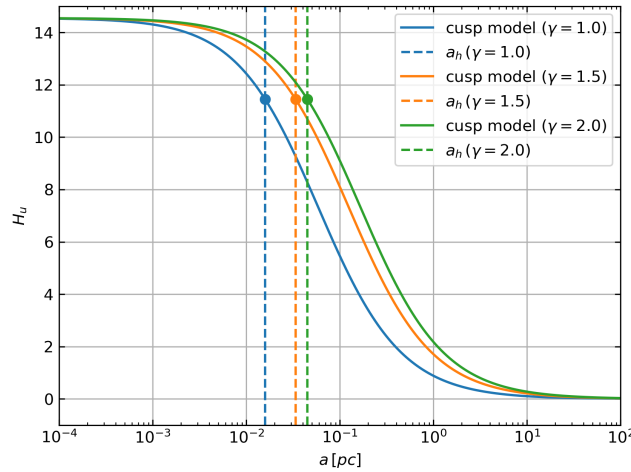
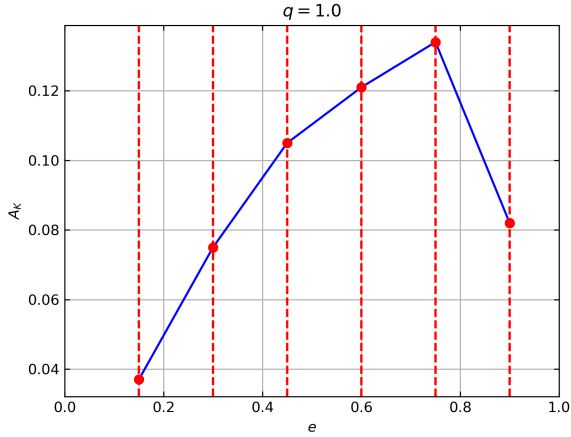


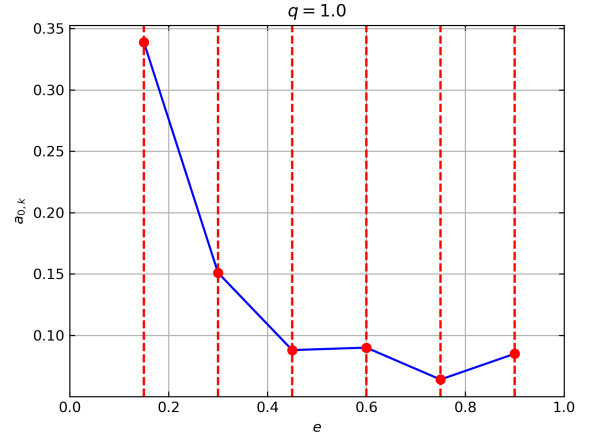
Figure 2.3: Dimensionless hardening rate H_u as a function of binary separation for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively), with the corresponding hardening radii a_h marked by vertical dashed lines of the same colour.

The final quantity required to numerically integrate Equations (2.20) is the dimensionless eccentricity growth function K_u . Unlike H_u , K_u depends not only on the binary semimajor axis but also on its eccentricity. This dependence on eccentricity is encoded in the best-fitting parameters of Equation (2.19), as reported in Table 2.2. Although the best-fitting parameters are provided only at a discrete set of eccentricities, namely $e = (0.15, 0.3, 0.45, 0.6, 0.75, 0.9)$, their values over the full interval $e \in [0, 1]$ can be obtained by linear interpolation in eccentricity,

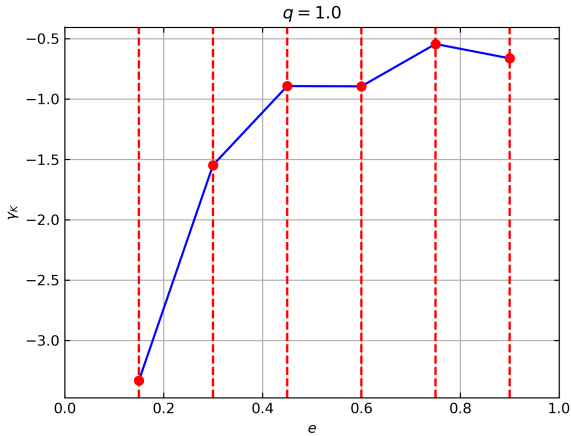
once the mass ratio q is specified (see, e.g., Sesana, Haardt, et al. (2006); Sesana (2010)). Figure 2.4 (a)-(b)-(c)-(d) illustrates this procedure.



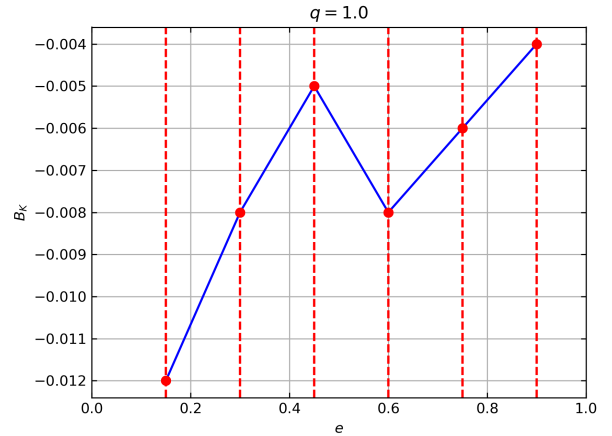
(a) Eccentricity growth rate parameter A_K (solid blue piecewise line) as a function of binary eccentricity e for binary masses ratio $q = 1$. Red dashed vertical lines mark the sampled eccentricities, with red dots corresponding to the values listed in Table 2.2.



(b) Eccentricity growth rate parameter $a_{0,K}$ (solid blue piecewise line), in a_h units, as a function of binary eccentricity e for binary masses ratio $q = 1$. Red dashed vertical lines mark the sampled eccentricities, with red dots corresponding to the values listed in Table 2.2.



(c) Eccentricity growth rate parameter γ_K (solid blue piecewise line) as a function of binary eccentricity e for binary masses ratio $q = 1$. Red dashed vertical lines mark the sampled eccentricities, with red dots corresponding to the values listed in Table 2.2.

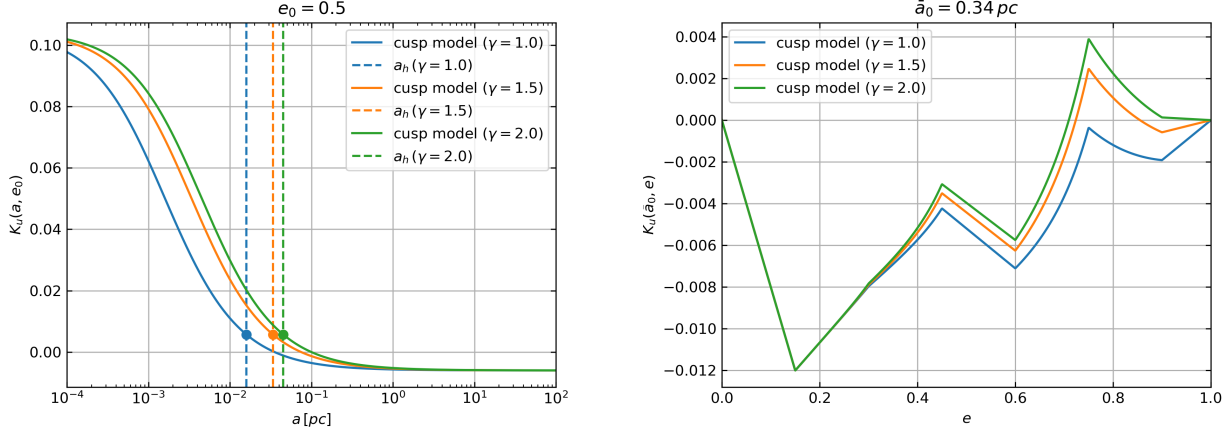


(d) Eccentricity growth rate parameter B_K (solid blue piecewise line) as a function of binary eccentricity e for binary masses ratio $q = 1$. Red dashed vertical lines mark the sampled eccentricities, with red dots corresponding to the values listed in Table 2.2.

Figure 2.4: Panels (a)–(b)–(c)–(d) show the best-fitting parameters A_K , $a_{0,K}$, γ_K , and B_K of the eccentricity growth-rate function K_u , respectively, as functions of the binary eccentricity e . Red points indicate the values obtained from three-body scattering experiments (see Table 2.2), while blue piecewise-linear curves show their linear interpolation. Vertical red dashed lines mark the sampled eccentricity values.

These interpolated best-fitting parameters can then be used to derive $K_u(a, e)$ and, in order to extend the interpolation to the boundaries of the physical eccentricity interval, two additional points are manually introduced at $e = 0$ and $e = 1$ in order to ensure $K_u(a, 0) = K_u(a, 1) = 0$. Figure 2.5 (a)–(b) show $K_u(a) = K_u(a, e_0)$ and $K_u(e) = K_u(\bar{a}_0, e)$, respectively, where $(\bar{a}_0, e_0) =$

$(0.34pc, 0.5)$ corresponds to the initial condition adopted in all three numerical simulations. As displayed, this two-variable function is smooth when evaluated at fixed e , while it is piecewise linear when evaluated at fixed a .



(a) Dimensionless eccentricity growth rate $K_u(a, e)$, at fixed eccentricity $e_0 = 0.5$, as a function of binary separation for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively), with the corresponding hardening radii a_h marked by vertical dashed lines of the same colour.

(b) Dimensionless eccentricity growth rate $K_u(a, e)$, at fixed binary semimajor axis $\bar{a}_0 = 0.34 pc$, as a function of binary eccentricity for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively). The boundary conditions $K_u(a, 0) = K_u(a, 1) = 0$ have been included.

Figure 2.5: 1-D slices of the dimensionless eccentricity growth rate $K_u(a, e)$, at fixed eccentricity (a) and binary separation (b), for three different stellar cusps slopes $\gamma = (1, 1.5, 2)$.

Having reported how the essential ingredients of the adopted hybrid model have been derived, we can now specify the setup adopted for the numerical integrations. In all the experiments presented in this Section, the integration starts from the initial conditions $(\bar{a}_0, e_0) = (0.34, pc, 0.5)$ and is terminated when the binary separation reaches the Schwarzschild radius of the binary, defined as

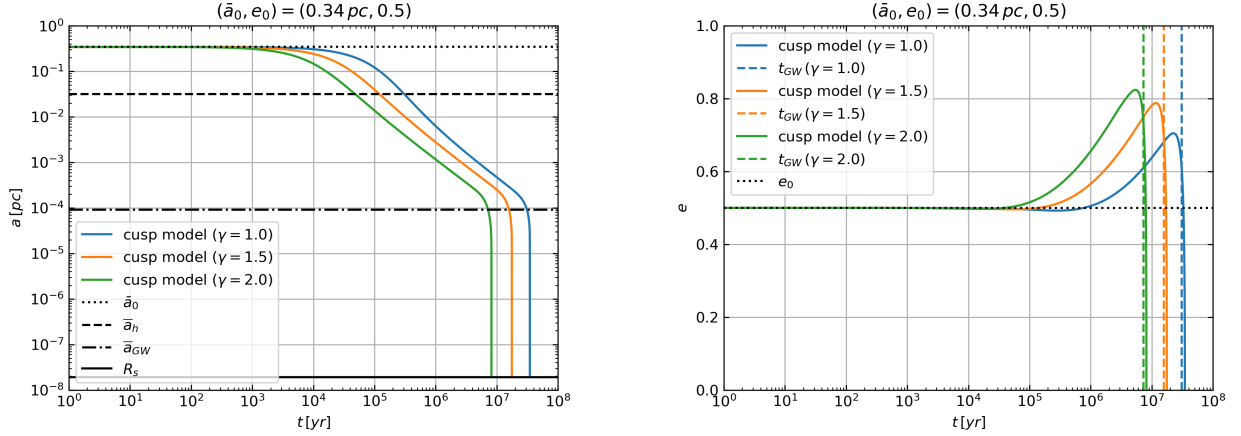
$$R_s = \frac{2GM_{12}}{c^2}, \quad (2.35)$$

which, for the total binary mass considered here, corresponds to $R_s = 5.9 \times 10^5 km$. Moreover, the integration time step Δt is chosen to ensure the numerical stability of the RK4 scheme according to the criterion given in Equation (2.22). For all numerical runs, we adopt

$$\Delta t = \epsilon \cdot 2.79 \min \left\{ \frac{1}{|\lambda_1|}; \frac{1}{|\lambda_2|} \right\} \quad (2.36)$$

where $\epsilon = 0.001$ is the chosen safety factor.

Figure 2.6 (a)–(b) show the time evolution of the binary semimajor axis, $a(t)$, and eccentricity, $e(t)$, respectively, for the three stellar cusp slopes considered, $\gamma = (1, 1.5, 2)$.



(a) Binary semimajor axis time evolution for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively).

Black horizontal lines indicate the mean values over the three stellar cusps of a_0 (dotted), a_h (dashed), a_{GW} (dash-dotted), and the binary R_s .

(b) Binary eccentricity evolution for three stellar cusp slopes, $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively).

Vertical dashed lines of the corresponding colours mark the times t_{GW} at which $a = a_{GW}$ and GW-driven circularization becomes dominant.

Figure 2.6: Evolutionary tracks $(t, a(t))$ in panel (a) and $(t, e(t))$ in panel (b) for three stellar cusp slopes, $\gamma = (1, 1.5, 2)$ and binary initial conditions $(a_0, e_0) = (0.34 pc, 0.5)$.

The first relevant result concerns the estimated coalescence timescales; they turn out to be $\tau_{\text{coal}} = (34.7, 17.6, 8.18) \text{ Myr}$. All three values are shorter than the Hubble time, $t_H = 13.8 \text{ Gyr}$, indicating that, for the considered initial conditions, an equal-mass IMBH binary with $m_1 = m_2 = 10^5 M_\odot$ may emit a potentially detectable GW signal in the LISA frequency band. Moreover, the coalescence time is found to decrease systematically with increasing cusp slope γ . This behaviour, shown in Figure 2.6 (a), can be attributed to the higher central stellar densities associated with steeper cusps, which provide a larger supply of stars interacting with the binary through loss-cone dynamics. The enhanced rate of stellar interactions accelerates the binary hardening, allowing it to reach the separation at which GW emission becomes the dominant shrinking mechanism, and consequently shortening the overall time to coalescence. Figure 2.6 (b), instead, shows that the binary eccentricity tends to increase during the phase dominated by stellar scattering. As the cusp becomes steeper, this dynamical regime becomes progressively shorter, allowing GW-driven circularization to take over at an earlier stage of the evolution. Consequently, the maximum eccentricity reached by the binary decreases with increasing γ .

These results motivate a more detailed investigation of the role played by the compactness of the stellar background in shaping the binary evolution and determining its corresponding coalescence time. This aspect will be explored in the next Section.

2.3 NGC 205 IMBH Binary evolution

The Local Group comprises several dozen galaxies, with the majority of its members being low-mass systems gravitationally associated with its two dominant galaxies, the Milky Way and M31 (Cimatti et al. 2019). Among them, M 32, NGC 205, NGC 5102, and NGC 5206, four nucleated low-mass early-type galaxies, are of particular interest because their central regions can be resolved in detail and their nuclear dynamics can be studied through high-resolution observations (Nguyen et al. 2018). In this Section, we investigate the evolution of an IMBH binary embedded in NGC 205, with particular emphasis on the role played by the binary

eccentricity e_0 at pairing on the resulting evolutionary tracks and coalescence timescales.

NGC 205 is a nucleated dwarf elliptical galaxy orbiting M 31, located at RA = 00^h40^m22.1^s and Dec = +41°41′07″. Its distance has been determined through the tip of the red giant branch method, yielding a distance modulus of $(m - M)_0 = 24.58 \pm 0.07$, corresponding to a distance of approximately 824 ± 27 kpc (McConnachie et al. 2005). HST/STIS spectroscopy of the Ca II triplet region revealed no significant nuclear rotation and a nearly constant velocity dispersion of ~ 20 km s⁻¹ within the central $\sim 0.3''$ (~ 1 pc). Dynamical modelling of the nuclear region yielded a 3σ upper limit of $3.8 \times 10^4 M_\odot$ for the mass of a possible central IMBH (see, e.g., Valluri et al. (2005); Geha et al. (2006)).

Following Khan and Holley-Bockelmann (2021), we adopt NGC 205 as a representative present-day, nucleated dwarf galaxy and use its observed properties to set up numerical experiments with the Quinlan–Sesana hybrid model. Its two-dimensional surface-brightness profile can be decomposed into two distinct structural components, corresponding to the bulge and the nuclear star cluster. We model each component with a three-dimensional deprojected Sersic density profile of the form (see, e.g., Prugniel and Simien (1997); Terzić and Graham (2005)):

$$\rho(r) = \rho_0 \left(\frac{r}{R_e} \right)^{-p} e^{-b \left(\frac{r}{R_e} \right)^{1/n}}. \quad (2.37)$$

The stellar density profile given in Equation (2.37) is characterized by three structural parameters, (ρ_0, R_e, n) , together with the model-dependent quantities $p(n)$ and $b(n)$. Here, R_e is the effective radius, enclosing half of the total luminosity of the stellar component, while n is the Sersic index controlling the shape of the profile. The normalization ρ_0 is determined by requiring the volume-integrated density to reproduce the total stellar mass of the component. The parameters p and b are functions of the Sersic index and can be approximated, for $0.5 < n < 10$, as $p = 1.0 - 0.6097/n + 0.05563/n^2$ and $b \approx 2n - 1/3 + 0.009876/n$ (Terzić and Graham 2005). Table 2.4 summarizes the main structural parameters of the bulge and NSC of NGC 205, which are required to specify the corresponding Sersic profiles.

Component	n	R_e [pc]	M_* [$10^7 M_\odot$]
NSC ...	1.6	1.3	0.18
Bulge ...	1.4	516	97.2

Table 2.4: NGC 205 structural parameters of stellar bulge and NSC. Column 1: Galaxy component. Column 2: Sersic index. Column 3: Effective radius. Column 4: Stellar mass. Values from Khan and Holley-Bockelmann (2021).

Since Chapter 3 is devoted to studying the dynamical evolution of IMBH binaries embedded in Sersic-like stellar backgrounds through direct N -body simulations, we briefly examine here the properties of this profile, defined by Equation (2.37), that are relevant for constructing equilibrium N -body initial conditions and interpreting the results presented in the following chapter.

Given the mass density profile in Equation (2.37) for a spherically symmetric stellar component of total mass M_* , the mass enclosed within a sphere of radius r is obtained by integrating the density profile over the corresponding volume:

$$M(< r) = 4\pi \int_0^r dx \rho(x) x^2. \quad (2.38)$$

By exploiting the standard substitution $Z \equiv b \left(\frac{r}{R_e} \right)^{1/n}$, the integral can be expressed in terms

of the lower incomplete gamma function,

$$\gamma(a, x) = \int_0^x dt e^{-t} t^{a-1}, \quad (2.39)$$

as

$$M(< r) = 4\pi\rho_0 R_e^3 n b^{n(p-3)} \gamma \left[n(3-p), b \left(\frac{r}{R_e} \right)^{1/n} \right]. \quad (2.40)$$

Imposing the normalization condition $M(< r) \rightarrow M_*$ as $r \rightarrow +\infty$, and using the fact that the lower incomplete gamma function reduces to the complete gamma function in this limit,

$$\Gamma(a) = \int_0^\infty dt e^{-t} t^{a-1}, \quad (2.41)$$

Equation (2.40) can be conveniently recast in terms of the total stellar mass as

$$M(< r) = M_* \frac{\gamma \left[n(3-p), b \left(\frac{r}{R_e} \right)^{1/n} \right]}{\Gamma[n(3-p)]} \quad (2.42)$$

and the normalizing density ρ_0 is unambiguously set by comparing Equations (2.40) and (2.42) to

$$\rho_0 = \frac{M_*}{4\pi R_e^3 n \Gamma[n(3-p)]} b^{n(3-p)}. \quad (2.43)$$

Lastly, the gravitational potential can therefore be written in the general form

$$\Phi(r) = \Phi(r_0) + \frac{GM(< r_0)}{r_0} - \frac{GM(< r)}{r} - 4\pi G \int_r^{r_0} dx \rho(x)x. \quad (2.44)$$

Although the stellar distribution is formally untruncated, its enclosed mass converges to the finite value M_* as $r \rightarrow +\infty$, allowing for the standard normalization $\Phi(\infty) = 0$ and $r_0 \rightarrow +\infty$. In this way, Equation (2.44) reduces to

$$\Phi(r) = -\frac{GM(< r)}{r} - 4\pi G \int_r^\infty dx \rho(x)x. \quad (2.45)$$

Substituting the Prugniel–Sérsic density profile reported in Equation (2.37) into Equation (2.45), the integral can be evaluated in terms of the upper incomplete gamma function,

$$\Gamma(a, x) = \int_x^\infty dt e^{-t} t^{a-1}, \quad (2.46)$$

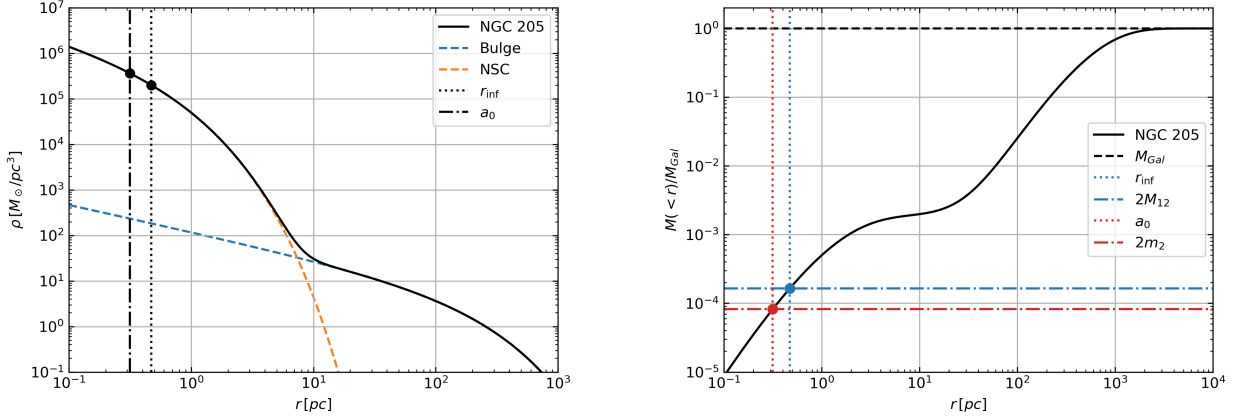
yielding

$$\Phi(r) = -\frac{GM_*}{r} \left\{ \frac{\gamma[n(3-p), Z]}{\Gamma[n(3-p)]} + Z^n \frac{\Gamma[n(2-p), Z]}{\Gamma[n(3-p)]} \right\}. \quad (2.47)$$

We can now turn to the suite of numerical experiments performed. Here, we present the evolutionary tracks obtained by evolving equal-mass IMBH binaries, with component masses $m_1 = m_2 = 4 \times 10^4 M_\odot$, within the stellar environment of NGC 205 (Khan and Holley-Bockelmann 2021). The IMBHs are initially placed at a separation a_0 , as defined in Equation (2.32), and the simulations are performed for five different initial eccentricities, $e = (0.01, 0.25, 0.50, 0.75, 0.99)$. At the end of § 2.1, we described the prescription proposed by Sesana and Khan (2015) for applying the self-consistent Quinlan–Sesana hybrid model to arbitrary spherically symmetric, isotropic stellar backgrounds. It requires only the additional computation of the stellar mass

density ρ_{inf} and velocity dispersion σ_{inf} at the binary influence radius r_{inf} , both set by the IMBH binary and stellar environment features.

Therefore, using Equation (2.43), the normalizing densities of the NGC 205 bulge and NSC are found to be $\rho_{0,Bulge} = 2.95 M_{\odot}/pc^r$ and $\rho_{0,NSC} = 4.80 \times 10^5 M_{\odot}/pc^3$, respectively. These values, along with the effective radius R_e and the Sersic index n , fully specify the corresponding Sersic stellar mass density and cumulative mass profiles, from which those of NGC 205 can then be reconstructed. These profiles are shown in Figure 2.7 (a) and (b).



(a) Mass density profiles of the NGC 205 stellar components: bulge (blue dashed), NSC (orange dashed), and the total galaxy (black solid). Black vertical lines mark the binary influence radius (dotted) and the initial binary separation a_0 (dash-dotted).

(b) Cumulative stellar mass profile of NGC 205, normalized to the total stellar mass M_{Gal} . Vertical dotted lines mark the characteristic scales r_{inf} (blue) and a_0 (red), while horizontal dash-dotted lines of the same colours indicate the corresponding mass values $2M_{12}$ and $2m_2$, respectively.

Figure 2.7: NGC 205 mass density profile, together with those of its stellar components, and cumulative mass profile.

As shown in Figure 2.7(b), the characteristic length scales r_{inf} and a_0 are determined from the cumulative mass profile by numerically solving equation

$$M(< r_{inf}) - 2M_{12} = 0 \quad (2.48)$$

and

$$M(< a_0) - 2m_2 = 0 \quad (2.49)$$

respectively. The former solution, i.e. r_{inf} , is then used to derive ρ_{inf} as highlighted in Figure 2.7 (a).

At this stage, the remaining quantities required to fully specify all the terms in Equations (2.20) are the stellar velocity dispersion at the binary influence radius, σ_{inf} , and the hardening radius, a_h , which is given by (Quinlan 1996)

$$a_h = \frac{Gm_2}{4\sigma_{inf}}. \quad (2.50)$$

A more detailed discussion about the stellar velocity dispersion, together with its application to the construction of equilibrium N -body initial conditions, is deferred to Chapter 3. In the present context, we limit ourselves to use the radial velocity dispersion formula

$$\sigma^2(r) = \frac{1}{\rho(r)} \int_r^\infty ds \frac{GM(< s)}{s^2} \rho(s), \quad (2.51)$$

holding for spherically symmetric and isotropic stellar system. By combining the previously reconstructed stellar mass density and cumulative mass profiles, Equation (2.51) returns the function $\sigma = \sigma_r$, which is displayed in Figure 2.8 normalized to the circular speed $V_c(r) = \sqrt{GM(< r)}/r$.

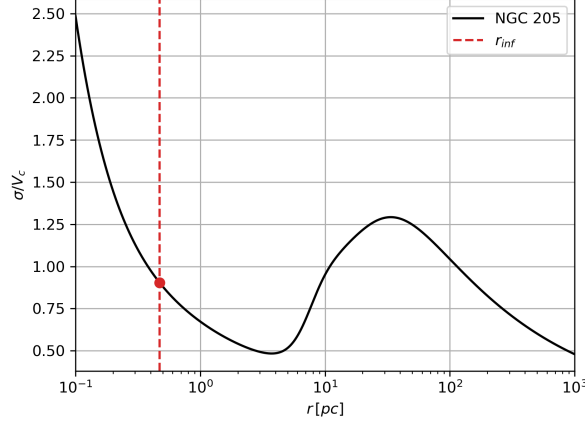


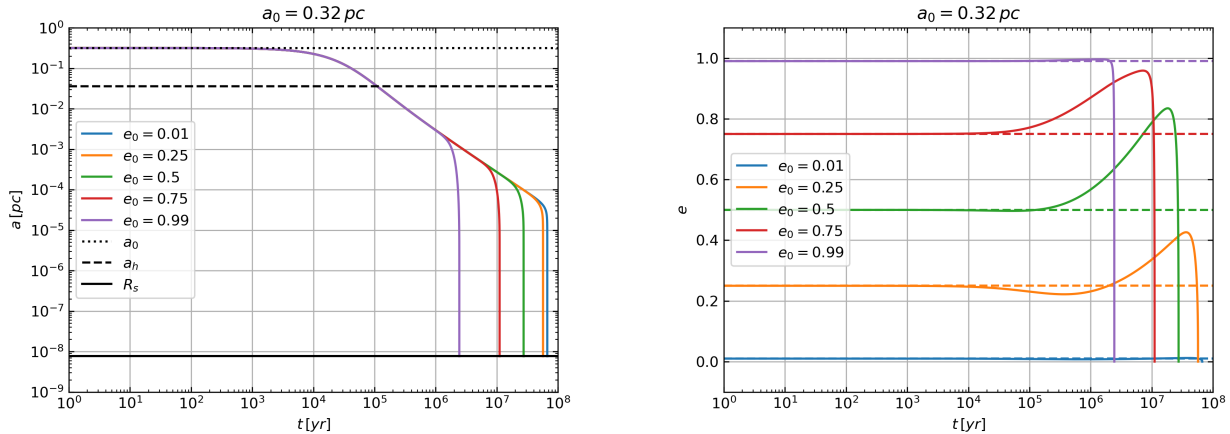
Figure 2.8: NGC 205 velocity dispersion profile (black solid line), normalized to the circular speed $V_c = \sqrt{GM(r)}/r$. The red vertical line marks the binary influence radius r_{inf} .

The main hybrid model parameters, obtained according the operations illustrated above and used for our numerical experiments, are conveniently listed in Table 2.5.

$r_{inf} [pc]$	$a_0 [pc]$	$a_h [pc]$	$\rho_{inf} [M_\odot/pc^3]$	$\sigma_{inf} [km/s]$
0.47	0.32	0.036	2.00	34.5

Table 2.5: NGC 205 hybrid model parameters. Column 1: Binary influence radius. Column 2: Binary initial separation. Column 3: Binary hardening radius. Column 4: Stellar mass density at binary influence radius. Column 5: Stellar velocity dispersion at binary influence radius.

Finally, the numerical integration of Equations (2.20), performed following the procedure described in § 2.2, yields the evolutionary tracks $(a(t), e(t))$ and the corresponding coalescence timescales, shown in Figure 2.9 (a)-(b), respectively.



(a) Binary semimajor axis time evolution for five different initial eccentricities $e_0 = (0.01, 0.25, 0.5, 0.75, 0.99)$ (blue, orange, green, red, and violet, blue solid lines, respectively). Black horizontal lines indicate the common binary initial separation (dotted), hardening radius (dashed) and the binary R_s (solid).

(b) Binary eccentricity time evolution for five, different initial eccentricities, $e_0 = (0.01, 0.25, 0.5, 0.75, 0.99)$ (blue, orange, green, red, and violet, blue solid lines, respectively). Horizontal dashed lines of the corresponding colours indicate the initial eccentricity of each the numerical experiment.

Figure 2.9: NGC 205 IMBH binary evolutionary tracks $(t, a(t))$ in panel (a) and $(t, e(t))$ in panel (b) for five initial eccentricities, $e_0 = (0.01, 0.25, 0.5, 0.75, 0.99)$ and binary initial separation $a_0 = 0.32 \text{ pc}$.

In excellent agreement with Sesana (2010) and Khan and Holley-Bockelmann (2021), our analysis shows that IMBH binaries forming with a higher initial eccentricity e_0 , together with the tendency of eccentric binaries to further increase their eccentricity during the stellar hardening phase, reach higher eccentricity peaks. As a result, the GW-driven regime takes over at an earlier stage, leading to a shorter coalescence timescale. Moreover, the presence of the NSC substantially increases the central stellar density of the galaxy, thereby providing a particularly favourable environment for the rapid hardening and coalescence of the IMBH binary. Specifically, it is found $\tau_{\text{coal}} = (67.0, 57.1, 27.5, 11.1, 2.4) \text{ Myr}$ for $e = (0.01, 0.25, 0.50, 0.75, 0.99)$, respectively.

2.4 Cosmic Size Evolution Down to NGC 205

In this final Section of Chapter 2, we investigate the effects of removing the NSC, allowing the IMBH binary to evolve within the stellar bulge alone, as well as those of cosmic size evolution on the bulge stellar distribution and of the initial eccentricity e_0 on the binary evolutionary tracks. This analysis allows us to combine them simultaneously in order to investigate their cumulative impact on the binary evolution. In particular, we determine which combinations of these effects may enable IMBH binaries hosted by non-nucleated dwarf galaxies at redshift z to become potential GW sources for LISA and ET, detectable in the present-day Universe.

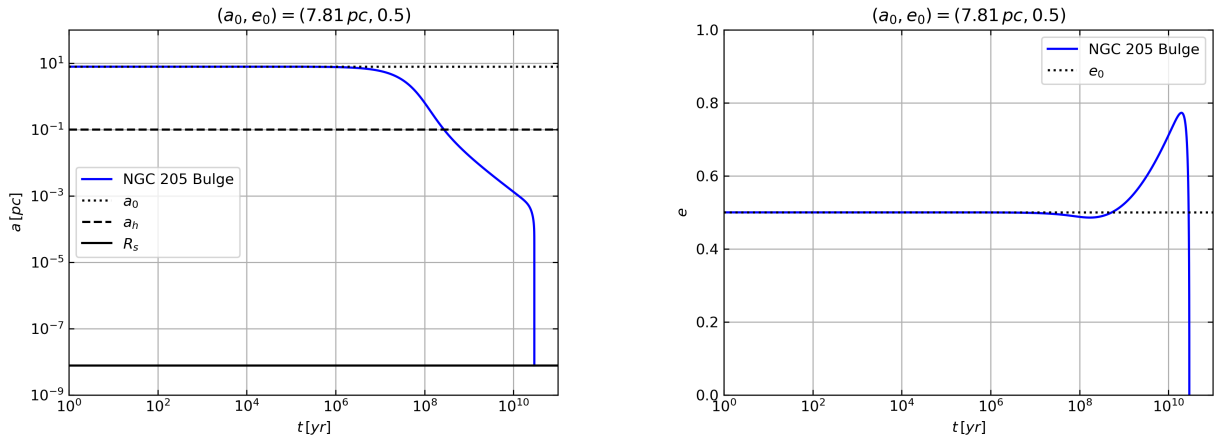
To this end, we first perform numerical experiments on the same IMBH binary analysed in § 2.3, this time evolving it within the stellar bulge of NGC 205 alone. Adopting the same prescription described extensively in the previous Sections, we integrate Equations (2.20) from the initial condition $(a_0, e_0) = (a_0, 0.5)$, with a_0 given by Equation (2.32), down to R_s using again RK4 scheme and the same integration timestep.

The relevant quantities entering the hybrid model for this numerical experiment are reported in Table 2.6.

$r_{inf} [pc]$	$a_0 [pc]$	$a_h [pc]$	$\rho_{inf} [M_\odot/pc^3]$	$\sigma_{inf} [km/s]$
10.5	7.81	0.10	25.4	20.8

Table 2.6: NGC 205 stellar bulge hybrid model parameters. Column 1: Binary influence radius. Column 2: Binary initial separation. Column 3: Binary hardening radius. Column 4: Stellar bulge mass density at binary influence radius. Column 5: Stellar bulge velocity dispersion at binary influence radius.

Figure 2.10 (a) and (b) show the predicted evolution of the IMBH binary semimajor axis and eccentricity, respectively, for this setup.



(a) NGC 205 IMBH binary semimajor axis time evolution (blue, solid line). Black horizontal lines mark the binary initial separation (dotted), hardening radius (dashed) and the binary R_s (solid). (b) NGC 205 IMBH binary eccentricity time evolution (blue, solid line). The black, horizontal line indicate the adopted binary initial eccentricity $e_0 = 0.5$.

Figure 2.10: Evolutionary tracks $(t, a(t))$ in panel (a) and $(t, e(t))$ in panel (b) of the IMBH binary evolving within NGC 205 stellar bulge with initial condition $(a_0, e_0) = (7.81 pc, 0.5)$.

The coalescence timescale estimated from this simulation is $t_{coal} = 22.5, Gyr$, significantly longer than the Hubble time, $t_H \approx 14, Gyr$, and in excellent agreement with the findings of Khan, Javed, et al. (2024). This result indicates that the IMBH binary hosted by this non-nucleated galaxy cannot reach coalescence within the current age of the Universe, even if the binary had formed at the earliest possible cosmic times.

While the analysis presented above appears to disfavour non-nucleated dwarf galaxies as potential hosts of GW sources detectable by LISA and ET, at least in the local Universe, such environments may still host coalescing IMBH binaries when the effects of cosmic size evolution are taken into account. In particular, galaxies of a given mass are expected to have had smaller effective radii at higher redshifts, hence, potentially compensating for the absence of a NSC. Here, we follow the cosmic size evolution empirical relation derived by Ono et al. (2023), according to which the effective radius $R_e = R_e(z)$ of a galaxy of fixed stellar mass evolves with redshift as

$$R_e(z) = R_e(z=0)(1+z)^{-1.22}. \quad (2.52)$$

Therefore, we perform a suite of numerical experiments using the Quinlan–Sesana hybrid model, in which the cosmic size evolution relation given in Equation (2.52) is applied by assuming the present-day structural properties of the NGC 205 stellar bulge to represent the endpoint of the redshift evolution of a galaxy. Specifically, all simulations consider an equal-mass IMBH

binary with $m_1 = m_2 = 4 \times 10^4 M_\odot$, initially separated by $a_0 = 10 pc$ and embedded in the corresponding redshift-evolved Sérsic stellar profile. The redshift range is sampled at 25 equally spaced values between $z = 0$ and $z = 12$, while the initial eccentricity is varied over the values $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$, resulting in 125 independent runs.

For each galaxy model, corresponding to one sampled point of the reshift space, the effective radius $R_e(z)$ is thus computing according to Equation (2.52), while the normalizing mass density ρ_0 is derived from Equation (2.43) by assuming the stellar mass uquals to $M_{bulge} = 97.2 \times 10^7 M_\odot$ reported in Table 2.4.

Figure 2.11 compares the stellar mass density profiles, together with the corresponding influence radii, of a subsample of the reconstructed galaxy models with that of the NGC 205 NSC, which predominantly determined the influence radius of the overall galaxy as discussed in § 2.3. Only reconstructed galaxies at $z \approx 10$, or higher, seem to have influence radii, and associated stellar density, comparable to those of the nuclear star cluster.

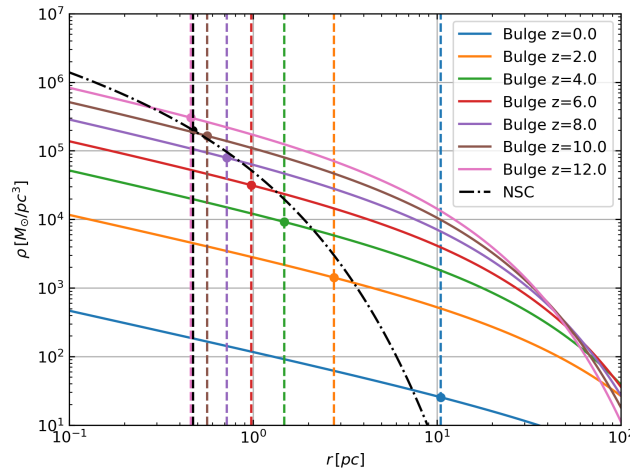


Figure 2.11: Mass density profiles of a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (solid blue, orange, green, red, violet, brown, and pink lines, respectively), together with the NGC 205 NSC profile (solid black line). The corresponding influence radii are indicated by vertical dashed lines of the same colours.

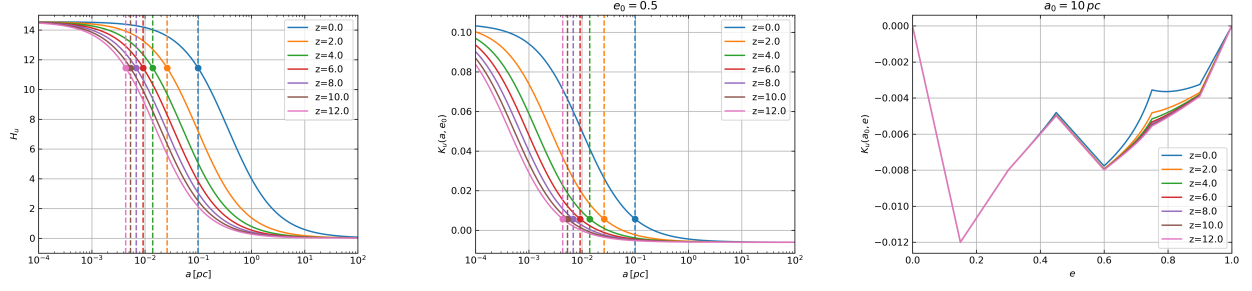
Moving forward, repeating the same procedure extensively described in the previous Sections, for each density profile we determine the cumulative mass profile $M(< r)$ by using Equation (2.42), the influence radius r_{inf} by numerically solving Equation (2.48), the mass density at the binary influence radius ρ_{inf} by plugging r_{inf} into the stellar density profile, the velocity dispersion at the binary influence radius σ_{inf} by evaluating Equation (2.51) at r_{inf} and finally substituting it within Equation (2.50) to derive the hardening radius a_h .

In Table 2.7 are listed all the parameters mentioned above for each galaxy model.

z	R_e [pc]	ρ_0 [$M_\odot/\text{pc}^3$]	n	r_{inf} [pc]	ρ_{inf} [$M_\odot/\text{pc}^3$]	a_h [pc]	σ_{inf} [km/s]
0	516	2.95	1.4	10.52	2.54×10^1	0.100	20.8
0.5	314.6	1.30×10^1	1.4	6.41	1.12×10^2	0.061	26.6
1	221.5	3.73×10^1	1.4	4.51	3.21×10^2	0.043	31.7
1.5	168.7	8.43×10^1	1.4	3.44	7.27×10^2	0.033	36.3
2	135.1	1.64×10^2	1.4	2.75	1.42×10^3	0.026	40.6
2.5	111.9	2.89×10^2	1.4	2.28	2.49×10^3	0.022	44.6
3	95.1	4.71×10^2	1.4	1.94	4.06×10^3	0.018	48.4
3.5	82.4	7.25×10^2	1.4	1.68	6.25×10^3	0.016	52.0
4	72.4	1.07×10^3	1.4	1.48	9.20×10^3	0.014	55.4
4.5	64.5	1.51×10^3	1.4	1.31	1.30×10^4	0.012	58.8
5	58.0	2.08×10^3	1.4	1.18	1.79×10^4	0.011	62.0
5.5	52.6	2.78×10^3	1.4	1.07	2.40×10^4	0.010	65.1
6	48.0	3.65×10^3	1.4	0.98	3.15×10^4	0.009	68.1
6.5	44.2	4.70×10^3	1.4	0.90	4.05×10^4	0.009	71.0
7	40.8	5.95×10^3	1.4	0.83	5.14×10^4	0.008	73.9
7.5	37.9	7.43×10^3	1.4	0.77	6.41×10^4	0.007	76.6
8	35.4	9.16×10^3	1.4	0.72	7.90×10^4	0.007	79.4
8.5	33.1	1.12×10^4	1.4	0.67	9.63×10^4	0.006	82.0
9	31.1	1.35×10^4	1.4	0.63	1.16×10^5	0.006	84.6
9.5	29.3	1.61×10^4	1.4	0.60	1.39×10^5	0.006	87.2
10	27.7	1.91×10^4	1.4	0.56	1.65×10^5	0.005	89.7
10.5	26.2	2.25×10^4	1.4	0.53	1.94×10^5	0.005	92.2
11	24.9	2.63×10^4	1.4	0.51	2.26×10^5	0.005	94.6
11.5	23.7	3.05×10^4	1.4	0.48	2.63×10^5	0.005	97.0
12	22.6	3.52×10^4	1.4	0.46	3.04×10^5	0.004	99.3

Table 2.7: Structural parameters (left of the double vertical line) and hybrid-model parameters (right of the double vertical line) for 25 galaxy models as a function of redshift. Column 1: Redshift. Column 2: Effective radius. Column 3: Sersic profile normalization density. Column 4: Sersic index. Column 5: Binary influence radius. Column 6: Stellar mass density at binary influence radius. Column 7: Binary hardening radius. Column 8: Stellar velocity dispersion at binary influence radius.

It is worth noting that the parameters $a_{0,H}$ and $a_{0,K}$ entering the parametrization of H_u and K_u in Equations (2.18) and (2.19) are expressed in units of the binary hardening radius. Therefore, despite the primary and secondary BH masses are fixed across all the galaxies considered, the shapes of $H_u(a)$ and $K_u(a, e)$ vary from one galaxy to another. This is because the galaxies have different velocity dispersions at the binary influence radius, and consequently different hardening radii. This behaviour is clearly illustrated in Figure 2.12(a)-(b)-(c). Panels (a) and (b) show $H_u(a)$ and $K_u(a, 0.5)$, respectively, along with the corresponding hardening radii, while panel (c) shows $K_u(10pc, e)$ for a subsample of the reconstructed galaxy models.



(a) Dimensionless stellar hardening rate H_u of a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (solid blue, orange, green, red, violet, brown, and pink lines, respectively). The corresponding hardening radii are indicated by vertical dashed lines of the same colours.

(b) Dimensionless eccentricity growth rate $K_u(a, 0.5)$ of a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (solid blue, orange, green, red, violet, brown, and pink lines, respectively). The corresponding hardening radii are indicated by vertical dashed lines of the same colours.

(c) Dimensionless eccentricity growth rate $K_u(a, e)$, at fixed $a = 10 \text{ pc}$, of a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (solid blue, orange, green, red, violet, brown, and pink lines, respectively). The boundary conditions $K_u(a, 0) = K_u(a, 1) = 0$ have been included.

Figure 2.12: Dimensionless stellar hardening and eccentricity growth rates for a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$. Panels (a), (b), and (c) show $H_u(a)$, $K_u(a, 0.5)$, and $K_u(10 \text{ pc}, e)$, respectively.

Finally, Figure 2.13 shows the evolutionary tracks $(t, a(t))$ and $(t, e(t))$ for the same representative subsample of galaxies considered above, exploring the full range of initial eccentricities adopted in this analysis, $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$, all starting at initial separation $a_0 = 10 \text{ pc}$.

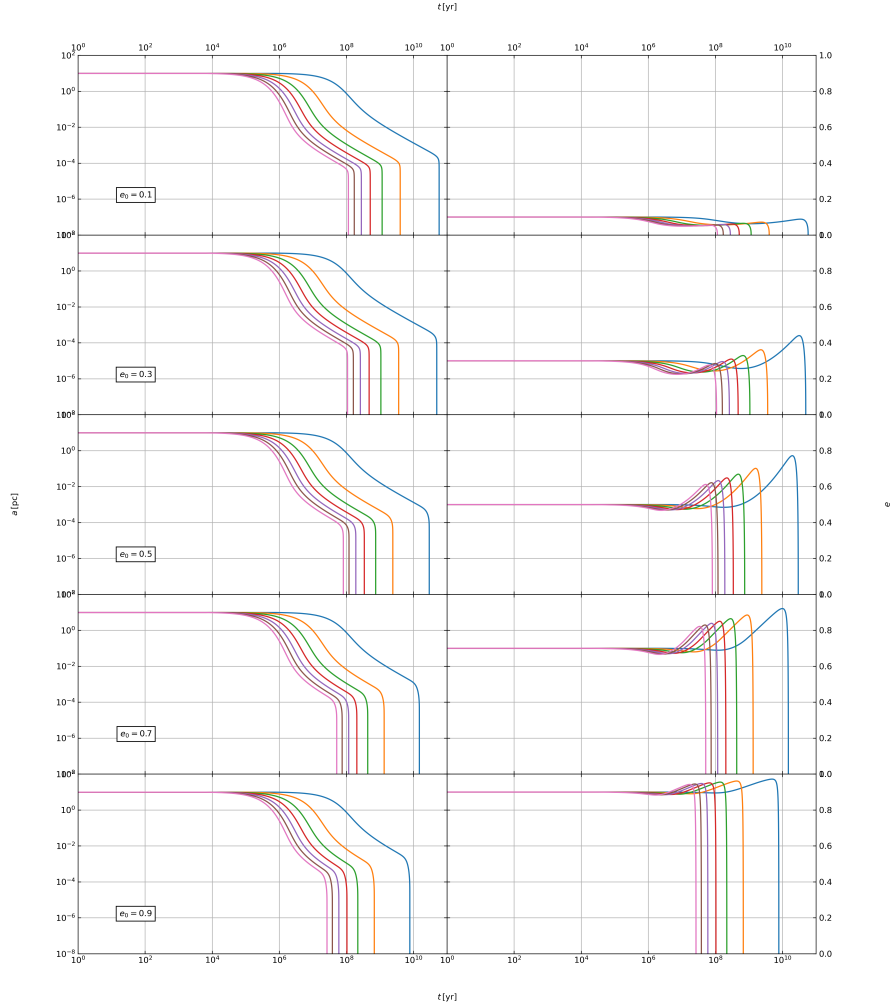


Figure 2.13: Evolutionary tracks of IMBH binaries evolving within a representative subsample of galaxy models. The left and right columns show the binary semimajor axis $a(t)$ and eccentricity $e(t)$, respectively, with each pair of panels in a given row corresponding to the evolution of the same galaxy model. The five rows, from top to bottom, correspond to initial eccentricities $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$, indicated by the panels on the left. Coloured curves correspond to galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (blue, orange, green, red, violet, brown, and pink, respectively).

The 125 runs performed to explore the redshift and initial eccentricity parameter spaces provide, in addition to an equal number of pairs of evolutionary tracks $(t, a(t))$ and $(t, e(t))$, an estimate of the coalescence time t_{coal} for each sampled configuration. Here, t_{coal} is defined as the time elapsed from the beginning of the numerical integration, corresponding to the initial binary conditions (a_0, e_0) , to the IMBH binary coalescence, assumed to occur at $a = R_s$. Whether a given galaxy model and set of binary initial conditions can host a GW-loud source detectable by LISA or ET in the present-day Universe depends not only on the estimated t_{coal} , but also on the cosmic time $t(z)$ corresponding to the redshift of the galaxy model (see, e.g. Coles and Lucchin (1995)).

In standard Λ CDM cosmology, within which the Friedmann–Lemaître–Robertson–Walker (FLRW) metric provides the appropriate description of the large-scale homogeneous and isotropic Universe, the Hubble function $H(z)$ describes the expansion rate of the Universe as a function of redshift and can be written as

$$H(z) = H_0 \sqrt{\Omega_k(1+z)^2 + \Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda}, \quad (2.53)$$

where H_0 is the present-day Hubble constant, Ω_k , Ω_m , Ω_r , and Ω_Λ are the present-day dimensionless density parameters associated with matter, radiation, spatial curvature, and dark energy, respectively. In the FLRW cosmological model, the cosmic time $t(z)$ corresponding to a given redshift z can be obtained from the Hubble function through the standard redshift–time relation

$$t(z) = \int_z^\infty \frac{dz'}{(1+z')H(z')}. \quad (2.54)$$

Adopting the cosmological parameters provided by Planck Collaboration et al. (2020), namely $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_m = 0.315$, and assuming a geometrically flat Λ CDM Universe, the cosmic time $t(z)$ is determined for each galaxy model from Equation (2.54). In this framework, $\Omega_\Lambda = 1 - \Omega_m = 0.685$, the radiation contribution is negligible at the redshifts considered and the age of the universe is estimated to be $t_{age} = t(z = 0) = 13.8 \text{ Gyr}$. This allows the coalescence times obtained from the numerical integrations, measured from the initial epoch of each model, to be converted into the corresponding cosmic times at which the IMBH binaries reach coalescence. Both the coalescence times and the corresponding cosmic times of binary coalescence are reported in Table 2.8 (a) and (b), respectively, for each of the 125 runs. Moreover, the predicted cosmic times of coalescence for these 125 models are displayed in Figure 2.14.

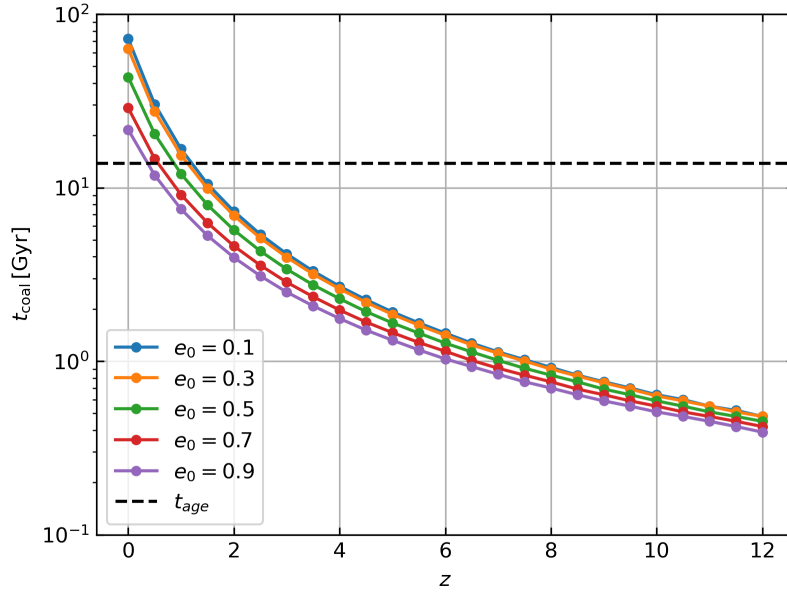


Figure 2.14: Cosmic coalescence time as a function of galaxy redshift for the 125 simulations. The solid blue, orange, green, red, and violet piecewise lines connect, respectively, the points corresponding to $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$ across the redshift sampling. The vertical black dashed line marks the present time.

$z \backslash e_0$	0.1	0.3	0.5	0.7	0.9
0	58.3	49.5	29.6	15.0	7.79
0.5	21.7	18.9	11.8	6.14	3.17
1	10.8	9.53	6.13	3.26	1.67
1.5	6.28	5.60	3.68	2.00	1.02
2	4.03	3.62	2.42	1.34	0.68
2.5	2.77	2.51	1.70	0.95	0.48
3	2.00	1.82	1.25	0.71	0.36
3.5	1.51	1.38	0.95	0.55	0.28
4	1.17	1.07	0.75	0.43	0.22
4.5	0.92	0.85	0.60	0.35	0.18
5	0.75	0.69	0.49	0.29	0.15
5.5	0.62	0.57	0.41	0.24	0.12
6	0.52	0.48	0.34	0.21	0.10
6.5	0.44	0.40	0.29	0.18	0.09
7	0.37	0.35	0.25	0.15	0.08
7.5	0.32	0.30	0.22	0.13	0.07
8	0.28	0.26	0.19	0.12	0.06
8.5	0.25	0.23	0.17	0.10	0.05
9	0.22	0.20	0.15	0.09	0.05
9.5	0.19	0.18	0.13	0.08	0.04
10	0.17	0.16	0.12	0.08	0.04
10.5	0.15	0.15	0.11	0.07	0.03
11	0.14	0.13	0.10	0.06	0.03
11.5	0.13	0.12	0.09	0.06	0.03
12	0.12	0.11	0.08	0.05	0.03

(a) Coalescence timescales, in Gyr, for the galaxy models evolved to different redshifts z and for different initial binary eccentricities e_0 . Each entry gives the time elapsed from the beginning of the simulation, $t_0 = 0$ s, to binary coalescence.

$z \backslash e_0$	0.1	0.3	0.5	0.7	0.9
0	72.1	63.3	43.4	28.8	21.6
0.5	30.3	27.5	20.4	14.7	11.8
1	16.7	15.4	12.0	9.11	7.52
1.5	10.5	9.87	7.95	6.27	5.29
2	7.31	6.90	5.70	4.61	3.96
2.5	5.38	5.12	4.31	3.57	3.10
3	4.15	3.97	3.39	2.85	2.50
3.5	3.30	3.17	2.75	2.35	2.08
4	2.70	2.61	2.29	1.97	1.76
4.5	2.26	2.18	1.93	1.68	1.51
5	1.92	1.86	1.66	1.46	1.32
5.5	1.66	1.61	1.45	1.28	1.16
6	1.45	1.41	1.27	1.14	1.03
6.5	1.27	1.24	1.13	1.01	0.93
7	1.13	1.11	1.01	0.91	0.84
7.5	1.02	1.00	0.91	0.83	0.76
8	0.92	0.90	0.83	0.76	0.70
8.5	0.83	0.82	0.76	0.69	0.64
9	0.76	0.75	0.69	0.64	0.59
9.5	0.70	0.69	0.64	0.59	0.55
10	0.64	0.63	0.59	0.55	0.51
10.5	0.60	0.59	0.55	0.51	0.48
11	0.55	0.55	0.51	0.48	0.45
11.5	0.52	0.51	0.48	0.45	0.42
12	0.48	0.48	0.45	0.42	0.39

(b) Cosmic coalescence times, in Gyr, for the galaxy models evolved to different redshifts z and for different initial binary eccentricities e_0 . Each entry gives the cosmic time at which the binary reaches coalescence, measured from the initial time $t_0 = t(z)$.

Table 2.8: Coalescence times of the IMBH binaries for the 25 galaxy models as a function of redshift z and initial binary eccentricity e_0 . Panel (a) reports the coalescence timescale measured from $t_0 = 0$ s, while panel (b) gives the corresponding cosmic time of coalescence.

In this chapter, we have investigated the self-consistent hybrid model developed by Quinlan (1996) and Sesana, Haardt, et al. (2006) to describe the dynamical evolution of massive black hole binaries within a fixed stellar background. Taking advantage of the versatility of this framework, we have performed a suite of numerical experiments aimed at identifying the main features of IMBH binary evolutionary tracks and systematically exploring the relevant parameter space. The collection of experiments presented in § 2.2-2.3-2.4, together with the analysis of the corresponding evolutionary tracks $(t, a(t))$ and $(t, e(t))$, allows us to draw the following main conclusions:

- Stellar environments characterised by enhanced central densities, such as nucleated galaxies, promote more efficient dynamical hardening of IMBH binaries, leading to a faster orbital decay.

- Eccentric IMBH binaries generally undergo further eccentricity growth during the stellar-scattering hardening phase. As a consequence, they can enter the GW-dominated regime at bigger eccentricity and reach coalescence on shorter timescales.
- Initially circular binaries tend to preserve their low eccentricity throughout both stellar-driven and GW-driven regimes.
- In sufficiently low-density stellar environments, the stellar-driven hardening can be too inefficient to bring the binary into the GW-dominated regime within a Hubble time t_H , potentially preventing its coalescence within the present time and detection by LISA or ET.
- Present-day, non-nucleated dwarf galaxies characterised by low central densities may have been sufficiently compact at higher redshifts, $z > 0$, to enable an IMBH binary hosted by them to coalesce within the present-day universe.

The results obtained in this chapter motivate the next step of our investigation, namely the transition from the semi-analytical hybrid description to direct N -body simulations. This approach will allow us to follow the evolution of the IMBH binary and its surrounding stellar system self-consistently, overcoming some of the main limitations inherent to the adopted hybrid model. In particular, direct N -body simulations will make it possible to follow the mutual dynamical evolution of the binary and the stellar background, rather than prescribing the latter as a fixed external potential, and will open the possibility of investigating the dynamical evolution and eventual merger of the stellar systems themselves.

3

IMBH BINARY EVOLUTION WITH KETJU N-BODY SIMULATIONS

The numerical experiments based on the Quinlan-Sesana hybrid model, presented in Chapter 2, have shown that, once an IMBH binary forms, its coalescence can occur within a Hubble time even in non-nucleated, high-redshift galaxies.

These results motivate a further investigation through self-consistent, direct N -body simulations, which allow the dynamical evolution of the stellar system and the black holes to be followed from the stages preceding binary formation through the subsequent evolution of the binary. This approach also makes it possible to relax some of the simplifying assumptions adopted in the hybrid model, in particular the assumption of a fully populated loss cone and the evolution within a fixed, spherically symmetric stellar background. In this context, KETJU provides a particularly suitable framework, as the interactions between the black holes and surrounding stars are resolved without gravitational softening. The chapter is structured as follows. Section 1 introduces the equilibrium theory of N -body systems and the procedure used to construct equilibrium initial conditions with OpOpGadget. Section 2 presents the results of high-resolution, direct N -body simulations of mergers between two identical systems, each consisting of an NGC 205-like stellar bulge hosting an IMBH. The final section explores a suite of numerical experiments designed to assess how the dynamical path toward binary coalescence is affected by the initial angular momentum of the merging systems.

3.1 Dynamical Equilibrium of Stellar Systems

This Section, the first of the final chapter, has two main purposes. First, it reviews the equilibrium theory of spherically symmetric, isotropic, collisionless stellar systems. Second, it introduces the main features of OpOpGadget, the numerical code used to generate the N -body initial conditions adopted in this thesis.

The Section is therefore divided into two Subsections. The former summarizes the fundamental equations governing collisionless stellar systems, including the collisionless Boltzmann equation, the Jeans equations, and the Eddington inversion formula. The latter focuses on the numerical aspects of OpOpGadget, describing the algorithm used by the code to generate N -body initial conditions and the analysis tools it provides.

3.1.1 Equilibrium Theory of Stellar Systems

The first step towards performing N -body experiments of merging stellar systems hosting IMBHs is the generation of reliable N -body initial conditions that capture the main properties of the system in isolation, when at equilibrium, and reproduce its relevant observables, such as the density profile and stellar velocity dispersion, consistently with those of the corresponding idealized model. This, in turn, requires a well-defined framework for describing the dynamical state of the system.

As discussed in § 1.2, provided that the stellar system is considered on timescales shorter than its two-body relaxation time t_{2b} , collisional effects are negligible, and the stars can be treated as collisionless particles moving in a smooth gravitational potential generated by a continuous mass density distribution ρ , smooth proxy of the actual stellar distribution. In this regime, when dealing with stars that are all identical to one another, the dynamical state of the stellar component can be described by a unique one-particle distribution function (DF) $f(\mathbf{x}, \mathbf{v}, t)$ of phase space (PhS), while the corresponding gravitational potential $\Phi(\mathbf{x})$ is self-consistently determined by the mass distribution $\rho(\mathbf{x})$ through the Poisson equation

$$\nabla^2 \Phi(\mathbf{x}) = 4\pi G \rho(\mathbf{x}). \quad (3.1)$$

For collisionless systems of this type, the corresponding DFs satisfy the so-called collisionless Boltzmann equation (CBE), which follows from the normalization of the DF and, consequently, from the conservation of the integrated probability, as it will be proved (Boltzmann 1995).

Building on the discussion in § 1.2, we begin by introducing the normalized DF in phase space. If we indicate with $\mathbf{w}$ an arbitrary set of canonical PhS coordinates, the DF of a system with total mass M can be set such that

$$\int_{\mathbb{R}^6} d\mathbf{w} f(\mathbf{w}) = M. \quad (3.2)$$

The CBE follows from requiring the DF to be conserved along the trajectories generated by the potential $\Phi(\mathbf{x}, t)$ associated with the density distribution $\rho(\mathbf{x}, t)$. Taking the total derivative with respect to the time of Equation (3.2) yields

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \dot{\mathbf{w}} \cdot \frac{\partial f}{\partial \mathbf{w}} = \frac{\partial f}{\partial t} + [f, H] = 0, \quad (3.3)$$

where D/Dt has been used to denote the Lagrangian derivative, the square bracket indicates the Poisson bracket

$$[f, H] = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right), \quad (3.4)$$

and H is the Hamiltonian function of the system (see, e.g. Binney and Tremaine 2011).

The arbitrariness of the canonical coordinates $\mathbf{w}$ enables us to specialize Equation (1.23) to the inertial Cartesian coordinates $(\mathbf{x}, \mathbf{v})$. With this choice, the Hamiltonian per unit mass becomes

$$H(\mathbf{x}, \mathbf{v}, t) = \frac{1}{2}v^2 + \Phi(\mathbf{x}, t) \quad (3.5)$$

and the Equation (3.3) turns into

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} - \frac{\partial \Phi}{\partial \mathbf{x}} \cdot \frac{\partial f}{\partial \mathbf{v}} = 0. \quad (3.6)$$

Equation (3.6) gives the CBE for a collisionless system in Cartesian coordinates, valid on timescales shorter than the corresponding t_{2b} and expresses the conservation of the phase-space probability density along the orbits generated by the potential $\Phi(\mathbf{x}, t)$. However, it is too general for our purposes, as we are interested in studying the equilibrium of stellar systems, and in particular of spherically symmetric equilibrium models. It is therefore useful to introduce the following definition: a system composed of N particles is said to be in equilibrium, or stationary, when any quantity $q(\mathbf{x}, t)$ describing the system, including its distribution function (DF), is time-independent, such that $\partial q / \partial t = 0$.

Under this condition, the DF is time-independent and satisfies the so-called stationary CBE, which is obtained from Equation (3.3) by setting $\partial f / \partial t = 0$:

$$\frac{Df}{Dt} = \dot{\mathbf{w}} \cdot \frac{\partial f}{\partial \mathbf{w}} = [f, H] = 0. \quad (3.7)$$

The solutions to Equation (3.7) are referred to as integrals of motion (Ciotti 2021). Specifically, the function $I(\mathbf{x}, \mathbf{v})$ is an integral of motion for the potential $\Phi(\mathbf{x}, \mathbf{v})$ if it satisfies

$$\frac{D}{Dt}[I(\mathbf{x}(t), \mathbf{v}(t))] = 0 \quad (3.8)$$

along any orbit of the potential $\Phi(\mathbf{x})$ and a relevant example is represented by the Hamiltonian itself of the system, provided

$$[H, H] = 0. \quad (3.9)$$

The comparison between Equation (3.7) and (3.8) reveals that solving the steady-state CBE is equivalent to being an integral of motion; this leads to the formulation of the Jeans theorem (see, e.g., Lynden-Bell 1962; Binney and Tremaine 2011).

Jeans Theorem. – *Any steady-state solution of the collisionless Boltzmann equation depends on the phase-space coordinates only through integrals of motion in the given potential, and any function of the integrals yields a steady-state solution of the collisionless Boltzmann equation.*

It follows from this theorem that, if $I_1, \dots, I_N$ is a set of N integrals of motion for a given collisionless and stationary system, its DF can be expressed as $f = f(I_1, \dots, I_N)$, rather than as a general function of phase space. Nevertheless, before examining the conditions under which a closed-form solution to Equation (3.7) can be obtained, let us introduce the main moments of the DF, which are directly related to kinematic observables and will prove useful for this purpose.

The first- and second-order velocity moments of the distribution function f , the mean velocity and the velocity dispersion tensor, respectively, are defined as

$$\overline{v_i}(\mathbf{x}, t) \equiv \frac{1}{\rho(\mathbf{x}, t)} \int_{\mathbb{R}^6} d^3\mathbf{v} f v_i \quad (3.10)$$

and

$$\sigma_{ij}^2(\mathbf{x}, t) \equiv \frac{1}{\rho(\mathbf{x}, t)} \int_{\mathbb{R}^6} d^3\mathbf{v} f(v_i - \bar{v}_i)(v_j - \bar{v}_j) = \bar{v}_i \bar{v}_j - \bar{v}_i \bar{v}_j. \quad (3.11)$$

These quantities enter the so-called Jeans equations, which follow from taking the first- and second-order velocity moments of Equation (3.3). In Cartesian coordinates they are

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho \bar{v}_i}{\partial x_i} = 0 \quad (3.12)$$

and

$$\frac{\partial \rho \bar{v}_i}{\partial t} + \frac{\partial \rho \bar{v}_i \bar{v}_j}{\partial x_j} = -\rho \frac{\partial \Phi}{\partial x_i} \quad (3.13)$$

respectively, where the Einstein summation convention over repeated indices is adopted.

Equations (3.12) and (3.13) generalize the mass continuity and momentum equations commonly used in fluid dynamics to collisionless stellar systems. For a spherically symmetric system, Equation (3.13) can be written in spherical coordinates (r, θ, ϕ) as

$$\frac{d(\rho \sigma_r^2)}{dr} + 2\frac{\rho}{r} \beta \sigma_r^2 = -\rho \frac{d\Phi}{dr}, \quad (3.14)$$

where $\beta = 1 - \sigma_t^2/(2\sigma_r^2)$ quantifies the degree of velocity anisotropy, with positive and negative values corresponding to radial and tangential anisotropy, respectively, while $\sigma_t^2 = 2\sigma_\theta^2 = 2\sigma_\phi^2 \equiv \sigma^2$ (see, e.g., W. C. Saslaw 1985, Binney and Tremaine 2011).

A stellar system is defined to be isotropic if $\sigma_r^2 = \sigma_\theta^2 = \sigma_\phi^2$ and thus $\beta = 0$. It is now clear from Equation (3.14) that, for a spherically symmetric and isotropic system, with $\rho = \rho(r)$ and $d\Phi/dr = GM(< r)/r^2$, the 1D velocity dispersion formula, mentioned in § 2.3, is given by Equation (2.51), which we reproduce here for convenience:

$$\sigma^2(r) = \frac{1}{\rho(r)} \int_r^\infty ds \frac{GM(< s)}{s^2} \rho(s). \quad (3.15)$$

We can now turn to the analysis of Equation (3.7) and its solutions. To this end, we recall the main properties of an important subclass of equilibrium models, namely ergodic systems, which are the type of systems we aim to initialize in our numerical experiments.

By definition, a system with Hamiltonian H is described by an ergodic DF if the latter depends solely on the Hamiltonian, i.e. $f = f(H)$. It is worth noting that, according to Equation (3.9), all stationary and collisionless systems admit H as an integral of motion. Therefore, the Jeans theorem implies that the DF can, for these systems, be written as $f = f(H, I_2, \dots, I_N)$, with the ergodic case representing a particular and highly restrictive subclass. A key property of ergodic systems follows from the first-order moments of the corresponding DF. In an inertial Cartesian frame.

In the inertial Cartesian frame where stars move under the action of a steady-state potential $\Phi(\mathbf{x})$, the Hamiltonian of the system is an even function of $\mathbf{v}$, causing $\bar{v}_x(\mathbf{x}) = \bar{v}_y(\mathbf{x}) = \bar{v}_z(\mathbf{x})$ to vanish, owing to Equation (3.10). Similarly, the velocity-dispersion tensor can be shown to be isotropic, since an ergodic DF depends on the velocity only through v^2 , that is

$$\sigma_{ij}^2(\mathbf{x}) = \sigma^2(\mathbf{x}) = \delta_{ij} \frac{4\pi}{3\rho(\mathbf{x})} \int_0^{+\infty} dv f\left(\frac{1}{2}v^2 + \Phi(\mathbf{x})\right) v^4. \quad (3.16)$$

Therefore, any system described by an ergodic DF is necessarily isotropic.

So far, we have focused on several theoretical aspects of N -body systems, with particular emphasis on their equilibrium properties, without yet discussing in detail how the DF is made. However, generating initial conditions (ICs) for equilibrium N -body models requires sampling

the DF directly, rather than relying solely on its moments. For globally isotropic stellar systems, the DF can be recovered from the density profile ρ through Eddington's inversion formula (Eddington 1916), which provides the corresponding phase-space distribution in terms of the density and gravitational potential. It reads

$$f(\varepsilon) = \frac{1}{\sqrt{8\pi^2}} \frac{d}{d\varepsilon} \int_{\varepsilon_t}^{\varepsilon} \frac{d\rho}{d\Psi} \frac{d\Psi}{\sqrt{\varepsilon - \Psi}}, \quad (3.17)$$

with $\varepsilon = -H$ and $\Psi = -\Phi$ being the relative energy and potential, respectively, and ε_t the truncation energy of the system.

The theoretical framework developed above allows us to turn a prescribed equilibrium density profile $\rho(r)$ into a self-consistent set of N -body initial conditions. In the following Subsection, we describe how this procedure is implemented for spherically symmetric, isotropic stellar systems, focusing in particular on the construction and sampling of the corresponding DF.

3.1.2 Generation of N-Body ICs with OpOpGadget

Having established the theoretical basis for describing dynamical equilibrium in stellar systems, we can now turn to the construction of N -body initial conditions. The goal is to generate a discrete stellar realization that consistently reproduces the prescribed total mass, density profile, and associated structural parameters, while satisfying the equilibrium conditions discussed above. For a given stellar distribution, stellar positions and velocities are generated by means of **OpOpGadget** (see Iorio et al. 2019). This numerical code provides a range of tools for generating N -body initial conditions and analyzing the outputs of N -body simulations. It also includes an extensive library of analytical models for spherically symmetric stellar systems, allowing canonical stellar profiles to be constructed from their specified structural parameters.

The first step in the workflow initialization is to specify the desired model, assign the values of the parameters required by the library model, and choose the number of particles to represent the stellar environment. After that, the code proceeds with assigning a set of spherical coordinates (r, θ, ϕ) to each particle by randomly picking the cumulative mass profile

$$M_*(< r) = 4\pi \int_0^r dy y^2 \rho(y) \quad (3.18)$$

using the variable $u = np.random.uniform(M_{min} = 0, M_{max} = M, N)$.

Since $M(< r)$ is a monotonically increasing function of radius, it can be inverted to obtain the corresponding radii, $r = M_*^{-1}(u)$, for the N sampled values of u . Yet, two additional independent random variables, uniformly distributed in the intervals $[-1, 1]$ and $[0, 2\pi]$ respectively, are used to sample the direction cosine and the azimuthal angle, thereby obtaining N sets of spherical coordinates (r, θ, ϕ) , which are then transformed into Cartesian coordinates.

While particle positions are drawn from the cumulative mass distribution, the velocity components are obtained by sampling the DF, assuming the gravitational potential Φ to be generated by the mass distribution associated with the previously initialized particles. Therefore, three random numbers $\mathbf{v}$, lying in the range $[-1, 1]$, allow to temporarily assign a normalized velocity to each particle, which will found to have the binding energy $\varepsilon = \Psi(1 - v^2)$. Then, the Von Neumann rejection method is implemented and the normalized velocity is accepted if

$$g \leq \frac{f(\varepsilon)}{f(\varepsilon = \Psi)}, \quad (3.19)$$

where g is another independent random number in $[0, 1]$. If the inequality in Equation (3.19) is satisfied, the final velocity vector is obtained by multiplying the normalized vector by the

factor $\sqrt{2\Psi}$. Once this procedure, which can be extended to multi-component stellar systems, is completed, each particle will have its own ID , mass $m = M_*/N$, position $\mathbf{x}$ and velocity $\mathbf{v}$ (see, e.g., Aarseth 2003).

All numerical experiments of merging between identical systems of stellar background and IMBH that will be presented in the next Sections, involve Sersic stellar distribution as described by Prugniel and Simien (1997). However, the Sersic analytic model implemented within OpOpGadget library takes as a reference Lima Neto et al. (1999) de-projection

$$\rho(r) = \rho_0 \left(\frac{r}{a}\right)^{-p} e^{-(r/a)^\nu}, \quad (3.20)$$

where $\nu = n^{-1}$ denotes the inverse of the Sérsic index, while a is the Sérsic scale length, defined such that $\Sigma(a) = \Sigma_0$. These quantities are referred to as the primary parameters and the remaining structural parameters can be derived from them, so that the same density distribution may be equivalently described by different sets of parameters.

This discrepancy provides a useful opportunity to illustrate both how the parameters are consistently converted and, more importantly, how the numerical code constructs and analyses a multi-component, spherically symmetric stellar system in equilibrium. Therefore, here, we analyse the generated N -body initial conditions for the same model considered in § 2.3, consisting of the stellar bulge and nuclear star cluster (NSC) of NGC 205, together with the central IMBH binary.

A consistent conversion between the primary parameters and the corresponding derived quantities is therefore required before generating N -body ICs. In particular, to convert from the scale length a to the effective radius R_e , we adopt the following analytical approximation (Lima Neto et al. 1999):

$$\ln R_e/a = \frac{0.6950 - \ln \nu}{\nu} - 0.1789. \quad (3.21)$$

Particular attention should be paid to the fact that the stellar particles to be initialized are required to be in dynamical equilibrium within the combined gravitational potential of both stars and central black hole. The presence of such a massive and compact object is expected to modify the equilibrium configuration of the stellar system compared to the isolated case. Although the IMBH is represented by a single object and would therefore generate an exactly Keplerian potential, we model it as a Plummer sphere (see, e.g., Plummer 1911) to avoid fixing its position. According to this, the density, the cumulative mass profile and the gravitational potential for a BH of mass M are

$$\rho(r) = \frac{3M}{4\pi a^3} \left(1 + \frac{r^2}{a^2}\right)^{-5/2}, \quad (3.22)$$

$$M(< r) = M \frac{r^3}{(r^2 + a^2)^{3/2}}; \quad (3.23)$$

and

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + a^2}}, \quad (3.24)$$

respectively, with a denoting the Plummer scalelength, linked to the half-mass radius $r_{1/2}$ of the distribution by

$$a \simeq 0.766 r_{1/2}. \quad (3.25)$$

The gravitational potential in Equation (3.24) approaches the Keplerian form for $r \gg a$, making the Plummer model a convenient representation of BH-like particles.

Once the structural parameters of the NGC 205 stellar bulge and NSC, namely (R_e, n, M_*) , and

the numbers of particles used to represent them, N_{bulge} and N_{NSC} , are specified, the model is naturally characterised by a gravitational softening length, or softening scale, ϵ . This parameter sets the intrinsic spatial resolution of the N -body realisation, below which the system cannot be reliably resolved. Since this quantity can only be computed by OpOpGadget after generating the model ICs, the Plummer scale radius a is set to the mean interparticle separation at the half-mass radius, R_h ,

$$\lambda = R_h \left(\frac{2}{N_h} \right)^{1/3}, \quad (3.26)$$

with N_h denoting the number of stars within R_h , which provides an estimate of the system gravitational softening length prior to generating its ICs.

OpOpGadget computes ϵ using the spline kernel of Gadget2 (Springel 2005), which is the same adopted in our suite of N -body simulations with KETJU (Mannerkoski et al. 2023). The softening procedure replaces the mass density of a point-like particle, $\rho(r) = m\delta(r)$, with a normalized spline kernel

$$W(r, h) = \frac{8}{\pi h^3} \begin{cases} 1 - 6q^2 + 6q^3, & 0 \leq q \leq \frac{1}{2}, \\ 2(1 - q)^3, & \frac{1}{2} < q \leq 1, \\ 0, & q > 1, \end{cases} \quad (3.27)$$

with $q \equiv r/h$ and $h = 2.8\epsilon$ setting the spline compact support, beyond which the gravitational potential reverts to the standard Newtonian form.

The positions, velocities, masses and IDs of each particle are stored within the object ***Nbody-Model***:, which can be then passed as argument of ***Analysis()***:, provided of the following useful methods:

1. ***Analysis.com()***: it returns Cartesian position and velocity of the center of mass for the generated N -body ICs.
2. ***Analysis.softening_scale(kernel=Gadget)***: it computes the system's softening length ϵ , which is used in Equation (3.27) and throughout KETJU N -body simulations employing softened gravity.
3. ***Analysis.tdyn(mq=50)***: It provides the dynamical time of the system as

$$T_{dyn} = 0.5\pi \sqrt{\frac{R_h^3}{GM_h}}, \quad (3.28)$$

with M_h being the mass inside R_h . This estimate has been adjusted to match the definition of the dynamical time at half mass radius adopted here, which is

$$t_{dyn} = \frac{1}{\sqrt{G\bar{\rho}}}, \quad (3.29)$$

with $\bar{\rho}$ indicating the average mass density within R_h .

Lastly, the ***NbodyModel*** object can be passed to ***Profile(NbodyModel, xmin, xmax)*** to extract the radial profiles of relevant quantities over the range $[x_{\min}, x_{\max}]$. This object provides two methods for characterising the spatial distribution of the generated N -body realisation:

1. ***Profile.dens()***: returns the radial grid and the corresponding density profile.
2. ***Profile.mass()***: returns the radial grid and the corresponding cumulative mass profile.

As a reference case and standard procedure adopted throughout the numerical experiments presented in the following sections, we now describe the generation and analysis of the N -body ICs for the NGC 205 stellar bulge, NSC, and IMBH model introduced in § 2.3.

First, the NGC 205 observational data reported in Table 2.4 are used to derive the Sersic scalelengths of the stellar bulge and NSC according to Equation (3.21). This yields $a_{bulge} = 1456 pc$ and $a_{NSC} = 0.24 pc$, respectively. Following the choice adopted by Khan and Holley-Bockelmann (2021), we then set the number of particles used to initialize the stellar bulge and NSC, N_{bulge} and N_{NSC} , such that the BH-to-stellar mass ratio is equal to 160. For an IMBH with mass $4 \times 10^4 M_\odot$, this results in $N_{bulge} = 3888000$ and $N_{NSC} = 7200$ generated stellar particles. These values are then used to estimate the mean interparticle separation, leading to $\lambda = 5.5 pc$, which in turn determines the Plummer sphere softening length. Therefore, once the masses, Sersic scalelengths, and Sersic indices of the two stellar components are provided to the OpOpGadget Sersic models, together with the IMBH mass and softening length for the Plummer sphere model and the corresponding number of particles for each component, the initial conditions are generated by assessing positions, velocities, masses, and particle IDs to all particles.

Figure 3.1 shows the spatial distribution of the stellar particles and the IMBH projected onto the x - y plane, on both the scale of the stellar bulge and that of the NSC.

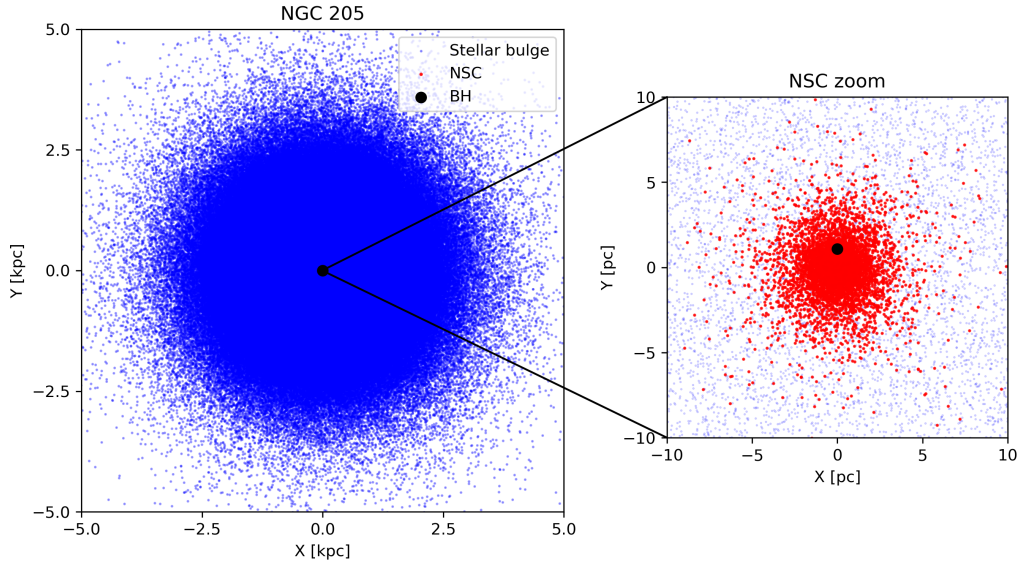


Figure 3.1: Positions of the N -body initial conditions for the NGC 205 stellar bulge, NSC, and hosted IMBH, projected onto the x - y plane. Blue and red dots represent bulge and NSC stars, respectively, while the black dot marks the IMBH. The main panel shows the system on the scale of the stellar bulge, whereas the inset provides a zoomed-in view of the NSC and IMBH.

Another way of visualizing the generated N -body ICs is provided in Figure 3.2, which shows the three Cartesian components of the position and velocity of the particles in six histograms. As expected, all distributions are centred around $0 kpc$ and $0 km/s$, respectively. While the bulge and NSC stars exhibit similar velocity distributions, their spatial distributions differ markedly, with a pronounced central peak arising from the compactness of the NSC relative to the stellar bulge.

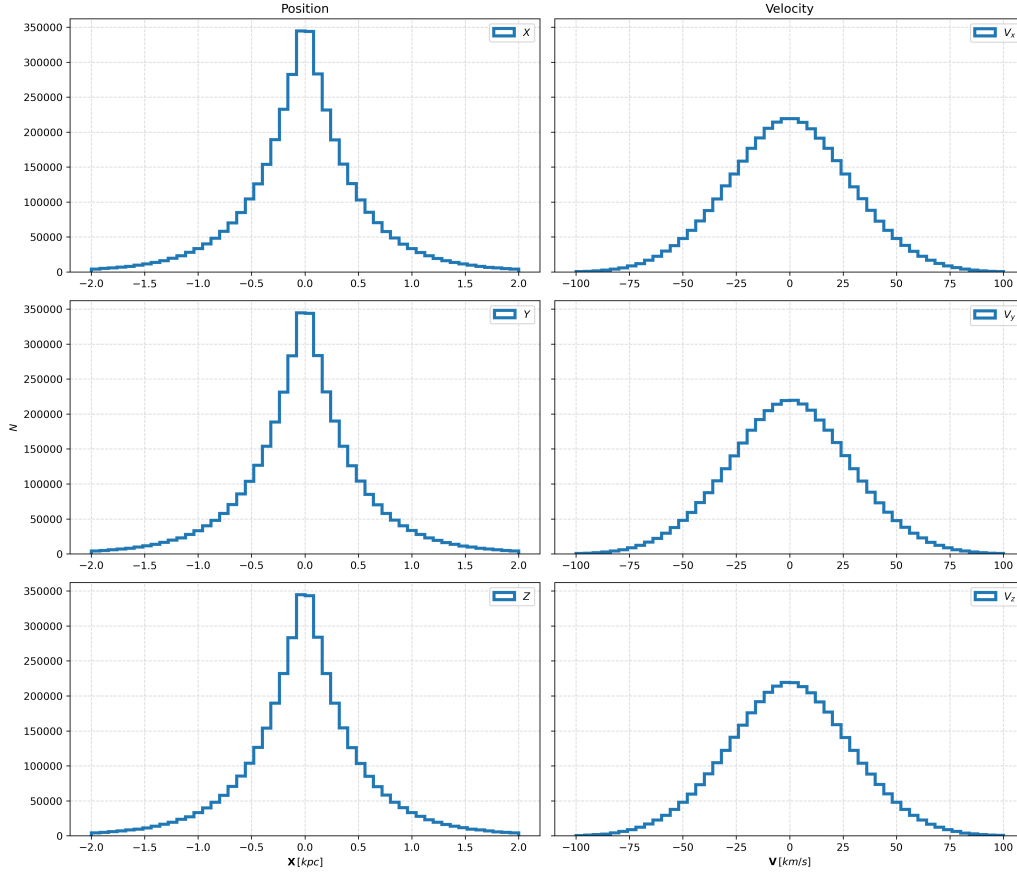
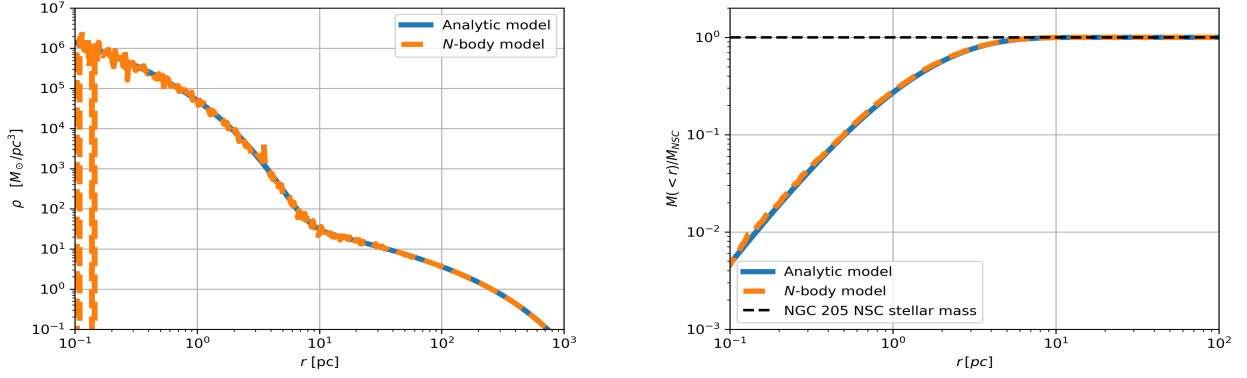


Figure 3.2: Histograms of the position and velocity distributions of the NGC 205, N -body generated, stellar components (bulge and NSC). Each row shows, on the left, the distribution of the i -th Cartesian component of the stellar positions and, on the right, the corresponding distribution of the i -th component of the stellar velocities. The position histograms span $[-2, 2] \text{ kpc}$ and are divided into 50 bins, each with a width of $\Delta x_i = 80 \text{ pc}$, while the velocity histograms span $[-100, 100] \text{ km/s}$ and are divided into 50 bins, each with a width of $\Delta v_i = 4 \text{ km/s}$.

The 0-th order moments of these six distributions, namely the system centre-of-mass position x_{CM} and velocity v_{CM} , are obtained using the `OpOpGadget Analysis.com()` method, while the corresponding first-order moments, namely the position and velocity dispersions, are computed through Python coding. The analysis yields $x_{CM} = [0.34 \pm 626, -0.37 \pm 627, 0.10 \pm 627] \text{ pc}$ and $v_{CM} = [-0.0086 \pm 28.9281, -0.0065 \pm 28.9258, 0.0243 \pm 28.8984] \text{ km/s}$, compatible with a spherically symmetric and isotropic stellar distribution whose centre of mass is located at the origin of the distribution's inertial frame S_{CM} .

Lastly, applying the `Profile.dens()` and `Profile.mass()` methods to the generated ICs provides the N -body representations of the stellar density and cumulative mass profiles of NGC 205. These profiles are shown in panels (a) and (b) of Figure 3.3, respectively, together with the corresponding analytical models.

Although the analytic and N -body models show good agreement, the finite number of stars used to construct the N -body ICs introduces numerical noise into the reconstructed distributions, in particular at small radii. This effect is quite evident in the mass density profile, whereas the cumulative mass profile is less affected.



(a) Comparison between the analytic stellar mass density profile of NGC 205 (blue dashed line), also shown in Figure 2.7a, and the corresponding profile reconstructed from the generated N -body initial conditions (orange dashed line) using `Profile.dens()`.

(b) Comparison between the analytic stellar cumulative mass profile of NGC 205 (blue dashed line), also shown in Figure 2.7b, and the corresponding profile reconstructed from the generated N -body initial conditions (orange dashed line) using `Profile.mass()`.

Figure 3.3: Comparison between the analytic and reconstructed stellar density (left) and cumulative mass (right) profiles of NGC 205 from the N -body ICs generated by OpOpGadget.

Before examining the results of our suite of numerical experiments, we conclude this Section with a Subsection devoted to reviewing the main features of the numerical code KETJU, employed to perform the simulations, and to assessing whether the initially equilibrium models remain in equilibrium when evolved with KETJU over several dynamical timescales, t_{dyn} .

3.1.3 Dynamical Evolution with KETJU

As anticipated, the numerical code employed in this thesis to simulate galaxy mergers and, consequently, the dynamical evolution of IMBH binaries is KETJU. Originally developed as an extension of GADGET-3, here we adopt KETJU publicly available version implemented in the GADGET-4 code (see, e.g., Springel et al. 2021; Mannerkoski et al. 2023). Before assessing the stability of the equilibrium N -body initial conditions when evolved with KETJU, we briefly review the main features of GADGET-4 code and KETJU module, and describe the numerical setup adopted for our simulations.

GADGET-4 is a highly parallelized N -body and hydrodynamical code that provides two methods for computing gravitational forces: the one-sided tree-based multipole expansion and the Fast Multipole Method (FMM). Unlike the tree-based approach, the FMM computes gravitational interactions symmetrically, thereby ensuring momentum conservation when combined with a suitable time-integration scheme. Besides, GADGET-4 provides two time-integration schemes: the leapfrog integrator and a hierarchical time-integration scheme based on a hierarchy of timebins. The former assigns to each particle with acceleration $\mathbf{a}$ an adaptive time step (Springel 2005),

$$\Delta t = \sqrt{\frac{2\eta\epsilon}{|\mathbf{a}|}}, \quad (3.30)$$

where η is a user-specified accuracy parameter. The latter instead relies on a decomposition of the Hamiltonian into fast and slow components, optimizing the time integration by identifying subsystems that evolve slowly or rapidly and integrating each of them on the appropriate timescale.

The main improvement introduced by KETJU is the regularized MSTAR integrator (Rantala

et al. 2020). By employing algorithmic regularization, this integrator treats BH–star interactions without gravitational softening when a star enters the corresponding regularized region. This allows KETJU to accurately resolve close encounters and, in particular, the dynamical interactions between BHs and surrounding stars at small separations (see, e.g., Mikkola and Tanikawa 1999; Mikkola and Tanikawa 1999). The regularized region surrounding each BH-like particle is required to extend beyond the support radius, 2.8ϵ , of the spline softening kernel introduced in § 3.1.2. This ensures that BH–star interactions are exactly Newtonian when the star is outside the regularized region. In all simulations presented in the following Sections, the physical radius of the regularized region is set to 3ϵ .

When two regularized regions overlap, KETJU can additionally account for relativistic corrections by including Post-Newtonian (PN) terms up to 3.5PN order (see, e.g., Thorne and Hartle 1985; Blanchet 2014). This treatment allows the evolution of a BH binary to be followed from the hardening phase to its final coalescence, defined by default to $12R_s$, where R_s is the Schwarzschild radius corresponding to the total binary mass.

Finally, we summarize the main numerical setup adopted throughout the experiments presented in the following Sections. The simulations are controlled through two sets of options: compile-time options, specified in the *Configuration file*, and run-time options, specified in the *Parameter file*. The main options adopted in each case are reported below.

The main options enabled in the *Configuration file* are:

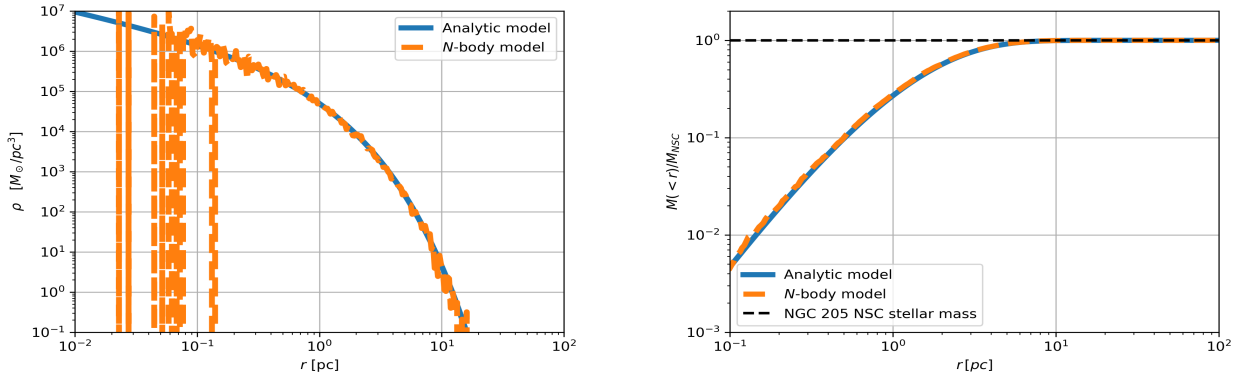
- *HIERARCHICAL_GRAVITY*: enables hierarchical time integration by splitting the Hamiltonian into fast and slow components. This scheme ensures momentum-conserving time integration.
- *SELFGRAVITY*: enables the computation of the self-gravity of the system when gravitational forces are evaluated.
- *FMM*: enables the Fast Multipole Method (FMM) for the computation of gravitational forces.
- *KETJU*: activates the KETJU functionality, including the MSTAR integration within the regularized regions.

The main options specified in the *Parameter file* are grouped in:

- *System of units*: the internal units are defined through `UnitLength_in_cm`, `UnitMass_in_g`, and `UnitVelocity_in_cm_per_s`, corresponding to $1\text{ pc}/h$, $1 M_\odot/h$, and 1 km s^{-1} , respectively. The gravitational constant is set to `GravityConstantInternal = 0`, so that G is expressed in internal units. With this choice, the corresponding internal time unit is approximately $9.8 \times 10^5\text{ yr}/h$.
- *Simulation time*: the beginning and end of the simulation are specified through `TimeBegin` and `TimeMax`, respectively, in code time unit.
- *KETJU regularization region*: the physical radius of the KETJU region is specified through `KetjuRegionPhysicalRadius`, set to three times the softening length of the corresponding particle class. `KetjuPNTerms` is set to 3.5.
- *Integration accuracy*: the relative tolerance of the KETJU integration is controlled by `KetjuIntegrationRelativeTolerance`, set to 10^{-7} . The relative tolerance used for the KETJU output is specified by `KetjuOutputTimeRelativeTolerance` and set to 10^{-6} .

Once the main KETJU features and the setup adopted for our numerical experiments have been described, we can assess whether an isolated N -body system generated in equilibrium remains in this configuration after several orbital periods of a typical star. Since the following sections will focus on high-resolution numerical experiments of merging stellar hosts, here we assess the stability of the system composed solely of the NGC 205 NSC and IMBH, as described in § 2.3. To this end, we let the system evolve for $50 \times t_{dyn}$ with the *KETJU* functionality enabled, and monitor the evolution of the stellar mass density and cumulative mass profiles, as well as the virial ratio Q , over this timescale.

Keeping the same BH-to-stellar mass ratio of 160 adopted in the previous Subsection, we initialize $N_{NSC} = 7200$ NSC stars, whose density and cumulative mass profiles are shown in Figure 3.4 and compared with their corresponding analytical models.



(a) Comparison between the analytic stellar mass density profile of NGC 205 NSC (blue solid line) and the corresponding profile reconstructed from the generated N -body initial conditions (orange dashed line) using `Profile.dens()`

(b) Comparison between the analytic cumulative mass profile of NGC 205 NSC (blue solid line) and the corresponding profile reconstructed from the generated N -body initial conditions (orange dashed line) using `Profile.mass()`

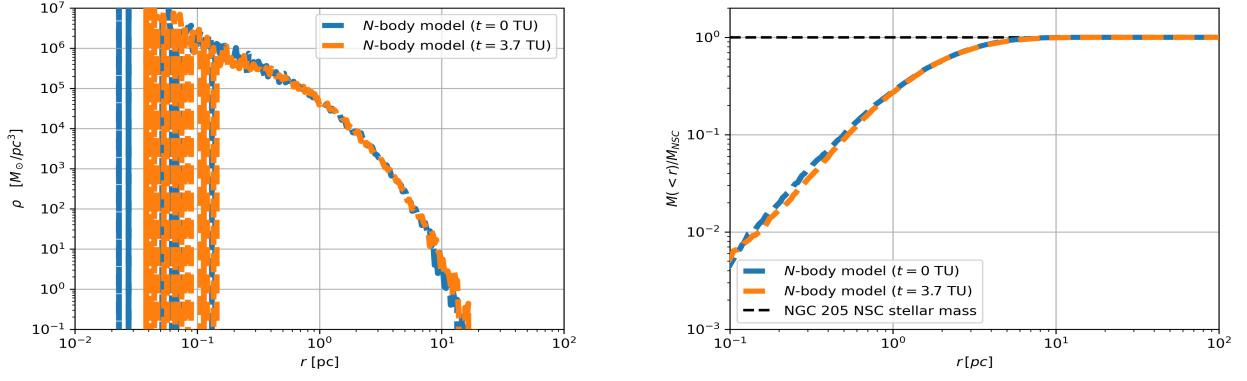
Figure 3.4: Comparison between the analytic and reconstructed stellar density (left) and cumulative mass (right) profiles of NGC 205 NSC from the N -body ICs generated by OpOpGadget.

As already discussed, although OpOpGadget constructs the system such that particle coordinates are defined in the inertial frame of the centre of mass, S_{CM} , a non-vanishing centre-of-mass velocity can arise from the discrete realization of the N -body system. This contribution must be accounted for when computing the virial ratio. In particular, while Equation (1.8) applies to the centre-of-mass frame, in an arbitrary inertial reference frame the virial ratio is more generally given by

$$Q = \frac{2(K - K_{CM})}{|W|}, \quad (3.31)$$

where K_{CM} is the center of mass kinetic energy. Therefore, at the beginning of the simulation, we find $Q \approx 1$, clearly indicating a dynamically equilibrated configuration, as widely discussed in § 1.1.

The dynamical time evaluated at the half-mass radius of the initialized system is found to be $t_{dyn} = aaa$. The system is therefore evolved over the interval $[0, \text{TimeMax}]$, with $\text{TimeMax} \approx 50 \times t_{dyn} = 3.7 TU$. Finally, analyzing the KETJU output allows us to reconstruct the same distributions shown in Figure 3.4, compare them with the corresponding initial profiles, and recompute the virial ratio. Applying the same procedure at the end of the simulation, we obtain $Q \approx 1$. The evolution of the system is further assessed in Figure 3.5, where the initial and final mass density and cumulative mass profiles are compared.



(a) Comparison between the reconstructed mass density profiles of the NGC 205 NSC *N*-body realization at the beginning (blue dashed line) and at the end (orange dashed line) of the simulation. (b) Comparison between the reconstructed cumulative mass profiles of the NGC 205 NSC *N*-body realization at the beginning (blue dashed line) and at the end (orange dashed line) of the simulation.

Figure 3.5: Comparison between the reconstructed mass density (left) and cumulative mass (right) profiles of the NGC 205 NSC *N*-body realization at the beginning and end of the simulation.

In conclusion, as demonstrated by Mannerkoski et al. (2023), activating the KETJU module preserves the ability of GADGET-4 to evolve initially at equilibrium systems without introducing spurious dynamical evolution of numerical origin. The analysis presented in this Section has allowed us to review the *N*-body theory, the main features of the numerical tools adopted in this thesis, establish the standard procedure used to generate and analyse the *N*-body initial conditions, and verify that the KETJU implementation is well suited to our purposes. In particular, the stability of the isolated system over several dynamical times provides a necessary validation of the high-resolution merger simulations, involving pairs of stellar host and IMBH, that will be presented in the following Sections.

3.2 Analysis of the Generated Initial Conditions

KETJU-UTILS

3.2.1 NCG 205 N-body Initial Conditions

COM data:

- position:
- velocity:

Numerically is found $r_{\text{half}} = \dots = R_{\text{eff}}$.

Yet, we get $t_{\text{dyn}} = \dots$, which agrees within $x\%$ the estimate obtained from the Analysis tool provided by OpOpgadget, which measures this quantity directly from the generated model. Waiting for evolution with Ketju.

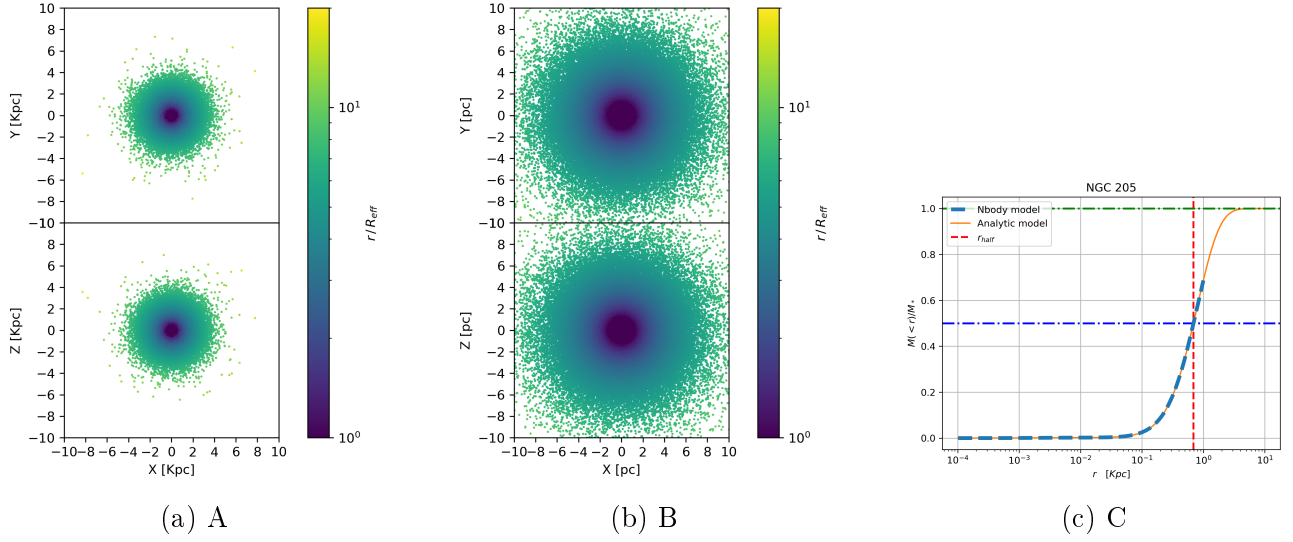


Figure 3.6: aaa

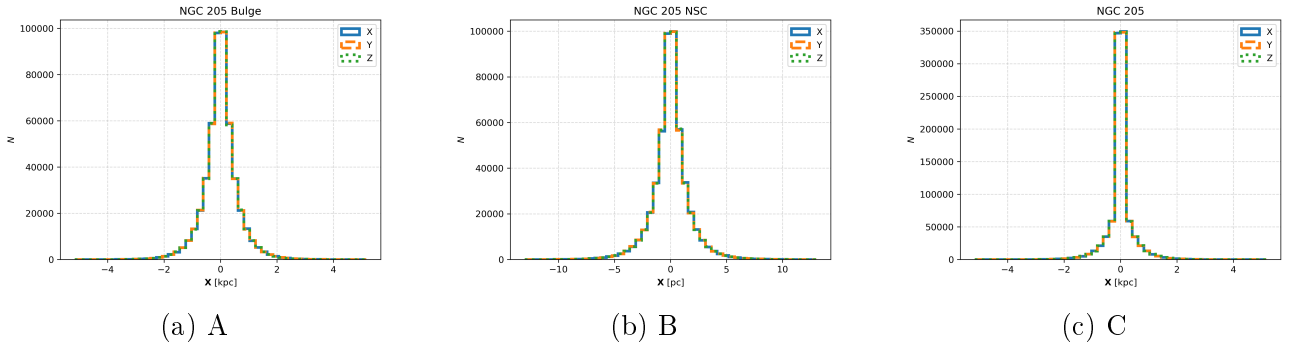


Figure 3.7: aaa

3.2.2 NCG 5206 N-body Initial Conditions

3.3 Ketju Numerical Simulations

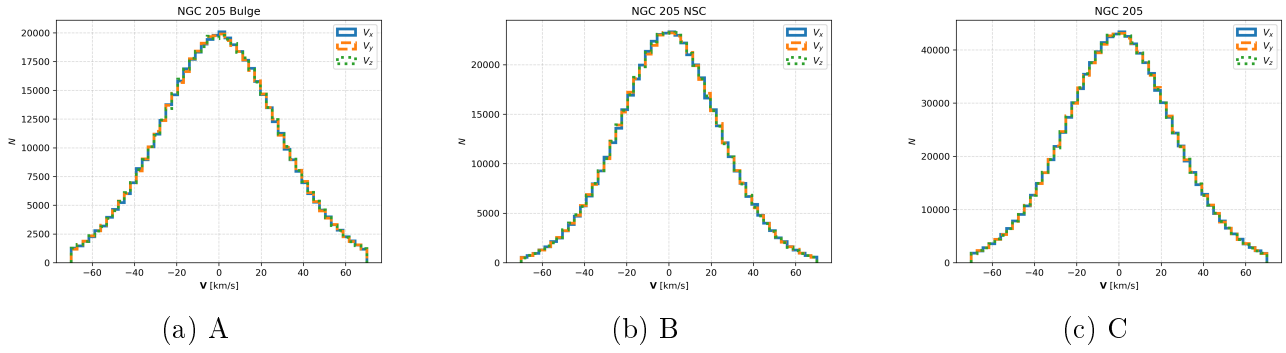


Figure 3.8: aaa

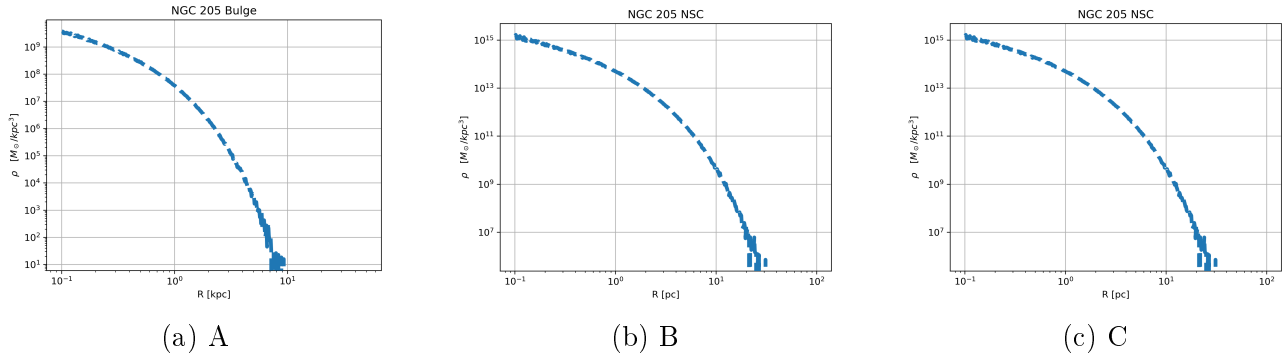


Figure 3.9: aaa

Things done:

- created double galaxy
- modified BHs position and velocity
- modified their position and velocity according to Nipoti paper (r_{90})
- built Hists to check position and velocity distribution
- verified dyn equilibrium
- computed the softening length in several ways
- computed dyn time
- ketju run
- BHs analysis with ketju-utils
- snapshot analysis with OpOpgadget
- method of shrinking spheres to get the density peak position

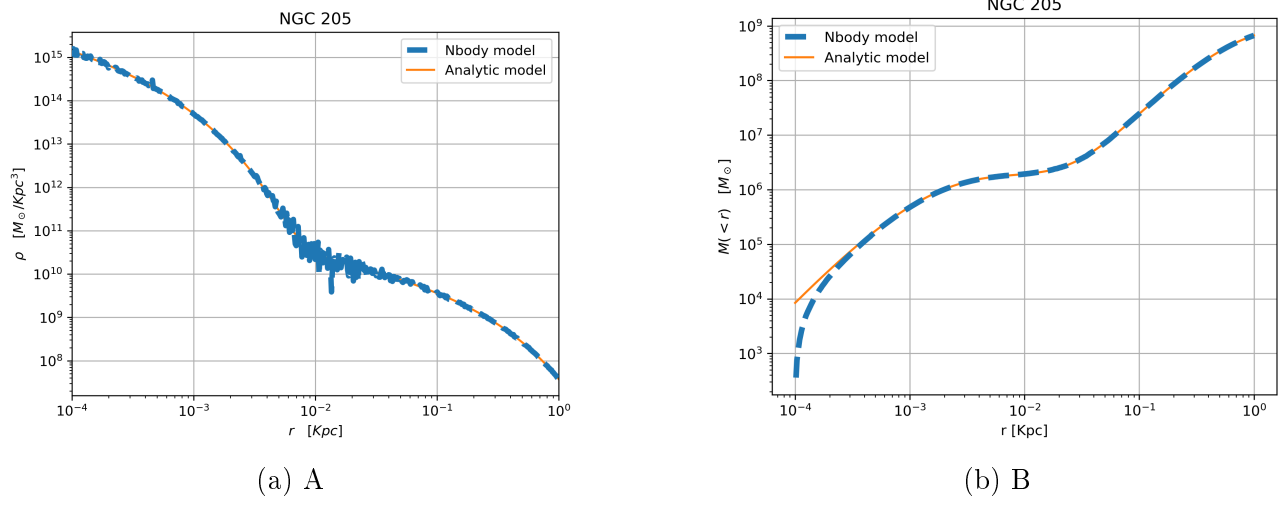


Figure 3.10: aaa

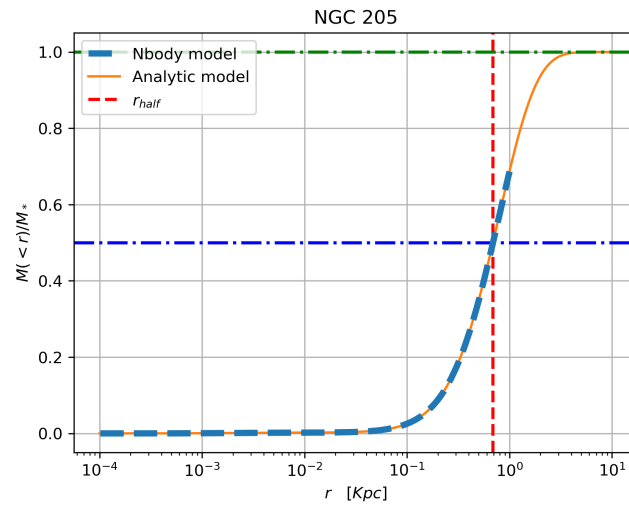


Figure 3.11: aaa

BIBLIOGRAPHY

- Aarseth, Sverre J (2003). *Gravitational N-body simulations*. Vol. 430.
- Abac, Adrian et al. (Mar. 2026). “The Science of the Einstein Telescope”. In: *JCAP* 2026.3, 081, p. 081. DOI: 10.1088/1475-7516/2026/03/081. arXiv: 2503.12263 [gr-qc].
- Abbott, B. P. et al. (Feb. 2016). “Observation of Gravitational Waves from a Binary Black Hole Merger”. In: *Phys. Rev. Lett.* 116 (6), p. 061102. DOI: 10.1103/PhysRevLett.116.061102. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.116.061102>.
- Amaro-Seoane, Pau, Jeff Andrews, et al. (Mar. 2023). “Astrophysics with the Laser Interferometer Space Antenna”. English. In: *Living Reviews in Relativity* 26.1. ISSN: 2367-3613. DOI: 10.1007/s41114-022-00041-y.
- Amaro-Seoane, Pau, Heather Audley, et al. (Feb. 2017). “Laser Interferometer Space Antenna”. In: *arXiv e-prints*, arXiv:1702.00786, arXiv:1702.00786. DOI: 10.48550/arXiv.1702.00786. arXiv: 1702.00786 [astro-ph.IM].
- Bañados, Eduardo et al. (Jan. 2018). “An 800-million-solar-mass black hole in a significantly neutral Universe at a redshift of 7.5”. In: *Nature* 553.7689, pp. 473–476. DOI: 10.1038/nature25180. arXiv: 1712.01860 [astro-ph.GA].
- Barnes, Joshua E. and Lars Hernquist (Dec. 1992). “Formation of dwarf galaxies in tidal tails”. In: *Nature* 360.6406, pp. 715–717. DOI: 10.1038/360715a0.
- Baumgardt, Alessia Gualandris, and Simon Portegies Zwart (2006). “Ejection of hypervelocity stars from the Galactic Centre by intermediate-mass black holes”. In: *Monthly Notices of the Royal Astronomical Society* 372.1, pp. 174–182.
- Baumgardt, H. (2017). “N-body modelling of globular clusters: masses, mass-to-light ratios and intermediate-mass black holes”. In: *Monthly Notices of the Royal Astronomical Society* 464.2, pp. 2174–2202. DOI: 10.1093/mnras/stw2488. arXiv: 1609.08794 [astro-ph.GA].
- Bekenstein, J. and M. Milgrom (Nov. 1984). “Does the missing mass problem signal the breakdown of Newtonian gravity?” In: *apj* 286, pp. 7–14. DOI: 10.1086/162570.
- Bennett, DP et al. (2002). “Gravitational microlensing events due to stellar-mass black holes”. In: *The Astrophysical Journal* 579.2, pp. 639–659.
- Binney, James and Scott Tremaine (2011). *Galactic dynamics*. Princeton university press.
- Blanchet, Luc (Dec. 2014). “Gravitational Radiation from Post-Newtonian Sources and Inspiral Compact Binaries”. In: *Living Reviews in Relativity* 17.1, 2, p. 2. DOI: 10.12942/lrr-2014-2. arXiv: 1310.1528 [gr-qc].
- Boltzmann, Ludwig (1995). *Lectures on Gas Theory*. Trans. by Stephen G. Brush. Originally published in German as *Vorlesungen über Gastheorie*, 1896–1898. New York: Dover Publications.
- Bradač, Maruša et al. (Dec. 2025). “Star Formation under a Cosmic Microscope: Highly Magnified $z = 11$ Galaxy behind the Bullet Cluster”. In: *The Astrophysical Journal Letters* 995.2, p. L74. DOI: 10.3847/2041-8213/ae27d2. URL: <https://doi.org/10.3847/2041-8213/ae27d2>.

- Butcher, John Charles (2016). *Numerical methods for ordinary differential equations*. John Wiley & Sons.
- Capozziello, Salvatore and Mariafelicia De Laurentis (2008). “Gravitational waves from stellar encounters”. In: *Astroparticle Physics* 30.2, pp. 105–112.
- Chandrasekhar, S. (Mar. 1943). “Dynamical Friction. I. General Considerations: the Coefficient of Dynamical Friction.” In: *The Astrophysical Journal* 97, p. 255. DOI: 10.1086/144517.
- Cimatti, Andrea, Filippo Fraternali, and Carlo Nipoti (2019). *Introduction to Galaxy Formation and Evolution: From Primordial Gas to Present-Day Galaxies*. Cambridge University Press. DOI: 10.1017/9781108610998.
- Ciotti, Luca (2021). *Introduction to Stellar Dynamics*. Cambridge: Cambridge University Press.
- Coles, Peter and Francesco Lucchin (1995). *Cosmology. The origin and evolution of cosmic structure*.
- Colpi, Monica et al. (Feb. 2024). “LISA Definition Study Report”. In: *arXiv e-prints*, arXiv:2402.07571, arXiv:2402.07571. DOI: 10.48550/arXiv.2402.07571. arXiv: 2402.07571 [astro-ph.CO].
- Devecchi, B. and M. Volonteri (Mar. 2009). “Formation of the First Nuclear Clusters and Massive Black Holes at High Redshift”. In: *The Astrophysical Journal* 694.1, pp. 302–313. DOI: 10.1088/0004-637X/694/1/302. arXiv: 0810.1057 [astro-ph].
- Eddington, A. S. (May 1916). “The distribution of stars in globular clusters”. In: *mnras* 76, pp. 572–585. DOI: 10.1093/mnras/76.7.572.
- Farmer, R et al. (2019). “Mind the gap: the location of the lower edge of the pair-instability supernova black hole mass gap”. In: *The Astrophysical Journal* 887.1, p. 53.
- Fryer, C. L., S. E. Woosley, and A. Heger (Mar. 2001). “Pair-Instability Supernovae, Gravity Waves, and Gamma-Ray Transients”. In: *The Astrophysical Journal* 550.1, pp. 372–382. DOI: 10.1086/319719. arXiv: astro-ph/0007176 [astro-ph].
- Gebhardt, Karl, R. M. Rich, and Luis C. Ho (Dec. 2005). “An Intermediate-Mass Black Hole in the Globular Cluster G1: Improved Significance from New Keck and Hubble Space Telescope Observations”. In: *The Astrophysical Journal* 634.2, pp. 1093–1102. ISSN: 1538-4357. DOI: 10.1086/497023.
- Geha, M. et al. (Jan. 2006). “Local Group Dwarf Elliptical Galaxies. I. Mapping the Dynamics of NGC 205 Beyond the Tidal Radius”. In: *The Astronomical Journal* 131.1, pp. 332–342. ISSN: 1538-3881. DOI: 10.1086/498686. URL: <http://dx.doi.org/10.1086/498686>.
- Grebel, Eva K, John S Gallagher III, and Daniel Harbeck (2003). “The progenitors of dwarf spheroidal galaxies”. In: *The Astronomical Journal* 125.4, pp. 1926–1939.
- Greene, Jenny E., Jay Strader, and Luis C. Ho (Aug. 2020). “Intermediate-Mass Black Holes”. In: *Annual Review of Astronomy and Astrophysics* 58, pp. 257–312. DOI: 10.1146/annurev-astro-032620-021835. arXiv: 1911.09678 [astro-ph.GA].
- Gualandris, Alessia et al. (Jan. 2022). “Eccentricity evolution of massive black hole binaries from formation to coalescence”. In: *Monthly Notices of the Royal Astronomical Society* 511.4, pp. 4753–4765. ISSN: 1365-2966. DOI: 10.1093/mnras/stac241. URL: <http://dx.doi.org/10.1093/mnras/stac241>.
- Heggie, D. C. (Dec. 1975). “Binary evolution in stellar dynamics.” In: *mnras* 173, pp. 729–787. DOI: 10.1093/mnras/173.3.729.
- Heggie, Douglas C. and Piet Hut (Apr. 1993). “Binary–Single-Star Scattering. IV. Analytic Approximations and Fitting Formulae for Cross Sections and Reaction Rates”. In: *apjs* 85, p. 347. DOI: 10.1086/191768.
- Hills, J. G. (Aug. 1983). “The effect of low-velocity, low-mass intruders (collisionless gas) on the dynamical evolution of a binary system”. In: *aj* 88, pp. 1269–1283. DOI: 10.1086/113418.
- Hilz, Michael et al. (Oct. 2012). “Relaxation and stripping – The evolution of sizes, dispersions and dark matter fractions in major and minor mergers of elliptical galaxies”. In: *Monthly Notices of the Royal Astronomical Society* 425.4, pp. 3119–3136. ISSN: 0035-8711. DOI: 10.

- 1111/j.1365-2966.2012.21541.x. eprint: <https://academic.oup.com/mnras/article-pdf/425/4/3119/4916245/425-4-3119.pdf>. URL: <https://doi.org/10.1111/j.1365-2966.2012.21541.x>.
- Iorio, G et al. (Aug. 2019). “The effect of tides on the Sculptor dwarf spheroidal galaxy”. In: *Monthly Notices of the Royal Astronomical Society* 487.4, pp. 5692–5710. ISSN: 0035-8711. DOI: 10.1093/mnras/stz1342. eprint: <https://academic.oup.com/mnras/article-pdf/487/4/5692/28901729/stz1342.pdf>. URL: <https://doi.org/10.1093/mnras/stz1342>.
- Jahnke, Knud (Jan. 2025). “The Soltan argument at redshift 6: UV-luminous quasars contribute less than 10% to early black hole mass growth”. In: *The Open Journal of Astrophysics* 8, 9, p. 9. DOI: 10.33232/001c.129063. arXiv: 2411.03184 [astro-ph.GA].
- Khan, Fazeel Mahmood and Kelly Holley-Bockelmann (Nov. 2021). “Extremely efficient mergers of intermediate-mass black hole binaries in nucleated dwarf galaxies”. In: *MNRAS* 508.1, pp. 1174–1188. DOI: 10.1093/mnras/stab2646. arXiv: 2109.12129 [astro-ph.GA].
- Khan, Fazeel Mahmood, Fiza Javed, et al. (Nov. 2024). “The Potential for Long-lived Intermediate-mass Black Hole Binaries in the Lowest Density Dwarf Galaxies”. In: *The Astrophysical Journal* 976.1, 22, p. 22. DOI: 10.3847/1538-4357/ad8082. arXiv: 2408.14541 [astro-ph.GA].
- Khochfar, S. and J. Silk (July 2009). “Dry mergers: a crucial test for galaxy formation”. In: *MNRAS* 397.1, pp. 506–510. DOI: 10.1111/j.1365-2966.2009.14958.x. arXiv: 0809.1734 [astro-ph].
- Kikuchihiro, Shotaro et al. (Apr. 2020). “Early Low-mass Galaxies and Star-cluster Candidates at $z \sim 6-9$ Identified by the Gravitational-lensing Technique and Deep Optical/Near-infrared Imaging”. In: *The Astrophysical Journal* 893.1, 60, p. 60. DOI: 10.3847/1538-4357/ab7dbe. arXiv: 1905.06927 [astro-ph.GA].
- Kormendy, John and Douglas Richstone (Jan. 1995). “Inward Bound—The Search For Super-massive Black Holes In Galactic Nuclei”. In: *Annual Review of Astronomy and Astrophysics* 33, p. 581. DOI: 10.1146/annurev.aa.33.090195.003053.
- Lightman, Alan P and Stuart L Shapiro (1977). “The distribution and consumption rate of stars around a massive, collapsed object”. In: *Astrophysical Journal*, vol. 211, Jan. 1, 1977, pt. 1, p. 244-262. 211, pp. 244–262.
- Lima Neto, G. B., D. Gerbal, and I. Marquez (Oct. 1999). “The specific entropy of elliptical galaxies: an explanation for profile-shape distance indicators?” In: *MNRAS* 309.2, pp. 481–495. ISSN: 1365-2966. DOI: 10.1046/j.1365-8711.1999.02849.x. URL: <http://dx.doi.org/10.1046/j.1365-8711.1999.02849.x>.
- Lodato, Giuseppe and Priyamvada Natarajan (Oct. 2006). “Supermassive black hole formation during the assembly of pre-galactic discs”. In: *mnras* 371.4, pp. 1813–1823. DOI: 10.1111/j.1365-2966.2006.10801.x. arXiv: astro-ph/0606159 [astro-ph].
- Lynden-Bell, D. (Feb. 1962). “Stellar Dynamics: Potentials with Isolating Integrals”. In: *Monthly Notices of the Royal Astronomical Society* 124.2, pp. 95–123. ISSN: 0035-8711. DOI: 10.1093/mnras/124.2.95. eprint: <https://academic.oup.com/mnras/article-pdf/124/2/95/8072797/mnras124-0095.pdf>. URL: <https://doi.org/10.1093/mnras/124.2.95>.
- (Jan. 1967). “Statistical mechanics of violent relaxation in stellar systems”. In: *mnras* 136, p. 101. DOI: 10.1093/mnras/136.1.101.
- Madigan, Ann-Marie, Clovis Hopman, and Yuri Levin (Aug. 2011). “SECULAR STELLAR DYNAMICS NEAR A MASSIVE BLACK HOLE”. In: *The Astrophysical Journal* 738.1, p. 99. DOI: 10.1088/0004-637X/738/1/99. URL: <https://doi.org/10.1088/0004-637X/738/1/99>.
- Maggiore, Michele (2007). *Gravitational Waves: Volume 1: Theory and Experiments*. DOI: 10.1093/acprof:oso/9780198570745.001.0001.

- Magorrian, John et al. (June 1998). “The Demography of Massive Dark Objects in Galaxy Centers”. In: *Astrophysical Journal* 115.6, pp. 2285–2305. DOI: 10.1086/300353. arXiv: astro-ph/9708072 [astro-ph].
- Maiolino, Roberto et al. (Nov. 2024). “JADES: The diverse population of infant black holes at $4 < z < 11$: Merging, tiny, poor, but mighty”. In: *aap* 691, A145, A145. DOI: 10.1051/0004-6361/202347640. arXiv: 2308.01230 [astro-ph.GA].
- Mannerkoski, Matias et al. (Sept. 2023). “KETJU - resolving small-scale supermassive black hole dynamics in GADGET-4”. In: *MNRAS* 524.3, pp. 4062–4082. DOI: 10.1093/mnras/stad2139. arXiv: 2306.04963 [astro-ph.IM].
- McConnachie, A. W. et al. (Jan. 2005). “Distances and metallicities for 17 Local Group galaxies”. In: *MNRAS* 356.3, pp. 979–997. DOI: 10.1111/j.1365-2966.2004.08514.x. arXiv: astro-ph/0410489 [astro-ph].
- Merritt, David (Nov. 2013). “Loss-cone dynamics”. In: *Classical and Quantum Gravity* 30.24, p. 244005. ISSN: 1361-6382. DOI: 10.1088/0264-9381/30/24/244005. URL: <http://dx.doi.org/10.1088/0264-9381/30/24/244005>.
- Merritt, David and Miloš Milosavljević (2005). “Massive Black Hole Binary Evolution”. In: *Living Reviews in Relativity* 8. ISSN: 1433-8351. DOI: 10.12942/lrr-2005-8. URL: <http://dx.doi.org/10.12942/lrr-2005-8>.
- Mikkola, Seppo and Kiyotaka Tanikawa (Dec. 1999a). “Algorithmic regularization of the few-body problem”. In: *mnras* 310.3, pp. 745–749. DOI: 10.1046/j.1365-8711.1999.02982.x.
- (Aug. 1999b). “Explicit Symplectic Algorithms For Time-Transformed Hamiltonians”. In: *Celestial Mechanics and Dynamical Astronomy* 74.4, pp. 287–295. DOI: 10.1023/A:1008368322547.
- Mikkola, Seppo and Mauri J. Valtonen (Nov. 1992). “Evolution of binaries in the field of light particles and the problem of two black holes”. In: *Monthly Notices of the Royal Astronomical Society* 259.1, pp. 115–120. ISSN: 0035-8711. DOI: 10.1093/mnras/259.1.115. eprint: <https://academic.oup.com/mnras/article-pdf/259/1/115/3513810/mnras259-0115.pdf>. URL: <https://doi.org/10.1093/mnras/259.1.115>.
- Milgrom, M. (July 1983). “A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis.” In: *apj* 270, pp. 365–370. DOI: 10.1086/161130.
- Milosavljević, Miloš (2003). “The Final Parsec Problem”. In: *AIP Conference Proceedings*. Vol. 686. AIP, pp. 201–210. DOI: 10.1063/1.1629432. URL: <http://dx.doi.org/10.1063/1.1629432>.
- Milosavljević, Miloš and David Merritt (Dec. 2001). “Formation of Galactic Nuclei”. In: *The Astrophysical Journal* 563.1, p. 34. DOI: 10.1086/323830. URL: <https://doi.org/10.1086/323830>.
- Nakane, Minami et al. (2026). *VENUS: A Strongly Lensed Clumpy Galaxy at $z \sim 11 - 12$ behind the Galaxy Cluster MACS J0257.1-2325*. arXiv: 2511.14483 [astro-ph.GA]. URL: <https://arxiv.org/abs/2511.14483>.
- Neumayer, Nadine, Anil Seth, and Torsten Böker (2020). “Nuclear star clusters”. In: *The Astronomy and Astrophysics Review* 28.1. ISSN: 1432-0754. DOI: 10.1007/s00159-020-00125-0. URL: <http://dx.doi.org/10.1007/s00159-020-00125-0>.
- Nguyen, Dieu D. et al. (May 2018). “Nearby Early-type Galactic Nuclei at High Resolution: Dynamical Black Hole and Nuclear Star Cluster Mass Measurements”. In: *The Astrophysical Journal* 858.2, p. 118. DOI: 10.3847/1538-4357/aabe28. URL: <https://doi.org/10.3847/1538-4357/aabe28>.
- Nipoti, Carlo, Pasquale Londrillo, and Luca Ciotti (Oct. 2007). “Galaxy merging in modified Newtonian dynamics”. In: *Monthly Notices of the Royal Astronomical Society: Letters* 381.1, pp. L104–L108. ISSN: 1745-3925. DOI: 10.1111/j.1745-3933.2007.00378.x. URL: <http://dx.doi.org/10.1111/j.1745-3933.2007.00378.x>.

- Ono, Yoshiaki et al. (July 2023). “Morphologies of Galaxies at $z \gtrsim 9$ Uncovered by JWST/NIRCam Imaging: Cosmic Size Evolution and an Identification of an Extremely Compact Bright Galaxy at $z \sim 12$ ”. In: *Astrophysical Journal* 951.1, 72, p. 72. DOI: 10.3847/1538-4357/acd44a. arXiv: 2208.13582 [astro-ph.GA].
- Perets, Hagai B. and Tal Alexander (Apr. 2008). “Massive Perturbors and the Efficient Merger of Binary Massive Black Holes”. In: *The Astrophysical Journal* 677.1, p. 146. DOI: 10.1086/527525. URL: <https://doi.org/10.1086/527525>.
- Perets, Hagai B., Clovis Hopman, and Tal Alexander (Feb. 2007). “Massive Perturber-driven Interactions between Stars and a Massive Black Hole”. In: *The Astrophysical Journal* 656.2, pp. 709–720. DOI: 10.1086/510377. arXiv: astro-ph/0606443 [astro-ph].
- Peters, P. C. and J. Mathews (July 1963). “Gravitational Radiation from Point Masses in a Keplerian Orbit”. In: *Physical Review* 131.1, pp. 435–440. DOI: 10.1103/PhysRev.131.435.
- Peterson, Bradley M (1997). *An introduction to active galactic nuclei*. Cambridge University Press.
- Planck Collaboration et al. (Sept. 2020). “Planck 2018 results. VI. Cosmological parameters”. In: *aap* 641, A6, A6. DOI: 10.1051/0004-6361/201833910. arXiv: 1807.06209 [astro-ph.CO].
- Plummer, H. C. (Mar. 1911). “On the Problem of Distribution in Globular Star Clusters: (Plate 8.)” In: *Monthly Notices of the Royal Astronomical Society* 71.5, pp. 460–470. ISSN: 0035-8711. DOI: 10.1093/mnras/71.5.460. eprint: <https://academic.oup.com/mnras/article-pdf/71/5/460/2937497/mnras71-0460.pdf>. URL: <https://doi.org/10.1093/mnras/71.5.460>.
- Prugniel, P. and F. Simien (May 1997). “The fundamental plane of early-type galaxies: non-homology of the spatial structure.” In: *Astronomy & Astrophysics* 321, pp. 111–122.
- Quinlan, Gerald D. (July 1996). “The dynamical evolution of massive black hole binaries I. Hardening in a fixed stellar background”. In: *New Astronomy* 1.1, pp. 35–56. DOI: 10.1016/S1384-1076(96)00003-6. arXiv: astro-ph/9601092 [astro-ph].
- Rantala, Antti et al. (Mar. 2020). “MSTAR - a fast parallelized algorithmically regularized integrator with minimum spanning tree coordinates”. In: *mnras* 492.3, pp. 4131–4148. DOI: 10.1093/mnras/staa084. arXiv: 2001.03180 [astro-ph.IM].
- Reines, Amy E., Jenny E. Greene, and Marla Geha (Oct. 2013). “Dwarf Galaxies with Optical Signatures of Active Massive Black Holes”. In: *Astrophysical Journal* 775.2, 116, p. 116. DOI: 10.1088/0004-637X/775/2/116. arXiv: 1308.0328 [astro-ph.CO].
- Rodriguez, Carl L. et al. (Aug. 2019). “Black holes: The next generation—repeated mergers in dense star clusters and their gravitational-wave properties”. In: *Physical Review D* 100.4. ISSN: 2470-0029. DOI: 10.1103/physrevd.100.043027. URL: <http://dx.doi.org/10.1103/PhysRevD.100.043027>.
- Saslaw, W. C. (1985). *Gravitational physics of stellar and galactic systems C. Saslaw*.
- Saslaw, William C. (Oct. 1970). “Violent Relaxation in Galaxies and Stellar Systems”. In: *Monthly Notices of the Royal Astronomical Society* 150.3, pp. 299–309. ISSN: 0035-8711. DOI: 10.1093/mnras/150.3.299. URL: <https://doi.org/10.1093/mnras/150.3.299>.
- Schutte, Zachary, Amy E. Reines, and Jenny E. Greene (Dec. 2019). “The Black Hole-Bulge Mass Relation Including Dwarf Galaxies Hosting Active Galactic Nuclei”. In: *The Astrophysical Journal* 887.2, 245, p. 245. DOI: 10.3847/1538-4357/ab35dd. arXiv: 1908.00020 [astro-ph.GA].
- Sesana, Alberto (Aug. 2010). “Self Consistent Model for the Evolution of Eccentric Massive Black Hole Binaries in Stellar Environments: Implications for Gravitational Wave Observations”. In: *Astrophysical Journal* 719.1, pp. 851–864. DOI: 10.1088/0004-637X/719/1/851. arXiv: 1006.0730 [astro-ph.CO].

- Sesana, Alberto, Francesco Haardt, and Piero Madau (Nov. 2006). “Interaction of Massive Black Hole Binaries with Their Stellar Environment. I. Ejection of Hypervelocity Stars”. In: *Astrophysical Journal* 651.1, pp. 392–400. DOI: 10.1086/507596. arXiv: astro-ph/0604299 [astro-ph].
- (Oct. 2008). “Interaction of Massive Black Hole Binaries with Their Stellar Environment. III. Scattering of Bound Stars”. In: *Astrophysical Journal* 686.1, pp. 432–447. DOI: 10.1086/590651. arXiv: 0710.4301 [astro-ph].
- Sesana, Alberto and Fazeel Mahmood Khan (Nov. 2015). “Scattering experiments meet N-body - I. A practical recipe for the evolution of massive black hole binaries in stellar environments”. In: *MNRAS* 454.1, pp. L66–L70. DOI: 10.1093/mnrasl/slv131. arXiv: 1505.02062 [astro-ph.GA].
- Springel, Volker (Dec. 2005). “The cosmological simulation code gadget-2”. In: *Monthly Notices of the Royal Astronomical Society* 364.4, pp. 1105–1134. ISSN: 0035-8711. DOI: 10.1111/j.1365-2966.2005.09655.x. eprint: <https://academic.oup.com/mnras/article-pdf/364/4/1105/18657201/364-4-1105.pdf>. URL: <https://doi.org/10.1111/j.1365-2966.2005.09655.x>.
- Springel, Volker et al. (Sept. 2021). “Simulating cosmic structure formation with the GADGET-4 code”. In: *mnras* 506.2, pp. 2871–2949. DOI: 10.1093/mnras/stab1855. arXiv: 2010.03567 [astro-ph.IM].
- Taylor, Stephen R. (2021). *The Nanohertz Gravitational Wave Astronomer*. arXiv: 2105.13270 [astro-ph.HE]. URL: <https://arxiv.org/abs/2105.13270>.
- Terzić, Balša and Alister W. Graham (Sept. 2005). “Density-potential pairs for spherical stellar systems with Sérsic light profiles and (optional) power-law cores”. In: *MNRAS* 362.1, pp. 197–212. DOI: 10.1111/j.1365-2966.2005.09269.x. arXiv: astro-ph/0506192 [astro-ph].
- Thorne, Kip S. and James B. Hartle (Apr. 1985). “Laws of motion and precession for black holes and other bodies”. In: *prd* 31.8, pp. 1815–1837. DOI: 10.1103/PhysRevD.31.1815.
- Toomre, Alar and Juri Toomre (Dec. 1972). “Galactic Bridges and Tails”. In: *ApJ* 178, pp. 623–666. DOI: 10.1086/151823.
- Tremaine, Scott et al. (Aug. 2002). “The Slope of the Black Hole Mass versus Velocity Dispersion Correlation”. In: *Astrophysical Journal* 574.2, pp. 740–753. DOI: 10.1086/341002. arXiv: astro-ph/0203468 [astro-ph].
- Valluri, Monica et al. (2005). “The Low End of the Supermassive Black Hole Mass Function: Constraining the Mass of a Nuclear Black Hole in NGC 205 via Stellar Kinematics”. In: *The Astrophysical Journal* 628.1, pp. 137–152. ISSN: 1538-4357. DOI: 10.1086/430752. URL: <http://dx.doi.org/10.1086/430752>.
- White, S. D. M. and M. J. Rees (May 1978). “Core condensation in heavy halos: a two-stage theory for galaxy formation and clustering.” In: *mnras* 183, pp. 341–358. DOI: 10.1093/mnras/183.3.341.